

Module 6: Chapters

- **Logarithm**
- **Sequence & Series**
- **Principles of Mathematical Induction**
- **Quadratic Equations**

LOGARITHM

1. DEFINITION :

The logarithm of a number N to a base ' a ' is an exponent indicating the power to which the base ' a ' must be raised to obtain the number N . This number is designated as $\log_a N$. (Read it "Log N on base a "). Here N is usually called argument of the Logarithm and ' a ' is called base of the Logarithm.

Hence : $\log_a N = x \Leftrightarrow a^x = N$, $a > 0$, $a \neq 1$ and $N > 0$

By the definition of logarithm, $\log_2 16$ is the exponent indicating the power to which 2 must be raised in order to obtain 16.

As $2^4 = 16$, hence $\log_2 16 = 4$.

Similarly

Exponential Form		Logarithmic Form
$3^5 = 243$	$\Leftrightarrow$	$\log_3 243 = 5$
$5^4 = 625$	$\Leftrightarrow$	$\log_5 625 = 4$
$2^{-3} = \frac{1}{8}$	$\Leftrightarrow$	$\log_2 \frac{1}{8} = -3$
$7^0 = 1$	$\Leftrightarrow$	$\log_7 1 = 0$

Note that the expressions $\log_3(-27)$, $\log_1 16$, $\log_0 5$ and $\log_2 0$ has no sense in real numbers since the equations $3^x = -27$, $1^x = 16$, $0^x = 5$, $2^x = 0$ are absurd for any real x , the reason being obvious that no such exponent x in real number could be found.

In general, the expression $\log_a N$ is meaningful if and only if, $a > 0$, $a \neq 1$ and $N > 0$.

The existence and uniqueness of the number $\log_a N$ follows from the properties of exponential functions.

Illustration 1 : If $\log_4 m = 1.5$, then find the value of m .

Solution : $\log_4 m = 1.5 \Rightarrow m = 4^{3/2} \Rightarrow m = 8$

Illustration 2 : If $\log_5 p = a$ and $\log_2 q = a$, then prove that $\frac{p^4 q^4}{100} = 100^{2a-1}$

Solution : $\log_5 p = a \Rightarrow p = 5^a$

$\log_2 q = a \Rightarrow q = 2^a$

$$\Rightarrow \frac{p^4 q^4}{100} = \frac{5^{4a} \cdot 2^{4a}}{100} = \frac{(10)^{4a}}{100} = \frac{(100)^{2a}}{100} = 100^{2a-1}$$

Illustration 3 : The value of N , satisfying $\log_a[1 + \log_b\{1 + \log_c(1 + \log_p N)\}] = 0$ is -

- (A) 4 (B) 3 (C) 2 (D) 1

Solution : $1 + \log_b\{1 + \log_c(1 + \log_p N)\} = a^0 = 1$

$$\Rightarrow \log_b\{1 + \log_c(1 + \log_p N)\} = 0 \quad \Rightarrow \quad 1 + \log_c(1 + \log_p N) = 1$$

$$\Rightarrow \log_c(1 + \log_p N) = 0 \quad \Rightarrow \quad 1 + \log_p N = 1$$

$$\Rightarrow \log_p N = 0 \quad \Rightarrow \quad N = 1$$

Ans. (D)

Do yourself - 1 :

1 Express the following in logarithmic form :

(a) $81 = 3^4$ (b) $0.001 = 10^{-3}$ (c) $2 = 128^{1/7}$

2 Express the following in exponential form :

(a) $\log_2 32 = 5$ (b) $\log_{\sqrt{2}} 4 = 4$ (c) $\log_{10} 0.01 = -2$

3 If $\log_{2\sqrt{3}} 1728 = x$, then find x .

4 Find the logarithms of the following numbers to the base 2 :

(i) 1 (ii) 2 (iii) 4 (iv) 8

(v) $\frac{1}{2}$ (vi) $\frac{1}{32}$ (vii) $\frac{1}{16}$ (viii) $\sqrt{2}$

(ix) $\sqrt[3]{8}$ (x) $2\sqrt{2}$ (xi) $\frac{1}{\sqrt[5]{2}}$ (xii) $\frac{1}{\sqrt[7]{8}}$

5 Find the logarithms of the following numbers to the base $\frac{1}{2}$:

(i) 1 (ii) $\frac{1}{2}$ (iii) $\frac{1}{8}$ (iv) 16

(v) $\sqrt{2}$ (vi) $\frac{1}{\sqrt{2}}$ (vii) $2\sqrt{2}$ (viii) $\frac{1}{4\sqrt{2}}$

6 Find the logarithms of the following numbers to the base 3 :

(i) 1 (ii) 3 (iii) 9 (iv) 81

(v) $\frac{1}{3}$ (vi) $\sqrt{3}$ (vii) $\frac{1}{3\sqrt{3}}$ (viii) $27\sqrt{3}$

(ix) $\sqrt[7]{9}$

7. Find the logarithms of the following numbers to the base $\frac{1}{3}$:
- (i) 1 (ii) $\frac{1}{3}$ (iii) $\frac{1}{9}$ (iv) 3
- (v) 9 (vi) 81 (vii) $\sqrt[3]{3}$ (viii) $\frac{1}{\sqrt[3]{3}}$
- (ix) $9\sqrt{3}$ (x) $\frac{1}{9\sqrt[4]{3}}$
8. Find the logarithms of the following numbers to the base 5.
- (i) 1 (ii) 5 (iii) 25 (iv) 625
- (v) $\frac{1}{5}$ (vi) $\frac{1}{25}$ (vii) $\frac{1}{\sqrt{5}}$ (viii) $\sqrt{\sqrt{5}}$
- (ix) $5^{\frac{1}{2}}$ (x) $5^{\frac{1}{3}}$ (xi) $\sqrt[4]{5\sqrt[3]{5}}$
9. Find all values of 'a' for which each of the following equalities hold true :
- (i) $\log_2 a = 2$ (ii) $\log_a 2 = 1$ (iii) $\log_a 1 = 0$ (iv) $\log_{10}(a(a+3)) = 1$
- (v) $\log_{1/3}(a^2 - 1) = -1$ (vi) $\log_2(a^2 - 5) = 2$ (vii) $\log_3 a = 2$ (viii) $\log_{1/3}(a) = 4$
- (ix) $\log_{1/3}(a) = 0$ (x) $\log_a(a+2) = 2$ (xi) $\log_3(a^2 + 1) = 1$
10. Evaluate the following :
- (i) $\log_5 \left(\frac{1}{\sqrt{5}} \right)$ (ii) $\log_{\sqrt{5}+1}(6 + 2\sqrt{5})$
- (iii) $\log_3(4\sin^2(x) + 4\cos^2(x) - 1)$ (iv) $\log_{\sqrt{3}-\sqrt{2}}(\sqrt{5-2\sqrt{6}})$
- (v) $\log_{1/3} \sqrt[4]{729 \cdot \sqrt[3]{9^{-1}} \cdot 27^{-4/3}}$

2. FUNDAMENTAL LOGARITHMIC IDENTITY :

From the definition of the logarithm of the number N to the base 'a', we have an identity :

$$a^{\log_a N} = N, \quad a > 0, \quad a \neq 1 \quad \text{and} \quad N > 0$$

This is known as the **FUNDAMENTAL LOGARITHMIC IDENTITY**.

Note :

Using the basic definition of logarithm we have 3 important deductions :

- (a) $\log_a 1 = 0$ i.e. logarithm of unity to any base is zero ($a > 0$; $a \neq 1$).
- (b) $\log_N N = 1$ i.e. logarithm of a number to the same base is 1.
($N > 0$; $N \neq 1$)
- (c) $\log_{\frac{1}{N}} N = -1 = \log_N \frac{1}{N}$ i.e. logarithm of a number to the base as its reciprocal is -1 .
($N > 0$; $N \neq 1$)

3. THE PRINCIPAL PROPERTIES OF LOGARITHMS :

If m, n are arbitrary positive numbers where $a > 0$, $a \neq 1$ and x is any real number, then

$$(a) \log_a mn = \log_a m + \log_a n \quad (b) \log_a \frac{m}{n} = \log_a m - \log_a n \quad (c) \log_a m^x = x \log_a m$$

Illustration 4 : Find the value of $2 \log \frac{2}{5} + 3 \log \frac{25}{8} - \log \frac{625}{128}$

Solution :

$$\begin{aligned} & 2 \log \frac{2}{5} + 3 \log \frac{25}{8} + \log \frac{128}{625} \\ &= \log \frac{2^2}{5^2} + \log \left(\frac{5^2}{2^3} \right)^3 + \log \frac{2^7}{5^4} \\ &= \log \frac{2^2}{5^2} \cdot \frac{5^6}{2^9} \cdot \frac{2^7}{5^4} = \log 1 = 0 \end{aligned}$$

Illustration 5 : If $\log_e x - \log_e y = a$, $\log_e y - \log_e z = b$ & $\log_e z - \log_e x = c$, then find the value of

$$\left(\frac{x}{y} \right)^{b-c} \times \left(\frac{y}{z} \right)^{c-a} \times \left(\frac{z}{x} \right)^{a-b}.$$

Solution : $\log_e x - \log_e y = a \Rightarrow \log_e \frac{x}{y} = a \Rightarrow \frac{x}{y} = e^a$

$$\log_e y - \log_e z = b \Rightarrow \log_e \frac{y}{z} = b \Rightarrow \frac{y}{z} = e^b$$

$$\log_e z - \log_e x = c \Rightarrow \log_e \frac{z}{x} = c \Rightarrow \frac{z}{x} = e^c$$

$$\begin{aligned} \therefore (e^a)^{b-c} \times (e^b)^{c-a} \times (e^c)^{a-b} \\ = e^{a(b-c)+b(c-a)+c(a-b)} = e^0 = 1 \end{aligned}$$

Illustration 6 : If $a^2 + b^2 = 23ab$, then prove that $\log \frac{(a+b)}{5} = \frac{1}{2}(\log a + \log b)$.

Solution : $a^2 + b^2 = (a+b)^2 - 2ab = 23ab$

$$\Rightarrow (a+b)^2 = 25ab \Rightarrow a+b = 5\sqrt{ab} \quad \dots(i)$$

Using (i)

$$\text{L.H.S.} = \log \frac{(a+b)}{5} = \log \frac{5\sqrt{ab}}{5} = \frac{1}{2} \log ab = \frac{1}{2}(\log a + \log b) = \text{R.H.S.}$$

Illustration 7 : If $\log_a x = p$ and $\log_b x^2 = q$, then $\log_x \sqrt{ab}$ is equal to (where $a, b, x \in \mathbb{R}^+ - \{1\}$)-

- (A) $\frac{1}{p} + \frac{1}{q}$ (B) $\frac{1}{2p} + \frac{1}{q}$ (C) $\frac{1}{p} + \frac{1}{2q}$ (D) $\frac{1}{2p} + \frac{1}{2q}$

Solution : $\log_a x = p \Rightarrow a^p = x \Rightarrow a = x^{1/p}$.
similarly $b^q = x^2 \Rightarrow b = x^{2/q}$

$$\text{Now, } \log_x \sqrt{ab} = \log_x \sqrt{x^{1/p} x^{2/q}} = \log_x x^{\left(\frac{1}{p} + \frac{2}{q}\right) \frac{1}{2}} = \frac{1}{2p} + \frac{1}{q}$$

Do yourself - 2 :

1 Find the value of the following :

(a) $\log_{1.43} \frac{43}{30}$ (b) $\left(\frac{1}{2}\right)^{\log_2 5}$

2. If $4^{\log_2 2x} = 36$, then find x .

3. Show that $\frac{1}{2} \log 9 + 2 \log 6 + \frac{1}{4} \log 81 - \log 12 = 3 \log 3$

4. Find the value of following

(i) $\log_{12} 8 + \log_{12} 3 + \log_{12} 6$ (ii) $\log_5 \frac{500}{3} - \log_5 \frac{4}{3}$
(iii) $\log_{39} \frac{15}{7} + \log_{39} \frac{13}{3} - \log_{39} \frac{5}{21}$ (iv) $2 \log_6 2 + 3 \log_6 3 + \log_6 12$

4. BASE CHANGING THEOREM :

Can be stated as "quotient of the logarithm of two numbers is independent of their common base."

Symbolically, $\log_b m = \frac{\log_a m}{\log_a b}$, where $a > 0, a \neq 1, b > 0, b \neq 1$

Note :

(i) $\log_b a \cdot \log_a b = \frac{\log a}{\log b} \cdot \frac{\log b}{\log a} = 1$; hence $\log_b a = \frac{1}{\log_a b}$.

(ii) $a^{\log_b c} = c^{\log_b a}$

(iii) **Base power formula :** $\log_{a^k} m = \frac{1}{k} \log_a m$

- (iv) The base of the logarithm can be any positive number other than 1, but in normal practice, only two bases are popular, these are 10 and $e (=2.718 \text{ approx})$. Logarithms of numbers to the base 10 are named as 'common logarithm' and the logarithms of numbers to the base e are named as Natural or Napierian logarithm. **We will consider $\log x$ as $\log_e x$ or $\ln x$.**
- (v) Conversion of base e to base 10 & viceversa :

$$\log_e a = \frac{\log_{10} a}{\log_{10} e} = 2.303 \times \log_{10} a ; \quad \log_{10} a = \frac{\log_e a}{\log_e 10} = \log_{10} e \times \log_e a = 0.434 \log_e a$$

- (vi) Some important values : $\log_{10} 2 \approx 0.3010$; $\log_{10} 3 \approx 0.4771$; $\ln 2 \approx 0.693$, $\ln 10 \approx 2.303$
- (vii) The positive real number 'n' is called the antilogarithm of a number 'm' to base 'a' if $\log_a n = m$
Thus, $\log_a n = m \Leftrightarrow n = \text{antilog}_a m$

Illustration 8 : If a, b, c are distinct positive real numbers different from 1 such that

$(\log_b a \cdot \log_c a - \log_a a) + (\log_a b \cdot \log_c b - \log_b b) + (\log_a c \cdot \log_b c - \log_c c) = 0$, then abc is equal to -

- (A) 0 (B) e (C) 1 (D) none of these

Solution : $(\log_b a \log_c a - 1) + (\log_a b \cdot \log_c b - 1) + (\log_a c \log_b c - 1) = 0$

$$\Rightarrow \frac{\log a}{\log b} \cdot \frac{\log a}{\log c} + \frac{\log b}{\log a} \cdot \frac{\log b}{\log c} + \frac{\log c}{\log a} \cdot \frac{\log c}{\log b} = 3$$

$$\Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 = 3 \log a \log b \log c$$

$$\Rightarrow (\log a + \log b + \log c) = 0 \quad [\because \text{If } a^3 + b^3 + c^3 - 3abc = 0, \text{ then } a + b + c = 0 \text{ if } a \neq b \neq c]$$

$$\Rightarrow \log abc = \log 1 \Rightarrow abc = 1$$

Illustration 9 : Evaluate : $81^{1/\log_5 3} + 27^{\log_9 36} + 3^{4/\log_7 9}$

Solution : $81^{\log_3 5} + 3^{3 \log_9 36} + 3^{4 \log_9 7}$

$$= 3^{4 \log_3 5} + 3^{\log_3 (36)^{3/2}} + 3^{\log_3 7^2} = 625 + 216 + 49 = 890.$$

Illustration 10 : Show that $\log_4 18$ is an irrational number.

Solution : $\log_4 18 = \log_4 (3^2 \times 2) = 2 \log_4 3 + \log_4 2 = 2 \frac{\log_2 3}{\log_2 4} + \frac{1}{\log_2 4} = \log_2 3 + \frac{1}{2}$

assume the contrary, that this number $\log_2 3$ is rational number.

$$\Rightarrow \log_2 3 = \frac{p}{q}. \text{ Since } \log_2 3 > 0 \text{ both numbers } p \text{ and } q \text{ may be regarded as natural number}$$

$$\Rightarrow 3 = 2^{p/q} \Rightarrow 2^p = 3^q$$

But this is not possible for any natural number p and q . The resulting contradiction completes the proof.

Illustration 11 : If in a right angled triangle, a and b are the lengths of sides and c is the length of hypotenuse and $c - b \neq 1$, $c + b \neq 1$, then show that

$$\log_{c+b} a + \log_{c-b} a = 2 \log_{c+b} a \cdot \log_{c-b} a.$$

Solution : We know that in a right angled triangle

$$c^2 = a^2 + b^2$$

$$c^2 - b^2 = a^2 \quad \dots\dots\dots (i)$$

$$\begin{aligned} \text{LHS} &= \frac{1}{\log_a (c+b)} + \frac{1}{\log_a (c-b)} = \frac{\log_a (c-b) + \log_a (c+b)}{\log_a (c+b) \cdot \log_a (c-b)} \\ &= \frac{\log_a (c^2 - b^2)}{\log_a (c+b) \cdot \log_a (c-b)} = \frac{\log_a a^2}{\log_a (c+b) \cdot \log_a (c-b)} \quad (\text{using (i)}) \\ &= \frac{2}{\log_a (c+b) \cdot \log_a (c-b)} = 2 \log_{(c+b)} a \cdot \log_{(c-b)} a = \text{RHS} \end{aligned}$$

Do yourself - 3 :

1. (i) Evaluate : $\frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$ (ii) Evaluate : $\log_9 27 - \log_{27} 9$
- (iii) Evaluate : $2^{\log_3 5} - 5^{\log_3 2}$ (iv) Evaluate : $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$
- (v) If $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} > x$, then x can be -
 (A) 2 (B) 3 (C) 3.5 (D) π
- (vi) If $\log_a 3 = 2$ and $\log_b 8 = 3$, then $\log_a b$ is -
 (A) $\log_3 2$ (B) $\log_2 3$ (C) $\log_3 4$ (D) $\log_4 3$
2. Find value of following :
 (i) $\log_7 3 \cdot \log_3 2 \cdot \log_2 7 \cdot \log_2 (125)$ (ii) $25^{\log_5 3}$
 (iii) $6^{\log_6 5} + 3^{\log_9 16}$ (iv) $\log_6 4 + \frac{1}{\log_9 6}$
3. Given that $\log_2 a = s$, $\log_4 b = s^2$ and $\log_{c^2} (8) = \frac{2}{s^3 + 1}$. Write $\log_2 \frac{a^2 b^5}{c^4}$ as a function of 's'
 (a, b, c > 0, c ≠ 1).

4. Prove that $a^x - b^y = 0$ where $x = \sqrt{\log_a b}$ & $y = \sqrt{\log_b a}$, $a > 0$, $b > 0$ & $a, b \neq 1$.
5. Compute the following : $a^{\frac{\log_b(\log_b N)}{\log_b a}}$
6. The value of $\log_5 9 \cdot \log_2(\sqrt{5} + 2) \cdot \log_{(\sqrt{5}-2)} 5 \cdot \log_3\left(\frac{1}{4}\right)$ is
7. Calculate : $4^{5 \log_4 \sqrt{2} (3-\sqrt{6}) - 6 \log_8 (\sqrt{3}-\sqrt{2})}$
8. Simplify : $\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409} \cdot \left((\sqrt{7})^{\frac{2}{\log_{25} 7}} - (125)^{\log_{25} 6} \right)$
9. Simplify : $5^{\log_{1/5}\left(\frac{1}{2}\right)} + \log_{\sqrt{2}} \frac{4}{\sqrt{7} + \sqrt{3}} + \log_{1/2} \frac{1}{10 + 2\sqrt{21}}$
10. Find the value of $49^{(1-\log_7 2)} + 5^{-\log_5 4}$.
11. Find the value of the expression $\frac{2}{\log_4(2000)^6} + \frac{3}{\log_5(2000)^6}$
12. Let $x = (\log_{1/3} 5) (\log_{125} 343) (\log_{49} 729)$ and $y = 25^{3 \log_{289} 11 \log_{28} \sqrt{17} \log_{1331} 784}$, then the value of $x^2 + y^2$ is
13. Find antilog of $\frac{5}{6}$ to the base 64.

5. LOGARITHMIC EQUATIONS

Illustration 12 : $\log_{\frac{1}{2}} (\log_2 \sqrt{2}x) = 1$, then find x ?

Solution : $\log_{\frac{1}{2}} (\log_2 \sqrt{2}x) = 1$

$$\Rightarrow \log_2 (\sqrt{2}x) = \frac{1}{2}$$

$$\Rightarrow \sqrt{2}x = 2^{\frac{1}{2}}$$

$$\Rightarrow x = 1$$

Illustration 13 : Solve the equation $2 \log_2 (\log_2 x) + \log_{1/2} \left(\frac{3}{2} + \log_2 x \right) = 1$.

Solution : Let $\log_2 x = t$

$$\Rightarrow 2 \log_2 (t) + \log_{\frac{1}{2}} \left(\frac{3}{2} + t \right) = 1$$

$$\Rightarrow 2 \log_2 t - \log_2 \left(\frac{3}{2} + t \right) = 1$$

$$\Rightarrow \log_2 \left(\frac{t^2}{\frac{3}{2} + t} \right) = 1$$

$$\Rightarrow \frac{2t^2}{3 + 2t} = 2$$

$$\Rightarrow t^2 - 2t - 3 = 0$$

$$\Rightarrow (t + 1)(t - 3) = 0$$

$$\Rightarrow t = 3 \because t > 0$$

$$\Rightarrow x = 8$$

Illustration 14 : Solve the equation $\log x^2 - \log (2x) = 3 \log 3 - \log 6$.

Solution : $\log x^2 - \log 2x = 3 \log 3 - \log 6$, $x > 0$

$$\Rightarrow 2 \log x - \log 2 - \log x = 3 \log 3 - \log 2 - \log 3 \Rightarrow \log x = 2 \log 3 \Rightarrow \log x = \log 9$$

$$\Rightarrow x = 9$$

Illustration 15 : Solve the equation $(\log_5 x)^2 + \log_5 x + 1 = \frac{7}{\log_5 x - 1}$

Solution : Put $\log_5 x = t$, we get $t^2 + t + 1 = \frac{7}{t - 1}$

$$(t - 1)(t^2 + t + 1) = 7 \Rightarrow t^3 + t^2 + t - t^2 - t - 1 = 7 \Rightarrow t^3 - 8 = 0$$

$$\Rightarrow (t - 2)(t^2 + 2t + 4) = 0 \Rightarrow t - 2 = 0; t^2 + 2t + 4 \neq 0 \Rightarrow t = 2$$

$$\text{Now, } t = \log_5 x, \text{ so } \log_5 x = 2$$

$$x = 5^2 \Rightarrow x = 25$$

Illustration 16 : Solve the equation $|x-1|^{\log_2 x^2 - 2\log_x 4} = (x-1)^7$

Solution : Obviously $x=2$ is a solution. Since, left side is positive, $x-1 > 0$.

The equation reduces $\log_2 x^2 - 2\log_x 4 = 7$

$$\Rightarrow 2t - \frac{4}{t} = 7, t = \log_2 x$$

$$\Rightarrow 2t^2 - 7t - 4 = 0 \Rightarrow t = 4, -\frac{1}{2}$$

But $t > 0$ since $x > 1$. $\therefore t = 4$

$$\Rightarrow x = 2^4 = 16$$

$$\therefore x = 2, 16$$

Illustration 17 : Solve the equation $x^{\log_3 x^2 + (\log_3 x)^2 - 10} = \frac{1}{x^2}$

Solution : Taking $\log_3$ on both sides, we get

$$(2t + t^2 - 10)t = -2t, \quad t = \log_3 x$$

$$\Rightarrow t(t^2 + 2t - 8) = 0 \Rightarrow t = 0, 2, -4$$

$$\Rightarrow x = 1, 9, \frac{1}{81}$$

Illustration 18 : Solve the equation $4^{\log_2 \ell n x} = \ell n x - (\ell n x)^2 + 1$

Solution : $4^{\log_2 \ell n x} = 2^{2\log_2 (\ell n x)} = 2^{\log_2 (\ell n x)^2} = (\ell n x)^2$

$$\Rightarrow (\ell n x)^2 = \ell n x - (\ell n x)^2 + 1 \Rightarrow 2(\ell n x)^2 - \ell n x - 1 = 0 \Rightarrow \ell n x = 1, -\frac{1}{2}$$

But $\ell n x > 0$

$$\therefore \ell n x = 1 \Rightarrow x = e.$$

Illustration 19 : Solve the equation $x : \log_{x+1}(x^2 + x - 6)^2 = 4$

Solution : We have,

$$\log_{x+1}(x^2 + x - 6)^2 = 4 \Rightarrow (x^2 + x - 6)^2 = (x+1)^4 = (x^2 + 2x + 1)^2$$

$$\Rightarrow (x^2 + x - 6 - x^2 - 2x - 1)(x^2 + x - 6 + x^2 + 2x + 1) = 0$$

$$\Rightarrow (-x-7)(2x^2+3x-5) = 0 \Rightarrow (x+7)(x-1)(2x+5) = 0 \Rightarrow x = -7, -5/2, 1$$

The values $x = -7$ and $x = -5/2$ are rejected because they make the base $x+1$ negative

Hence, $x = 1$ is the only solution of the given equation.

Do yourself - 4 :

1. Find all values of x for which the following equalities hold true.

- (i) $\log_2 x^2 = 1$ (ii) $\log_3 x = \log_3(2-x)$ (iii) $\log_4 x^2 = \log_4 x$
 (iv) $\log_{1/2}(2x+1) = \log_{1/2}(x+1)$ (v) $\log_{1/3}(x^2+8) = -2$

2. Find all the values of x for which the following equalities hold true.

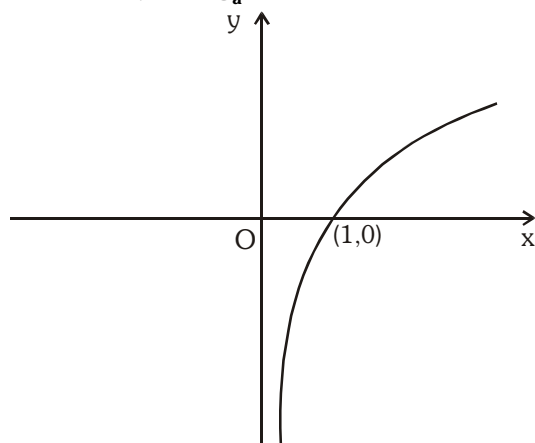
- (i) $\log_2 x^2 = 2$ (ii) $\log_{1/4} x^2 = 1$
 (iii) $\log_{1/2} x - \log_{1/2}(3-x) = 0$ (iv) $\log_2(x+1) - \log_2(2x-3) = 0$

Solve the following equations :

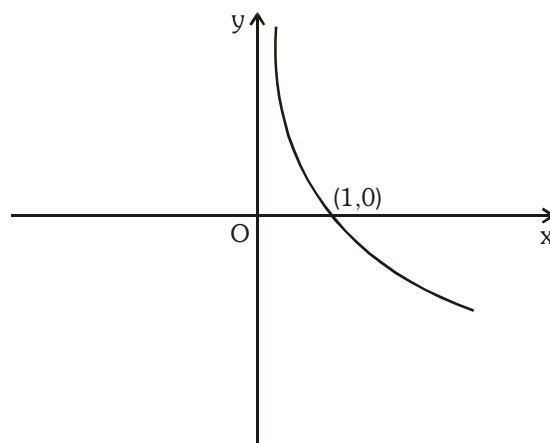
3. $\log_3(4^x + 15 \cdot 2^x + 27) - 2\log_3(4 \cdot 2^x - 3) = 0$ 4. $\frac{\log_8(8/x^2)}{(\log_8 x)^2} = 3$
 5. $\log_x 3 \cdot \log_{x/3} 3 + \log_{x/81} 3 = 0$ 6. $1 + 2 \log_x 2 \cdot \log_4(10-x) = 2/\log_4 x$
 7. $x^{x+1} = x$ 8. $x^{\log_x(x+3)^2} = 16$
 9. $\sqrt{x^{\log_{10} \sqrt{x}}} = 10$ 10. $3^{\log_a x} + 3 \cdot x^{\log_a 3} = 2$
 11. $\log_a(1 - \sqrt{1+x}) = \log_{a^2}(3 - \sqrt{1+x})$ 12. $2\log_8(2x) + \log_8(x^2+1-2x) = 4/3$
 13. $|\log_2 x| = 3$ 14. $\log_{x-1} 3 = 2$
 15. $\log_4(2\log_3(1 + \log_2(1 + 3\log_3 x))) = \frac{1}{2}$ 16. $\log_3(1 + \log_3(2^x - 7)) = 1$
 17. $\log_3(3^x - 8) = 2 - x$ 18. $\frac{\log_2(9 - 2^x)}{3 - x} = 1$
 19. $\log_{5-x}(x^2 - 2x + 65) = 2$
 20. $\log_{10} 5 + \log_{10}(x+10) - 1 = \log_{10}(21x-20) - \log_{10}(2x-1)$
 21. $x^{1+\log_{10} x} = 10x$
 22. $2(\log_x \sqrt{5})^2 - 3\log_x \sqrt{5} + 1 = 0$
 23. $3 + 2\log_{x+1} 3 = 2\log_3(x+1)$

6. GRAPHS OF LOGARITHMIC FUNCTION AND ITS INEQUALITIES

Graph of $y = \log_a x$:



when $a > 1$



When $0 < a < 1$

$$(i) \quad \log_a x < \log_a y \Leftrightarrow \begin{cases} x < y & \text{if } a > 1 \\ x > y & \text{if } 0 < a < 1 \end{cases}$$

$$(ii) \quad \text{If } a > 1, \text{ then } \log_a x < p \Rightarrow 0 < x < a^p \quad \text{and} \quad \log_a x > p \Rightarrow x > a^p$$

$$(iii) \quad \text{If } 0 < a < 1, \text{ then } \log_a x < p \Rightarrow x > a^p \quad \text{and} \quad \log_a x > p \Rightarrow 0 < x < a^p$$

Note :

(i) If base of logarithm is greater than 1 then logarithm of greater number is greater.

i.e. $\log_2 8 = 3$, $\log_2 4 = 2$ etc. and if base of logarithm is between 0 and 1 then on that base logarithm of greater number is smaller. i.e. $\log_{1/2} 8 = -3$, $\log_{1/2} 4 = -2$ etc.

(ii) It must be noted that whenever the number and the base are on the same side of unity then logarithm of that number to that base is positive, however if the number and the base are located on different side of unity then logarithm of that number to that base is negative.

$$\text{e.g. } \log_{10} \sqrt[3]{10} = \frac{1}{3}; \log_{\sqrt{7}} 49 = 4; \log_{\frac{1}{2}} \left(\frac{1}{8} \right) = 3; \log_2 \left(\frac{1}{32} \right) = -5; \log_{10} (0.001) = -3$$

Illustration 20 : Solve for x : $x^{\log_5 x} > 5$

Solution : as $x > 0$ (for existence)
now solving inequality

$$x^{\log_5 x} > 5. \text{ Taking 'log' with base '5' we have } \log_5 x \cdot \log_5 x > 1$$

$$\Rightarrow (\log_5 x - 1)(\log_5 x + 1) > 0 \Rightarrow \log_5 x > 1 \text{ or } \log_5 x < -1$$

$$\Rightarrow x > 5 \text{ or } x < 1/5. \text{ Also we must have } x > 0$$

$$\text{Thus, } x \in (0, 1/5) \text{ or } x \in (5, \infty)$$

Illustration 21 : Solve for $x : \log_3(2x + 1) < \log_3 5$.

Solution : Checking existence

$$2x + 1 > 0 \Rightarrow x > -\frac{1}{2}$$

Now solving inequality we have $2x + 1 < 5$

$$\Rightarrow 2x < 4$$

$$\Rightarrow x < 2$$

$$\Rightarrow x \in (-1/2, 2)$$

Illustration 22 : Solve for $x : (\log_{10} 100x)^2 + (\log_{10} 10x)^2 + \log_{10} x \leq 14$.

Solution : Checking existence

$$x > 0$$

Now solving inequality,

$$\text{Let } u = \log_{10} x$$

$$(2 + u)^2 + (1 + u)^2 + u \leq 14 \Rightarrow u^2 + 4u + 4 + u^2 + 2u + 1 + u \leq 14$$

$$\Rightarrow 2u^2 + 7u - 9 \leq 0 \Rightarrow 2u^2 + 9u - 2u - 9 \leq 0$$

$$\Rightarrow u(2u + 9) - 1(2u + 9) \leq 0 \Rightarrow (2u + 9)(u - 1) \leq 0$$

$$\Rightarrow \frac{-9}{2} \leq u \leq 1 \Rightarrow \frac{-9}{2} \leq \log_{10} x \leq 1 \Rightarrow 10^{-\frac{9}{2}} \leq x \leq 10$$

Illustration 23 : Solve for $x : \log_3((x + 2)(x + 4)) + \log_{1/3}(x + 2) < \frac{1}{2} \log_{\sqrt{3}} 7$.

Solution : Checking existence,

$$(x + 2)(x + 4) > 0 \text{ and } (x + 2) > 0$$

$$x < -4 \text{ or } x > -2 \text{ and } x > -2$$

Now solving inequality.

$$\log_3((x + 2)(x + 4)) + \log_{1/3}(x + 2) < \frac{1}{2} \log_{\sqrt{3}} 7$$

$$\Rightarrow \log_3(x + 2)(x + 4) - \log_3(x + 2) < \log_3 7$$

$$\Rightarrow \log_3(x + 4) < \log_3 7$$

$$\Rightarrow x + 4 < 7 \Rightarrow x < 3$$

$$\Rightarrow -2 < x < 3 \Rightarrow x \in (-2, 3)$$

Illustration 24 : Solve for $x : \log_{1/3} \log_4(x^2 - 5) > 0$

Solution : Checking existence,

$$(i) \quad \log_4(x^2 - 5) > 0 \Rightarrow x^2 - 5 > 1$$

$$\Rightarrow (x - \sqrt{6})(x + \sqrt{6}) > 0$$

$$\Rightarrow x \in (-\infty, -\sqrt{6}) \cup (\sqrt{6}, \infty)$$

$$(ii) \quad x^2 - 5 > 0 \Rightarrow x \in (-\infty, -\sqrt{5}) \cup (\sqrt{5}, \infty)$$

solving inequality,

$$\log_{1/3} \log_4(x^2 - 5) > 0$$

$$\Rightarrow \log_4(x^2 - 5) < 1$$

$$\Rightarrow x^2 - 5 < 4 \Rightarrow x^2 - 9 < 0$$

$$\Rightarrow (x - 3)(x + 3) < 0$$

$$\Rightarrow x \in (-3, 3) \quad \dots(iii)$$

$\therefore$ Answer : $(i) \cap (ii) \cap (iii)$

$$\Rightarrow x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$$

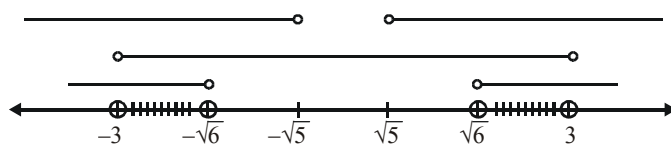


Illustration 25 : Solve for $x : \frac{\log_2(4x^2 - x - 1)}{\log_2(x^2 + 1)} > 1$

Solution : Checking existence

$$(i) \quad x^2 + 1 \neq 1 \Rightarrow x \neq 0$$

$$(ii) \quad 4x^2 - x - 1 > 0$$

Now solving inequality,

$$\frac{\log_2(4x^2 - x - 1)}{\log_2(x^2 + 1)} > 1$$

$$\Rightarrow \log_2(4x^2 - x - 1) > \log_2(x^2 + 1) \Rightarrow \log_2(4x^2 - x - 1) - \log_2(x^2 + 1) > 0$$

$$\Rightarrow \log_2 \frac{4x^2 - x - 1}{x^2 + 1} > 0$$

$$\Rightarrow 4x^2 - x - 1 - x^2 - 1 > 0 \Rightarrow 3x^2 - x - 2 > 0 \Rightarrow 3x^2 - 3x + 2x - 2 > 0$$

$$\Rightarrow 3x(x - 1) + 2(x - 1) > 0 \Rightarrow (x - 1)(3x + 2) > 0 \Rightarrow x < -2/3 \text{ or } x > 1$$

$$\Rightarrow x \in (-\infty, -2/3) \text{ or } x \in (1, \infty)$$

Note : $x < -\frac{2}{3}$ and $x > 1$; $4x^2 - x - 1 > 0$

Illustration 26 : Solve for x : $\log_4(3^x - 1) \log_{1/4} \left(\frac{3^x - 1}{16} \right) \leq \frac{3}{4}$

Solution : Checking existence $3^x - 1 > 0 \Rightarrow x > 0$,

Now solving inequality

$$\log_4(3^x - 1) \log_{1/4} \left(\frac{3^x - 1}{16} \right) \leq \frac{3}{4}$$

$$\Rightarrow \log_4(3^x - 1) \cdot [-\log_4(3^x - 1) + \log_4 16] \leq \frac{3}{4}$$

$$\Rightarrow \log_4(3^x - 1) [-\log_4(3^x - 1) + 2] \leq \frac{3}{4} \Rightarrow -[\log_4(3^x - 1)]^2 + 2[\log_4(3^x - 1)] \leq \frac{3}{4}$$

Put $\log_4(3^x - 1) = t \Rightarrow -t^2 + 2t \leq \frac{3}{4}$

$$\Rightarrow -4t^2 + 8t - 3 \leq 0 \Rightarrow 4t^2 - 8t + 3 \geq 0 \Rightarrow 4t^2 - 6t - 2t + 3 \geq 0$$

$$\Rightarrow 2t(2t - 3) - 1(2t - 3) \geq 0 \Rightarrow (2t - 3)(2t - 1) \geq 0$$

$$\Rightarrow \log_4(3^x - 1) \leq \frac{1}{2} \text{ or } \log_4(3^x - 1) \geq \frac{3}{2}$$

$$\Rightarrow 0 < 3^x - 1 \leq 4^{1/2} \text{ or } 3^x - 1 \geq 4^{3/2}$$

$$\Rightarrow 1 < 3^x \leq 3 \text{ or } 3^x \geq 9$$

$$\Rightarrow 0 < x \leq 1 \text{ or } x \geq 2$$

$$\Rightarrow x \in (0, 1] \cup [2, \infty)$$

Illustration 27 : Solve for x : $\log_{1/3}(x^2 - 6x + 18) - 2\log_{1/3}(x - 4) < 0$

Solution : Checking existence

$$(1) x^2 - 6x + 18 > 0 \Rightarrow x \in \mathbb{R}$$

$$(2) x - 4 > 0 \Rightarrow x > 4$$

Now solving inequality

$$\log_{1/3}(x^2 - 6x + 18) - \log_{1/3}(x - 4)^2 < 0$$

$$\Rightarrow \log_{1/3} \frac{(x^2 - 6x + 18)}{(x - 4)^2} < 0 \text{ and } 2 \log_{1/3}(x - 4) = \log_{1/3}(x - 4)^2$$

only when $x - 4 > 0$, so we get

$$\log_{1/3} \left(\frac{x^2 - 6x + 18}{(x - 4)^2} \right) < 0 \text{ and } x - 4 > 0 \Rightarrow x > 4 \quad \dots(1)$$

$$\Rightarrow x^2 - 6x + 18 > (x - 4)^2 \Rightarrow x^2 - 6x + 18 > x^2 - 8x + 16$$

$$2x + 2 > 0 \Rightarrow x > -1 \Rightarrow x \in (-1, \infty) \quad \dots(2)$$

from equation (1) and (2), we get $x \in (4, \infty)$

Illustration 28 : Solve for x : $\log_e(x^2 - 2x - 2) \leq 0$

Solution : The values of x satisfying the inequality $\log_e(x^2 - 2x - 2) \leq 0$ must be such that

$$0 < x^2 - 2x - 2 \leq 1$$

$$\text{we have, } x^2 - 2x - 2 > 0 \Rightarrow (x - 1)^2 > 3$$

$$\Rightarrow |x - 1|^2 > 3 \Rightarrow |x - 1| > \sqrt{3} \Rightarrow x - 1 > \sqrt{3} \text{ or } x - 1 < -\sqrt{3}$$

$$\Rightarrow x > 1 + \sqrt{3} \text{ or } x < 1 - \sqrt{3} \quad \dots(1)$$

$$\text{Again } x^2 - 2x - 2 \leq 1 \Rightarrow x^2 - 2x \leq 3 \Rightarrow (x - 1)^2 \leq 4 \Rightarrow |x - 1| \leq 2$$

$$\Rightarrow |x - 1| \leq 2 \Rightarrow -2 \leq x - 1 \leq 2 \Rightarrow -1 \leq x \leq 3 \quad \dots(2)$$

The value of x satisfying both the inequalities equation (1) and (2) are given by;

$$\text{Hence, } x \in [-1, 1 - \sqrt{3}) \cup (1 + \sqrt{3}, 3]$$

Illustration 29 : Solve for $x : \log_x \left(2x - \frac{3}{4} \right) > 2$

Solution : For existance of logarithm

$$2x - \frac{3}{4} > 0 \text{ and } x > 0 \text{ and } x \neq 1$$

$$\text{so, } x \in \left(\frac{3}{8}, \infty \right) - \{1\}$$

To find the value of x satisfying the inequality $\log_x [2x - (3/4)] > 2$

Case I. Let $0 < x < 1$

$$\text{Then, } \log_x [2x - (3/4)] > 2 \Rightarrow [2x - (3/4)] < x^2$$

$$\Rightarrow x^2 - 2x + (3/4) > 0 \Rightarrow 4x^2 - 8x + 3 > 0 \Rightarrow (2x - 1)(2x - 3) > 0$$

$$\Rightarrow \left(x - \frac{1}{2} \right) \left[x - \left(\frac{3}{2} \right) \right] > 0 \Rightarrow x > 3/2 \text{ or } x < 1/2$$

$$\Rightarrow x < 1/2 \text{ because we have } 0 < x < 1.$$

$\therefore$ But for $\log[2x - (3/4)]$ to be meaningful, we must have

$$2x - (3/4) > 0 \Rightarrow x > 3/8$$

Therefore, if $0 < x < 1$, the values of x satisfying the given inequality are given by :

$$3/8 < x < 1/2$$

Case II. Let $x > 1$

$$\text{Then, } \log_x [2x - (3/4)] > 2 \Rightarrow [2x - (3/4)] > x^2$$

$$\Rightarrow x^2 - 2x + (3/4) < 0 \Rightarrow 4x^2 - 8x + 3 < 0 \Rightarrow (2x - 1)(2x - 3) < 0$$

$$\Rightarrow \left(x - \frac{1}{2} \right) \left[x - \left(\frac{3}{2} \right) \right] < 0 \Rightarrow 1/2 < x < 3/2$$

But we have $x > 1$

$\therefore$ We must have $1 < x < 3/2$ and obviously these values of x make $2x - (3/4) > 0$

Thereofre, if $x > 1$, the values of x satisfying the given inequality are given by, $1 < x < 3/2$

$$x \in \left(\frac{3}{8}, \frac{1}{2} \right) \cup \left(1, \frac{3}{2} \right)$$

Illustration 30 : Solve for $x : \log_{0.5}(x^2 - 5x + 6) \geq -1$

Solution : Checking existence $x^2 - 5x + 6 > 0$,

$$\Rightarrow x \in (-\infty, 2) \cup (3, \infty)$$

Now solving inequality

$$\log_{0.5}(x^2 - 5x + 6) \geq -1 \Rightarrow 0 < x^2 - 5x + 6 \leq (0.5)^{-1}$$

$$\Rightarrow x^2 - 5x + 6 \leq 2$$

$$\begin{cases} x^2 - 5x + 6 > 0 \\ x^2 - 5x + 6 \leq 2 \end{cases} \Rightarrow x \in [1, 2) \cup (3, 4]$$

Hence, solution set of original inequation : $x \in [1, 2) \cup (3, 4]$

Illustration 31 : Solve for $x : \log_2 x \leq \frac{2}{\log_2 x - 1}$.

Solution : Let $\log_2 x = t$

$$t \leq \frac{2}{t-1} \Rightarrow t - \frac{2}{t-1} \leq 0$$

$$\Rightarrow \frac{t^2 - t - 2}{t-1} \leq 0 \Rightarrow \frac{(t-2)(t+1)}{(t-1)} \leq 0$$

$$\Rightarrow t \in (-\infty, -1] \cup (1, 2]$$

$$\text{or } \log_2 x \in (-\infty, -1] \cup (1, 2]$$

$$\text{or } x \in \left(0, \frac{1}{2}\right] \cup (2, 4]$$

Illustration 32 : Find all x such that $\log_{1/2} x > \log_{1/3} x$.

Solution : We have $\log_{1/2} x > \log_{1/3} x$.

$$\Rightarrow -\log_2 x > -\log_3 x \Rightarrow \log_2 x < \log_3 x$$

$$\Rightarrow \log_2 x < \frac{\log_2 x}{\log_2 3}$$

$$\Rightarrow \log_2 3 \log_2 x < \log_2 x \quad (\text{as } \log_2 3 > 0)$$

$$\Rightarrow \log_2 x (\log_2 3 - 1) < 0$$

Since $\log_2 3 - 1 > 0$, from the latter inequality we obtain $\log_2 x < 0$, hence $x < 1$. But the original inequality is meaningful only when $x > 0$. Therefore all x that satisfy the original inequality lie in the interval $0 < x < 1$.

Answer : $x \in (0, 1)$

Illustration 33 : Solve the inequation : $\log_{2x+3} x^2 < \log_{2x+3} (2x+3)$

Solution : For existence of logarithm

$$(i) \quad x^2 > 0 \quad (ii) \quad 2x+3 > 0 \quad (iii) \quad 2x+3 \neq 1 \Rightarrow x \in \left(-\frac{3}{2}, \infty\right) - \{-1, 0\} \quad \dots(i)$$

Now solving inequality

$$\text{Case I : } 0 < 2x+3 < 1 \Rightarrow -\frac{3}{2} < x < -1$$

$$\because \log_{2x+3} x^2 < \log_{2x+3} 2x+3$$

$$\Rightarrow x^2 > 2x+3$$

$$\Rightarrow (x-3)(x+1) > 0$$

$$\Rightarrow x \in (-\infty, -1) \cup (3, \infty); \text{ but } -\frac{3}{2} < x < -1$$

$$\Rightarrow x \in \left(-\frac{3}{2}, -1\right) \text{ intersection with (i)} \Rightarrow x \in \left(-\frac{3}{2}, -1\right)$$

$$\text{Case II : } 2x+3 > 1 \Rightarrow x > -1$$

$$\because \log_{2x+3} x^2 < \log_{2x+3} 2x+3$$

$$\Rightarrow x^2 < 2x+3$$

$$\Rightarrow (x-3)(x+1) < 0$$

$$\Rightarrow x \in (-1, 3); \text{ but } x > -1$$

$$\Rightarrow x \in (-1, 3) \text{ intersection with (i)} \Rightarrow x \in (-1, 3) - \{0\}$$

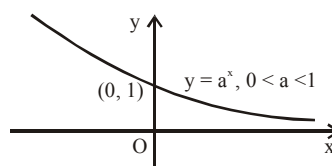
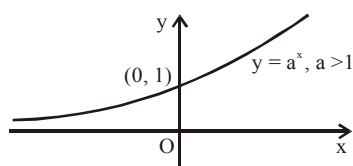
$$\therefore x \in \text{case I} \cup \text{case II}$$

$$\Rightarrow x \in \left(-\frac{3}{2}, -1\right) \cup (-1, 0) \cup (0, 3)$$

Do yourself-5 :

Solve the following inequalities.

1. $\log_{0.3}(x^2 + 8) > \log_{0.3}(9x)$
2. $\log_7\left(\frac{2x-6}{2x-1}\right) > 0$
3. $\log_{1/3}\sqrt{x^2 - 2x} > -\frac{1}{2}$
4. $(\log 100x)^2 + (\log 10x)^2 + \log x \leq 14$ (Here Base of log is 10)
5. $\log_{1/2}(x+1) > \log_2(2-x)$
6. $\log_x 2 \cdot \log_{2x} 2 \cdot \log_2 4x > 1$
7. $\log_{1/5}(2x^2 + 5x + 1) < 0$
8. $\log_x \frac{4x+5}{6-5x} < -1$
9. $\log_{x^2}(2+x) < 1$
10. $\log_x \frac{4x+5}{6-5x} < -1$

7. GRAPH OF EXPONENTIAL FUNCTION, ITS EQUATION AND INEQUALITIES

$$\text{If } a^{f(x)} > b \Rightarrow \begin{cases} f(x) > \log_a b & \text{when } a > 1 \\ f(x) < \log_a b & \text{when } 0 < a < 1 \end{cases}$$

Note : Domain of a^x is $\mathbb{R}$ and Range is $(0, \infty)$.**Illustration 34 :** Solve the equation $3^x \cdot 8^{\frac{x}{x+2}} = 6$.**Solution :** Some students solved this equation thus : rewriting it as

$$3^x \cdot 2^{\frac{3x}{x+2}} = 3^1 \cdot 2^1$$

They chose a root x so that the exponents of the respective bases were the same :

$$x = 1, \frac{3x}{x+2} = 1$$

hence the “answer” $x = 1$.

But this “answer” is incorrect in the sense that only one root of the equation is found and nothing has been said about any other roots. Actually, if the exponents on the appropriate bases are equal, then the products of these powers are equal, however the converse is not in any way implied and is simply incorrect. For instance, the equation

$$3^1 \cdot 2^1 = 3^2 \cdot 2^{\log_2(2/3)}$$

is valid, but $1 \neq 2$ and $1 \neq \log_2(2/3)$. Therefore, the foregoing reasoning may lead to a loss of roots, and this is exactly what occurred in the equation at hand.

Taking logarithms of both members of the original to the base 10, we get

$$x \log_{10} 3 + \frac{3x}{x+2} \log_{10} 2 = \log_{10} 6$$

$$\text{or} \quad x^2 \log_{10} 3 + x(3 \log_{10} 2 + 2 \log_{10} 3 - \log_{10} 6) - 2 \log_{10} 6 = 0$$

We now have to solve this quadratic equation. This can be done using a familiar formula, but we will try to simplify the solution by an ingenious device, since we have already seen, by trial and error, that $x_1 = 1$ is a root of the original equation and, consequently, satisfies the equivalent quadratic equation. For this reason, by Viète's theorem the second root of the quadratic equation is $x_2 = (-2 \log_{10} 6) / \log_{10} 3 = -2 \log_3 6$ and so the original equation has two roots;

$$x_1 = 1, x_2 = -2 \log_3 6.$$

Thus, it is useful to be able to guess a root, but never consider the guessing as the whole solution.

Illustration 35 :
$$\begin{cases} 2^{y-x}(x+y) = 1, \\ (x+y)^{x-y} = 2. \end{cases}$$

Solution : The domain of definition is $x+y > 0$.

$$\begin{cases} x+y = \frac{1}{2^{y-x}} = 2^{x-y}, \\ (x+y)^{x-y} = 2. \end{cases}$$

From the first equation we find $x+y = 2^{x-y}$ and substitute it into the second equation.

$$\text{Then } (2^{x-y})^{x-y} = 2 \Rightarrow 2^{(x-y)^2} = 2 \Rightarrow (x-y)^2 = 1.$$

The solution of the given system is the solution of the collection of systems

$$\begin{cases} x-y=1, \\ x+y=2, \end{cases} \quad \text{or} \quad \begin{cases} x-y=-1, \\ x+y=\frac{1}{2}. \end{cases}$$

Answer : $\left(\frac{3}{2}, \frac{1}{2}\right), \left(-\frac{1}{4}, \frac{3}{4}\right).$

Illustration 36 : Solve for $x : 2^{x+2} > \left(\frac{1}{4}\right)^{\frac{1}{x}}$

Solution : We have $2^{x+2} > 2^{-2/x}$. Since the base $2 > 1$, we have $x + 2 > -\frac{2}{x}$

(the sign of the inequality is retained).

$$\text{Now } x + 2 + \frac{2}{x} > 0 \quad \Rightarrow \quad \frac{x^2 + 2x + 2}{x} > 0$$

$$\Rightarrow \frac{(x+1)^2 + 1}{x} > 0 \quad \Rightarrow \quad x \in (0, \infty)$$

Illustration 37 : Solve for $x : (1.25)^{1-x} < (0.64)^{2(1+\sqrt{x})}$

Solution : We have $\left(\frac{5}{4}\right)^{1-x} < \left(\frac{16}{25}\right)^{2(1+\sqrt{x})}$ or $\left(\frac{4}{5}\right)^{x-1} < \left(\frac{4}{5}\right)^{4(1+\sqrt{x})}$

Since the base $0 < \frac{4}{5} < 1$, the inequality is equivalent to the inequality $x - 1 > 4(1 + \sqrt{x})$

$$\Rightarrow \frac{x-5}{4} > \sqrt{x}$$

Now, R.H.S. is positive

$$\Rightarrow \frac{x-5}{4} > 0 \quad \Rightarrow \quad x > 5 \quad \dots\dots(i)$$

$$\text{we have } \frac{x-5}{4} > \sqrt{x}$$

both sides are positive, so squaring both sides

$$\Rightarrow \frac{(x-5)^2}{16} > x \quad \text{or} \quad \frac{(x-5)^2}{16} - x > 0$$

$$\text{or } x^2 - 26x + 25 > 0 \quad \text{or } (x-25)(x-1) > 0$$

$$\Rightarrow x \in (-\infty, 1) \cup (25, \infty) \quad \dots\dots(ii)$$

intersection (i) & (ii) gives $x \in (25, \infty)$

Do yourself - 6 :

Solve for $x \in \mathbb{R}$

1. $4^x - 10 \cdot 2^{x-1} = 24$

2. $4 \cdot 2^{2x} - 6^x = 18 \cdot 3^{2x}$

3. $3^{2x-3} - 9^{x-1} + 27^{2x/3} = 675$

4. $7^{x+2} - \frac{1}{7} \cdot 7^{x+1} - 14 \cdot 7^{x-1} + 2 \cdot 7^x = 48$

5. $\left(\frac{5}{3}\right)^{x+1} \cdot \left(\frac{9}{25}\right)^{x^2+2x-11} = \left(\frac{5}{3}\right)^9$

6. $(3^{x^2-7.2x+3.9} - 9\sqrt{3}) \log(7-x) = 0$

7. $5^{2x} = 3^{2x} + 2(5^x) + 2(3^x)$

8. $\left(\frac{2}{3}\right)^{\frac{|x|-1}{|x|+1}} > 1$

9. $(1.25)^{1-(\log_2 x)^2} < (0.64)^{2+\log_{\sqrt{2}} x}$

10. $\left(\frac{1}{2}\right)^{\log_3 \log_{1/5} \left(x^2 - \frac{4}{5}\right)} < 1$

11. $4^x < 2^{x+1} + 3$

12. $e^{\frac{x^2+2}{x^2-1}} < \frac{1}{e^2}$

8. CHARACTERISTIC AND MANTISSA :

For any given number N , logarithm can be expressed as $\log_a N = \text{Integer} + \text{Fraction}$

The integer part is called characteristic and the fractional part (always taken non negative) is called mantissa.

When the value of $\log_{10} N$ is given, then to find digits of 'N' we use only the mantissa part. The characteristic is used only in determining the number of digits in the integral part (if $N \geq 1$) or the number of zeros after decimal & before first non-zero digit in the number (if $0 < N < 1$).

Note :

- (i) The mantissa part of logarithm of a number is always non-negative ($0 \leq m < 1$)
- (ii) If the characteristic of $\log_{10} N$ be 'C' and $C \geq 0$, then the number of digits in N is $(C + 1)$
- (iii) If the characteristic of $\log_{10} N$ be '- C' and $C > 0$, then there exist $(C - 1)$ zeros after decimal in N.

In summary, if characteristic of $\log_{10} N$ is 'C' then number of digits ($N \geq 1, N \in \mathbb{N}$) or number of zeros after decimal in N ($0 < N < 1$) = $|C + 1|$.

Do yourself - 7 :

1. Evaluate : $\log_{10}(0.06)^6$
2. Find number of digits in 18^{20}
3. Determine number of cyphers (zeros) between decimal & first significant digit in $\left(\frac{1}{6}\right)^{200}$
4. Let 'L' denotes the antilog of 0.4 to the base 1024.
and 'M' denotes the number of digits in 6^{10} (Given $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$)
and 'N' denotes the number of positive integers which have the characteristic 2, when base of the logarithm is 6.
Find the value of LMN.
5. If characteristic of $\log_{10}(0.00006)$ is A & characteristic $\log_3 750$ is B, then A + B is
6. If $\log_{10} 2 = .3010$, $\log_{10} 3 = .4771$, then number of digits in $4^8 \cdot 3^7 \cdot 5^3$ is 'P', then $\frac{P-1}{2}$ is
7. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, then find sum of the number of digits in 6^{15} and the number of zeros immediately after the decimal in 3^{-100} .

EXERCISE (O-1)

Straight Objective Type

1. If $a^4 \cdot b^5 = 1$ then the value of $\log_a(a^5 b^4)$ equals
 (A) $9/5$ (B) 4 (C) 5 (D) $8/5$ **LG0142**
2. The expression $\log_p \underbrace{\sqrt[p]{\sqrt[p]{\sqrt[p]{\dots \sqrt[p]{p}}}}}_{n \text{ radical sign}}$, where $p \geq 2, p \in \mathbb{N}; n \in \mathbb{N}$ when simplified is
 (A) independent of p (B) independent of p and of n
 (C) dependent on both p and n (D) positive **LG0143**
3. The number $\log_2 7$ is
 (A) an integer (B) a rational number
 (C) an irrational number (D) a prime number **LG0015**
4. If a, b, c are positive real numbers such that $a^{\log_3 7} = 27$; $b^{\log_7 11} = 49$ and $c^{\log_{11} 25} = \sqrt{11}$. The value of $\left(a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}\right)$ equals
 (A) 489 (B) 469 (C) 464 (D) 400 **LG0074**
5. If $2^a = 3$ and $9^b = 4$ then value of (ab) is -
 (A) 1 (B) 2 (C) 3 (D) 4 **LG0001**
6. $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \log_{10}(\log_4 5) + \dots + \log_{10}(\log_{1023} 1024)$ simplifies to
 (A) a composite (B) a prime number
 (C) rational which is not an integer (D) an integer **LG0144**
7. If $\log_2(x^2 + 1) + \log_{13}(x^2 + 1) = \log_2(x^2 + 1) \log_{13}(x^2 + 1)$, ($x \neq 0$), then $\log_7(x^2 + 24)$ is equal to
 (A) 1 (B) 2 (C) 3 (D) 4 **LG0006**
8. Given $\log_3 a = p = \log_b c$ and $\log_b 9 = \frac{2}{p^2}$. If $\log_9 \left(\frac{a^4 b^3}{c}\right) = \alpha p^3 + \beta p^2 + \gamma p + \delta$ ($\forall p \in \mathbb{R} - \{0\}$), then $(\alpha + \beta + \gamma + \delta)$ equals
 (A) 1 (B) 2 (C) 3 (D) 4 **LG0007**
9. The greatest value of $(4\log_{10} x - \log_x(.0001))$ for $0 < x < 1$ is -
 (A) 4 (B) -4 (C) 8 (D) -8 **LG0014**

10. If $60^a = 3$ and $60^b = 5$ then the value of $12^{\frac{1-a-b}{2(1-b)}}$ equals
 (A) 2 (B) 3 (C) $\sqrt{3}$ (D) $\sqrt{12}$ **LG0017**
11. Let ABC be a triangle right angled at C. The value of $\frac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ ($b + c \neq 1, c - b \neq 1$) equals
 (A) 1 (B) 2 (C) 3 (D) $1/2$ **LG0018**
12. If α and β are the roots of the equation $(\log_2 x)^2 + 4(\log_2 x) - 1 = 0$ then the value of $\log_\beta \alpha + \log_\alpha \beta$ equals
 (A) 18 (B) -16 (C) 14 (D) -18 **LG0019**
13. If $\log_2 (4 + \log_3 (x)) = 3$, then sum of digits of x is -
 (A) 3 (B) 6 (C) 9 (D) 18 **LG0002**
14. Sum of all the solution(s) of the equation $\log_{10}(x) + \log_{10}(x+2) - \log_{10}(5x+4) = 0$ is -
 (A) -1 (B) 3 (C) 4 (D) 5 **LG0003**
15. The product of all the solutions of the equation $x^{1+\log_{10} x} = 100000x$ is -
 (A) 10 (B) 10^5 (C) 10^{-5} (D) 1 **LG0004**
16. If x_1 and x_2 are the roots of equation $e^{3/2} \cdot x^{2\log x} = x^4$, then the product of the roots of the equation is -
 (A) e^2 (B) e (C) $e^{3/2}$ (D) e^{-2} **LG0005**
17. If $\log_a (1 - \sqrt{1+x}) = \log_a (3 - \sqrt{1+x})$, then number of solutions of the equation is -
 (A) 0 (B) 1 (C) 2 (D) infinitely many **LG0008**
18. The number of solution(s) of $\sqrt{\log_3 (3x^2)} \cdot \log_9 (81x) = \log_9 x^3$ is -
 (A) 0 (B) 1 (C) 2 (D) 3 **LG0009**
19. If $x, y \in 2^n$ when $n \in \mathbb{I}$ and $1 + \log_x y = \log_2 y$, then the value of $(x+y)$ is -
 (A) 2 (B) 4 (C) 6 (D) 8 **LG0011**
20. The number of integral solutions of $|\log_5 x^2 - 4| = 2 + |\log_5 x - 3|$ is -
 (A) 1 (B) 2 (C) 3 (D) 0 **LG0016**

21. If $\log_{10} \left(\frac{1}{2^x + x - 1} \right) = x(\log_{10} 5 - 1)$, then $x =$
 (A) 4 (B) 3 (C) 2 (D) 1
LG0021
22. The number of solutions of the equation $\log_{x-3}(x^3 - 3x^2 - 4x + 8) = 3$ is equal to
 (A) 4 (B) 3 (C) 2 (D) 1
LG0022
23. Sum of the roots of the equation $9^{\log_3(\log_2 x)} = \log_2 x - (\log_2 x)^2 + 1$ is equal to
 (A) 2 (B) 4 (C) 6 (D) 8
LG0023
24. If 1, $\log_9(3^{1-x} + 2)$ and $\log_3(4 \cdot 3^x - 1)$ are in A.P. then x can be
 (A) $\log_4 3$ (B) $\log_3 4$ (C) $1 - \log_3 4$ (D) $\log_3 0.25$
LG0025
25. Number of real solutions of the equation $\sqrt{\log_{10}(-x)} = \log_{10} \sqrt{x^2}$ is :
 (A) zero (B) exactly 1 (C) exactly 2 (D) 4
LG0145
26. If x_1 & x_2 are the two values of x satisfying the equation $7^{2x^2} - 2(7^{x^2+x+12}) + 7^{2x+24} = 0$, then $(x_1 + x_2)$ equals -
 (A) 0 (B) 1 (C) -1 (D) 7
LG0010
27. If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval
 (A) $(2, \infty)$ (B) $(1, 2)$ (C) $(1, \infty)$ (D) none of these
LG0020
28. If x satisfies the inequality $\log_{25} x^2 + (\log_5 x)^2 < 2$, then $x \in$
 (A) $\left(\frac{1}{25}, 5 \right)$ (B) $(1, 2)$ (C) $(4, 5)$ (D) $(0, 1)$
LG0024
29. If $x = \log_2 \left(\sqrt{56 + \sqrt{56 + \sqrt{56 + \sqrt{56 + \dots \infty}}} \right)$, then which of the following statements holds good ?
 (A) $x < 0$ (B) $0 < x < 2$ (C) $2 < x < 4$ (D) $3 < x < 4$
LG0013
30. If $n \in \mathbb{N}$ such that characteristic of n^2 to the base 8 is 2, then number of possible values of n is -
 (A) 14 (B) 15 (C) 448 (D) infinite
LG0012

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $\log_a x = b$ for permissible values of a and x then identify the statement(s) which can be correct ?
 (A) If a and b are two irrational numbers then x can be rational.
 (B) If a rational and b irrational then x can be rational.
 (C) If a irrational and b rational then x can be rational.
 (D) If a rational and b rational then x can be rational. LG0146
2. Which of the following statements is(are) correct ?
 (A) $7^{1/7} > (42)^{1/14} > 1$ (B) $\log_3(5) \log_7(9) \log_{11}(13) > -2$
 (C) $\sqrt{99+70\sqrt{2}} + \sqrt{99-70\sqrt{2}}$ is rational (D) $\frac{1}{\log_4 3} + \frac{1}{\log_7 3} > 3$
LG0027
3. Let $N = \frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2}$. Then N is : LG0147
 (A) a natural number (B) a prime number (C) a rational number (D) an integer
4. If $(\log_\beta \alpha)^2 + (\log_\alpha \beta)^2 = 79$, ($\alpha > 0$, $\beta > 0$, $\alpha \neq 1$, $\beta \neq 1$) then value of $(\log_\beta \alpha) + (\log_\alpha \beta)$ can be -
 (A) 7 (B) -9 (C) 9 (D) -7
LG0026
5. For $a > 0$, $\neq 1$ the roots of the equation $\log_{ax} a + \log_x a^2 + \log_{a^2x} a^3 = 0$ can be
 (A) $a^{-3/4}$ (B) $a^{-4/3}$ (C) $a^{-1/2}$ (D) none of these
LG0028
6. The solution of the equation $5^{\log_a x} + 5x^{\log_a 5} = 3$, ($a > 0$, $a \neq 1$) is
 (A) $a^{-\log_5 2}$ (B) $a^{\log_5 2}$ (C) $2^{-\log_5 a}$ (D) $2^{\log_5 a}$
LG0029
7. The equation $x^{(3/4)(\log_2 x)^2 + \log_2 x - 5/4} = \sqrt{2}$ has
 (A) at least one real solution (B) exactly three real solutions
 (C) exactly one irrational solution (D) real + non real roots
LG0031
8. The equation $\log_{x^2} 16 + \log_{2x} 64 = 3$ has :
 (A) one irrational solution (B) no prime solution
 (C) two real solutions (D) one integral solution LG0148

9. The equation $x^{\left[(\log_3 x)^2 - \frac{9}{2} \log_3 x + 5\right]} = 3\sqrt{3}$ has
 (A) exactly three real solution (B) at least one real solution
 (C) exactly one irrational solution (D) complex roots **LG0149**
10. Let a and b be real numbers greater than 1 for which there exists a positive real number c , different from 1, such that $2(\log_a c + \log_b c) = 9 \log_{ab} c$, then the possible value of $\log_a b$.
 (A) $1/2$ (B) 3 (C) $1/3$ (D) 2 **LG0071**
11. If x_1, x_2 are the value(s) of x satisfying the equation $\log_2^2(x-2) + \log_2(x-2) \log_2\left(\frac{3}{x}\right) - 2 \log_2^2\left(\frac{3}{x}\right) = 0$, then $x_1 + x_2$ is equal to
 (A) $x_1 + x_2 = 9$ (B) $x_1 x_2 = 18$ (C) $|x_1 - x_2| = 3$ (D) $x_1^2 + x_2^2 = 45$ **LG0040**
12. The solution set of the system of equations $\log_3 x + \log_3 y = 2 + \log_3 2$ and $\log_{27}(x+y) = \frac{2}{3}$ is :
 (A) $\{6, 3\}$ (B) $\{3, 6\}$ (C) $\{6, 12\}$ (D) $\{12, 6\}$ **LG0151**
13. If $2^{x+y} = 6^y$ and $3^{x-1} = 2^{y+1}$, then the value of $(\log 3 - \log 2)/(x-y)$ is
 (A) 1 (B) $\log_2 3 - \log_3 2$ (C) $\log(3/2)$ (D) $\log 3 - \log 2$ **LG0030**
14. The inequality $-1 \leq \left(\frac{1}{3}\right)^x < 2$ is satisfied by
 (A) $x \in [0, 1]$ (B) $x > -\log_3 2$ (C) $x < -\log_3 2$ (D) $x < -1$ **LG0032**
15. Which of the following statements are true
 (A) $\log_2 3 < \log_{12} 10$ (B) $\log_6 5 < \log_7 8$
 (C) $\log_3 26 < \log_2 9$ (D) $\log_{16} 15 > \log_{10} 11 > \log_7 6$ **LG0150**

EXERCISE (O-3)**Linked Comprehension Type****Paragraph for Question 1 and 2**

Let $\log_2 5 = a$; $\log_5 3 = b$

1. The value of $\log_{50} 30$ is equal to

(A) $\frac{b+ab+1}{b+2}$ (B) $\frac{a+ab+1}{2a+1}$ (C) $\frac{a+ab+1}{a+2}$ (D) $\frac{a+ab+b}{2a+1}$

LG0033

2. Roots of the equation $\log_{10} 2 \cdot \log_{10} 5x^2 - (\log_{10} 5 + \log_{10} 15 \cdot \log_{10} 2)x + \log_{10} 15 = 0$ are

(A) **a, b** (B) **1 + a, 1 + b** (C) **a - 1, 1 - b** (D) **a + 2; b + 2**

LG0033**Paragraph for Question 3 and 4**

Let $\log_5 N = I_1 + f_1$ and $\log_3 N = I_2 + f_2$, where I_1, I_2 are integers and $f_1, f_2 \in [0, 1)$.

On the basis of above information, answer the following questions :

3. If $I_1 = 2$ and $I_2 = 3$, then maximum integral value of N is greater than or equal to -

(A) 79 (B) 80 (C) 81 (D) 82 **LG0152**

4. If $I_1 = 3$ and $I_2 = 4$, then number of possible integral values of N is less than or equal to -

(A) 119 (B) 116 (C) 117 (D) 118 **LG0153**

Paragraph for Question 5 and 6

If (x_1, y_1) and (x_2, y_2) are the solution of the system of equation.

$$\log_{225}(x) + \log_{64}(y) = 4$$

$$\log_x(225) - \log_y(64) = 1,$$

5. The value of $\log_{30}(x_1 y_1 x_2 y_2)$ is

LG0154

6. Number of digits in $(x_1 y_1 x_2 y_2)$ is

LG0097

Matrix Match Type

7. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) The value of x for which

(1) 2

$$2\ln(2^x - 5) = \ln 2 + \ln\left(2^x - \frac{7}{2}\right) \text{ is}$$

(2) 4

(Q) The sum of the real roots of the equation

(3) 3

$$x^{\log_{10}\left(\frac{5x}{2}\right)} = 10^{\log_{10} x} \text{ is}$$

(4) 5

(R) If $\alpha = 2^{\frac{(\log_3(\log_3 5))}{\log_3 2}}$ and $\beta^\alpha = 9$, then value

(5) $\frac{1}{2}$

$$\frac{1}{5}(\beta^{\alpha^2}) \text{ is equal to}$$

(S) $\log_{\sqrt{3}}\left(\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots\infty}}}\right)$ is equal to

The correct option is

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 5$

(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 5$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 4; S \rightarrow 1$

LG0155

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) Positive value of x satisfying the equation

(1) 5

$$(25)^{\frac{2x^2 - 5x - 9}{4}} = (\sqrt{11})^{3\log_{11} 5} \text{ is}$$

(Q) Sum of values of x which satisfy the equation

(2) 12

$$\sqrt{|x - 4|^{x-2}} = \sqrt[4]{|x - 4|^{x+2}} \text{ is}$$

(R) The expression $\sqrt{\log_{0.5}^2 32}$ has the value equal to

(3) 4

(S) The value of 3k, where

(4) 18

$$\log(\log 9) + \log(\log 49) = \log k + \log(\log 3) + \log(\log 7), \text{ is}$$

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

LG0156

9. In the following matrix match **Column-I** has some quantities and **Column-II** has some comments or other quantities

Match the each element in **Column-I** with corresponding element(s) in **Column-II**

Column-I	Column-II
(A) $\log_{\sin \pi/6} e$	(P) $x^{\log_x 2} ; x > 0 ; x \neq 1$
(B) 2	(Q) $2^{\log_5 7}$
(C) $\log_{\tan \pi/3} \pi$	(R) negative
(D) $7^{\log_5 2}$	(S) positive

LG0157

10. Column-I	Column-II
(A) Anti logarithm of $(0.\overline{6})$ to the base 27 has the value equal to	(p) 5
(B) Characteristic of the logarithm of 2008 to the base 2 is	
(C) The value of b satisfying the equation, $\log_e 2 \cdot \log_b 625 = \log_{10} 16 \cdot \log_e 10$ is	(q) 7
(D) Number of naughts after decimal before a significant figure comes in the number $\left(\frac{5}{6}\right)^{100}$, is	(r) 9 (s) 10

(Given $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$)

EXERCISE (O-4)

Numerical Grid Type

1. Let A denotes the value of $\log_{10} \left(\frac{ab + \sqrt{(ab)^2 - 4(a+b)}}{2} \right) + \log_{10} \left(\frac{ab - \sqrt{(ab)^2 - 4(a+b)}}{2} \right)$

when $a = 43$ and $b = 57$ and B denotes the value of the expression $(2^{\log_6 18}) \cdot (3^{\log_6 3})$.

Find the value of (A.B).

LG0045

2. Let $y = \sqrt{\log, 3.\log, 12.\log, 48.\log, 192+16} - \log, 12.\log, 48+10$. Find $y \in \mathbb{N}$.

LG0089

3. There exist positive integers A, B and C with no common factor greater than 1 such that $A \log_{200} 5 + B \log_{200} 2 = C$, then the value of A + B + C is

LG0043

4. If $a = \log_{12} 18$ & $b = \log_{24} 54$ then find the value of $ab + 5(a - b)$.

LG0077

5. Number of solutions of $\log_{(1+x)}\left(\frac{1-x}{1+x}\right)-2\log_{(1-x)}(1+x)=0$ is

LG0038

6. The minimum possible real x which satisfy the equation, $2\log_2 \log_2 x + \log_{1/2} \log_2 (2\sqrt{2}x) = 1$.

LG0044

7. Find the square of the sum of the solution of the equation

$$\log_3 x \cdot \log_4 x \cdot \log_5 x = \log_3 x \cdot \log_4 x + \log_4 x \cdot \log_5 x + \log_5 x \cdot \log_3 x.$$

LG0048

8. If 'x' and 'y' are real numbers such that $2\log(2y - 3x) = \log x + \log y$, find $\frac{y+x}{y-x}$.

LG0075

9. If $4^A + 9^B = 10^C$, where $A = \log_{16} 4$, $B = \log_3 9$ & $C = \log_x 83$, then find x.

LG0158

10. Find the number of positive integers not satisfying the inequality $\log_3(4^x - 2 \cdot 2^x + 17) > 5$. **LG0159**

EXERCISE (JA)

1. Number of solutions of $\log_4(x-1) = \log_2(x-3)$ is [JEE 2001 (Screening)]
 (A) 3 (B) 1 (C) 2 (D) 0

LG0106

2. Let (x_0, y_0) be the solution of the following equations [JEE 2011, 3 (-1)]

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

$$3^{\ln x} = 2^{\ln y}$$

Then x_0 is

- (A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 6

LG0107

3. The value of $6 + \log_{\frac{3}{2}} \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \dots \right)$ is [JEE 2012, 4M]

LG0108

4. If $3^x = 4^{x-1}$, then $x =$ [JEE-Advanced 2013, 4, (-1)]

- (A) $\frac{2\log_3 2}{2\log_3 2 - 1}$ (B) $\frac{2}{2 - \log_2 3}$ (C) $\frac{1}{1 - \log_4 3}$ (D) $\frac{2\log_2 3}{2\log_2 3 - 1}$

LG0109

5. The value of $\left((\log_2 9)^2 \right)^{\frac{1}{\log_2 (\log_2 9)}} \times (\sqrt{7})^{\frac{1}{\log_4 7}}$ is _____ [JEE(Advanced)-2018]

LG0110

6. The product of all positive real values of x satisfying the equation $x^{(16(\log_5 x)^3 - 68\log_5 x)} = 5^{-16}$ is _____. [JEE(Advanced)-2022]

LG0160

ANSWER KEY

Do yourself - 1

1. (a) $\log_3 81 = 4$ (b) $\log_{10}(0.001) = -3$ (c) $\log_{128} 2 = 1/7$
2. (a) $32 = 2^5$ (b) $4 = (\sqrt{2})^4$ (c) $0.01 = 10^{-2}$
3. 6
4. (i) 0 (ii) 1 (iii) 2 (iv) 3 (v) -1
(vi) -5 (vii) -4 (viii) 1/2 (ix) 1 (x) 3/2
(xi) -1/5 (xii) -3/7
5. (i) 0 (ii) 1 (iii) 3 (iv) -4 (v) -1/2
(vi) 1/2 (vii) -3/2 (viii) 9/4
6. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) -1
(vi) 1/2 (vii) -3/2 (viii) 7/2 (ix) 2/7
7. (i) 0 (ii) 1 (iii) 2 (iv) -1 (v) -2
(vi) -4 (vii) -1/3 (viii) 1/7 (ix) -5/2 (x) 9/4
8. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) -1
(vi) -2 (vii) -1/2 (viii) 1/4 (ix) 1/2 (x) 1/3
(xi) 1/3
9. (i) 4 (ii) 2 (iii) $a > 0, a \neq 1$ (iv) -5, 2 (v) -2, 2
(vi) -3, 3 (vii) 9 (viii) 1/81 (ix) 1 (x) 2
(xi) $\sqrt{2}, -\sqrt{2}$
10. (i) $-\frac{1}{2}$ (ii) 2 (iii) 1 (iv) 1 (v) -1

Do yourself - 2

1. (a) 1 (b) $\frac{1}{5}$ 2. 3
4. (i) 2 (ii) 3 (iii) 1 (iv) 4

Do yourself - 3

1. (i) 3 (ii) 5/6 (iii) 0 (iv) 2 (v) (A)
(vi) (C)
2. (i) 3 (ii) 9 (iii) 9 (vi) 2
3. $2s + 10s^2 - 3(s^3 + 1)$ 5. $\log_b N$ 6. 4 7. 9
8. 1 9. 6 10. $\frac{25}{2}$ 11. 1/6 12. 34
13. 32

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	C	B	A	D	B	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	C	C	D	A	A	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	D	A	C	C	B	A	A	C	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B,D	A,B,C,D	B,C	B,C	A,C	A,B,C,D	A,B,C,D	A,B,C,D	A,D
Que.	11	12	13	14	15					
Ans.	A,B,C,D	A,B	C,D	A,B	B,C					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8		
Ans.	B	B	A,B	A,D	12.00	18.00	D	A		
Q.9	A	B	C	D	Q.10	A	B	C	D	
	R	P,S	S	Q,S		r	s	p	q	

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	12	6	6	1	0	8.00	3721.00	2.60	10.00	2.00

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6				
Ans.	B	C	4	A,B,C	8	1				

Important Notes

SEQUENCE & SERIES

1. DEFINITION :

Sequence :

A succession of terms $a_1, a_2, a_3, a_4, \dots$ formed according to some rule or law.

Examples are : 1, 4, 9, 16, 25

$-1, 1, -1, 1, \dots$

$$\frac{x}{1!}, \frac{x^2}{2!}, \frac{x^3}{3!}, \frac{x^4}{4!}, \dots$$

A finite sequence has a finite (i.e. limited) number of terms, as in the first example above. An infinite sequence has an unlimited number of terms, i.e. there is no last term, as in the second and third examples.

Series :

The indicated sum of the terms of a sequence. In the case of a finite sequence $a_1, a_2, a_3, \dots$,

a_n the corresponding series is $a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$. This series has a finite or limited number of terms and is called a finite series.

2. ARITHMETIC PROGRESSION (A.P.) :

A.P. is a sequence whose terms differ by a fixed number. This fixed number is called the common difference. If a is the first term & d the common difference, then A.P. can be written as

$$a, a + d, a + 2d, \dots, a + (n-1)d, \dots$$

(a) n^{th} term of AP $T_n = a + (n-1)d$, where $d = t_n - t_{n-1}$

(b) The sum of the first n terms : $S_n = \frac{n}{2}[a + \ell] = \frac{n}{2}[2a + (n-1)d]$

where ℓ is n^{th} term.

Note :

- (i) n^{th} term of an A.P. is of the form $An + B$ i.e. a linear expression in ' n ', in such a case the coefficient of n is the common difference of the A.P. i.e. A .
- (ii) Sum of first ' n ' terms of an A.P. is of the form $An^2 + Bn$ i.e. a quadratic expression in ' n ', in such case the common difference is twice the coefficient of n^2 . i.e. $2A$
- (iii) Also n^{th} term $T_n = S_n - S_{n-1}$

Illustration 1 : If $(x + 1)$, $3x$ and $(4x + 2)$ are first three terms of an A.P. then its 5th term is -

- (A) 14 (B) 19 (C) 24 (D) 28

Solution : $(x + 1)$, $3x$, $(4x + 2)$ are in AP

$$\Rightarrow 3x - (x + 1) = (4x + 2) - 3x \quad \Rightarrow x = 3$$

$$\therefore a = 4, d = 9 - 4 = 5 \quad \Rightarrow T_5 = 4 + (4)5 = 24 \quad \text{Ans. (C)}$$

Illustration 2 : The sum of first four terms of an A.P. is 56 and the sum of its last four terms is 112. If its first term is 11 then find the number of terms in the A.P.

Solution : $a + a + d + a + 2d + a + 3d = 56$

$$4a + 6d = 56$$

$$44 + 6d = 56 \quad (\text{as } a = 11)$$

$$6d = 12 \quad \text{hence } d = 2$$

Let total number of terms = n

Now sum of last four terms.

$$a + (n - 1)d + a + (n - 2)d + a + (n - 3)d + a + (n - 4)d = 112$$

$$\Rightarrow 4a + (4n - 10)d = 112 \quad \Rightarrow 44 + (4n - 10)2 = 112$$

$$\Rightarrow 4n - 10 = 34$$

$$\Rightarrow n = 11 \quad \text{Ans.}$$

Illustration 3 : The sum of first n terms of two A.P.s. are in ratio $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms.

Solution : Let a_1 and a_2 be the first terms and d_1 and d_2 be the common differences of two A.P.s respectively then

$$\frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{7n+1}{4n+27} \Rightarrow \frac{a_1 + \left(\frac{n-1}{2}\right)d_1}{a_2 + \left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27}$$

For ratio of 11th terms

$$\frac{n-1}{2} = 10 \quad \Rightarrow n = 21$$

$$\text{so ratio of 11th terms is } \frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \frac{4}{3} \quad \text{Ans.}$$

Do yourself - 1 :

- Write down the sequence whose n^{th} terms is : (a) $\frac{2^n}{n}$ (b) $\frac{3+(-1)^n}{3^n}$
- For an A.P, show that $t_m + t_{2n+m} = 2t_{m+n}$
- If the sum of first p terms of an A.P. is q and the sum of its first q terms is p , then find the sum of its first $(p+q)$ term.

Find the

- sum 49, 44, 39.... to 17 terms.
- sum $\frac{3}{4}, \frac{2}{3}, \frac{7}{12}, \dots$ to 19 terms.
- sum 1.3, -3.1, -7.5,.... to 10 terms
- sum $a - 3b, 2a - 5b, 3a - 7b, \dots$ to 40 terms.
- In an AP of which 'a' is the 1st term, if the sum of the 1st p terms is equal to zero, show that the sum of the next q terms is $-\left(\frac{aq(p+q)}{p-1}\right)$

3. PROPERTIES OF A.P. :

- If each term of an A.P. is increased, decreased, multiplied or divided by the some nonzero number, then the resulting sequence is also an A.P.
- Three numbers in A.P. : $a-d, a, a+d$
Four numbers in A.P. : $a-3d, a-d, a+d, a+3d$
Five numbers in A.P. : $a-2d, a-d, a, a+d, a+2d$
Six numbers in A.P. : $a-5d, a-3d, a-d, a+d, a+3d, a+5d$ etc.
- The common difference can be zero, positive or negative.
- k^{th} term from the last $= (n-k+1)^{\text{th}}$ term from the beginning (If total number of terms $= n$).
- The sum of the two terms of an AP equidistant from the beginning & end is constant and equal to the sum of first & last terms. $\Rightarrow T_k + T_{n-k+1} = \text{constant} = a + \ell$.
- Any term of an AP (except the first) is equal to half the sum of terms which are equidistant from it. $a_n = (1/2)(a_{n-k} + a_{n+k})$, $k < n$
For $k=1$, $a_n = (1/2)(a_{n-1} + a_{n+1})$; For $k=2$, $a_n = (1/2)(a_{n-2} + a_{n+2})$ and so on.
- If a, b, c are in AP, then $2b = a + c$.

Illustration 4 : Four numbers are in A.P. If their sum is 20 and the sum of their squares is 120, then the middle terms are -

- (A) 2, 4 (B) 4, 6 (C) 6, 8 (D) 8, 10

Solution : Let the numbers are $a - 3d, a - d, a + d, a + 3d$

$$\text{given, } a - 3d + a - d + a + d + a + 3d = 20 \quad \Rightarrow 4a = 20 \Rightarrow a = 5$$

$$\text{and } (a - 3d)^2 + (a - d)^2 + (a + d)^2 + (a + 3d)^2 = 120 \quad \Rightarrow 4a^2 + 20d^2 = 120$$

$$\Rightarrow 4 \times 5^2 + 20d^2 = 120 \quad \Rightarrow d^2 = 1 \Rightarrow d = \pm 1$$

Hence numbers are 2, 4, 6, 8 or 8, 6, 4, 2 **Ans. (B)**

Illustration 5 : If $a_1, a_2, a_3, \dots, a_n$ are in A.P. where $a_i > 0$ for all i , show that :

$$\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} = \frac{(n-1)}{\sqrt{a_1} + \sqrt{a_n}}$$

Solution :

$$\text{L.H.S.} = \frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}}$$

$$= \frac{1}{\sqrt{a_2} + \sqrt{a_1}} + \frac{1}{\sqrt{a_3} + \sqrt{a_2}} + \dots + \frac{1}{\sqrt{a_n} + \sqrt{a_{n-1}}}$$

$$= \frac{\sqrt{a_2} - \sqrt{a_1}}{(a_2 - a_1)} + \frac{\sqrt{a_3} - \sqrt{a_2}}{(a_3 - a_2)} + \dots + \frac{\sqrt{a_n} - \sqrt{a_{n-1}}}{a_n - a_{n-1}}$$

Let 'd' is the common difference of this A.P.

$$\text{then } a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1} = d$$

Now L.H.S.

$$= \frac{1}{d} \{ \sqrt{a_2} - \sqrt{a_1} + \sqrt{a_3} - \sqrt{a_2} + \dots + \sqrt{a_{n-1}} - \sqrt{a_{n-2}} + \sqrt{a_n} - \sqrt{a_{n-1}} \} = \frac{1}{d} \{ \sqrt{a_n} - \sqrt{a_1} \}$$

$$= \frac{a_n - a_1}{d(\sqrt{a_n} + \sqrt{a_1})} = \frac{a_1 + (n-1)d - a_1}{d(\sqrt{a_n} + \sqrt{a_1})} = \frac{1}{d} \frac{(n-1)d}{(\sqrt{a_n} + \sqrt{a_1})} = \frac{n-1}{\sqrt{a_n} + \sqrt{a_1}} = \text{R.H.S.}$$

Illustration 6 : The series of natural numbers is divided into groups (1), (2, 3, 4), (5, 6, 7, 8, 9) and so on. Show that the sum of the numbers in n^{th} group is $n^3 + (n - 1)^3$

Solution : The groups are (1), (2, 3, 4), (5, 6, 7, 8, 9)

The number of terms in the groups are 1, 3, 5.....

$\therefore$ The number of terms in the n^{th} group = $(2n - 1)$

the last term of the n^{th} group is n^2

If we count from last term common difference should be -1

$$\text{So the sum of numbers in the } n^{\text{th}} \text{ group} = \left(\frac{2n-1}{2} \right) \{ 2n^2 + (2n-2)(-1) \}$$

$$= (2n-1)(n^2 - n + 1) = 2n^3 - 3n^2 + 3n - 1 = n^3 + (n-1)^3$$

Do yourself - 2 :

- Find the sum of first 24 terms of the A.P. $a_1, a_2, a_3, \dots$, if it is known that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$.
- Find the number of terms common to the two A.P.'s 3, 7, 11, 407 and 2, 9, 16, 709
- In an A.P. the first term is 2, the last term is 29, the sum is 155; find the common difference.
- The sum of 15 terms of an A.P. is 600, and the common difference is 5; find the first term.
- The third term of an A.P. is 18, and the seventh term is 30; find the sum of 17 terms.
- The sum of three numbers in A.P. is 27, and their product is 504; find them.
- The sum of three numbers in A.P. is 12, and the sum of their cubes is 408; find them.
- Find the sum of 15 terms of the series whose n^{th} term is $4n + 1$.
- In an AP if the sum of 7 terms is 49, and the sum of 17 terms is 289, find the sum of n terms.
- The sum of four integers in A.P. is 24, and their product is 945; find them.
- If the sum of n terms of an A.P. is $2n + 3n^2$, find the r^{th} term.
- If the sum of m terms of an A.P. is to the sum of n terms as m^2 to n^2 , show that the m^{th} term is to the n^{th} term as $2m - 1$ is to $2n - 1$.
- The sum of n terms of two arithmetic series are in the ratio of $(7n + 1) : (4n + 27)$. Find the ratio of their n^{th} term.
- If the sum of n terms of an AP is $Pn + Qn^2$, where P, Q are constants, then its common difference is

4. GEOMETRIC PROGRESSION (G.P.) :

G.P. is a sequence of non zero numbers each of the succeeding term is equal to the preceeding term multiplied by a constant. Thus in a GP the ratio of successive terms is constant. This constant factor is called the COMMON RATIO of the sequence & is obtained by dividing any term by the immediately previous term. Therefore $a, ar, ar^2, ar^3, ar^4, \dots$ is a GP with 'a' as the first term & 'r' as common ratio.

(a) n^{th} term ; $T_n = a r^{n-1}$

(b) Sum of the first n terms; $S_n = \frac{a(r^n - 1)}{r - 1}$, if $r \neq 1$

(c) Sum of infinite G.P. , $S_\infty = \frac{a}{1 - r}$; $0 < |r| < 1$

5. PROPERTIES OF GP :

- If each term of a G.P. be multiplied or divided by the some non-zero quantity, then the resulting sequence is also a G.P.
- Three consecutive terms of a GP : $a/r, a, ar$;
Four consecutive terms of a GP : $a/r^3, a/r, ar, ar^3$ & so on.
- If a, b, c are in G.P. then $b^2 = ac$.
- If in a G.P, the product of two terms which are equidistant from the first and the last term, is constant and is equal to the product of first and last term. $\Rightarrow T_k \cdot T_{n-k+1} = \text{constant} = a \cdot \ell$
- If each term of a G.P. be raised to the same power, then resulting sequence is also a G.P.
- In a G.P., $T_r^2 = T_{r-k} \cdot T_{r+k}$, $k < r$, $r \neq 1$
- If the terms of a given G.P. are chosen at regular intervals, then the new sequence is also a G.P.
- If $a_1, a_2, a_3, \dots, a_n$ is a G.P. of positive terms, then $\log a_1, \log a_2, \dots, \log a_n$ is an A.P. and vice-versa.
- If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are two G.P.'s then $a_1 b_1, a_2 b_2, a_3 b_3, \dots$ & $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots$ is also in G.P.

Illustration 7 : If a, b, c, d and p are distinct real numbers such that

$$(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cd) + (b^2 + c^2 + d^2) \leq 0 \text{ then } a, b, c, d \text{ are in}$$

- (A) A.P. (B) G.P. (C) H.P. (D) None of these

Solution : Here, the given condition $(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cd) + b^2 + c^2 + d^2 \leq 0$

$$\Rightarrow (ap - b)^2 + (bp - c)^2 + (cp - d)^2 \leq 0$$

$\therefore$ a square can not be negative

$$\therefore ap - b = 0, bp - c = 0, cp - d = 0 \Rightarrow p = \frac{b}{a} = \frac{c}{b} = \frac{d}{c} \Rightarrow a, b, c, d \text{ are in G.P.}$$

Ans. (B)

Illustration 8 : If positive real numbers a, b, c are in G.P., then the equations $ax^2 + 2bx + c = 0$ and

$dx^2 + 2ex + f = 0$ have a common root if $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in -

- (A) A.P. (B) G.P. (C) H.P. (D) None of these

Solution : a, b, c are in G.P $\Rightarrow b^2 = ac$

Now the equation $ax^2 + 2bx + c = 0$ can be rewritten as $ax^2 + 2\sqrt{ac}x + c = 0$

$$\Rightarrow (\sqrt{a}x + \sqrt{c})^2 = 0 \Rightarrow x = -\sqrt{\frac{c}{a}}, -\sqrt{\frac{c}{a}}$$

If the two given equations have a common root, then this root must be $-\sqrt{\frac{c}{a}}$.

$$\text{Thus } d\frac{c}{a} - 2e\sqrt{\frac{c}{a}} + f = 0 \Rightarrow \frac{d}{a} + \frac{f}{c} = \frac{2e}{c}\sqrt{\frac{c}{a}} = \frac{2e}{\sqrt{ac}} = \frac{2e}{b} \Rightarrow \frac{d}{a}, \frac{e}{b}, \frac{f}{c} \text{ are in A.P.}$$

Ans. (A)

Illustration 9 : A number consists of three digits which are in G.P. the sum of the right hand and left hand digits exceeds twice the middle digit by 1 and the sum of the left hand and middle digits is two third of the sum of the middle and right hand digits. Find the numbers.

Solution : Let the three digits be a, ar and ar^2 then number is

$$100a + 10ar + ar^2 \quad \dots(i)$$

$$\text{Given, } a + ar^2 = 2ar + 1$$

$$\text{or } a(r^2 - 2r + 1) = 1$$

$$\text{or } a(r - 1)^2 = 1 \quad \dots(ii)$$

$$\text{Also given } a + ar = \frac{2}{3}(ar + ar^2)$$

$$\Rightarrow 3 + 3r = 2r + 2r^2 \Rightarrow 2r^2 - r - 3 = 0 \Rightarrow (r + 1)(2r - 3) = 0$$

$$\therefore r = -1, 3/2$$

$$\text{for } r = -1, a = \frac{1}{(r-1)^2} = \frac{1}{4} \notin \mathbb{I} \quad \therefore r \neq -1$$

$$\text{for } r = 3/2, a = \frac{1}{\left(\frac{3}{2}-1\right)^2} = 4 \quad \{\text{from (ii)}\}$$

$$\text{From (i), number is } 400 + 10 \cdot 4 \cdot \frac{3}{2} + 4 \cdot \frac{9}{4} = 469$$

Ans.

Illustration 10 : Find the value of $0.32\overline{58}$

Solution : Let $R = 0.32\overline{58} \Rightarrow R = 0.32585858\dots$ (i)

Here number of figures which are not recurring is 2 and number of figures which are recurring is also 2.

$$\text{then } 100 R = 32.585858\dots \quad \dots\dots(ii)$$

$$\text{and } 10000 R = 3258.5858\dots \quad \dots\dots(iii)$$

Subtracting (ii) from (iii), we get

$$9900 R = 3226 \Rightarrow R = \frac{1613}{4950}$$

Aliter Method : $R = .32 + .0058 + .000058 + .00000058 + \dots\dots\dots$

$$= .32 + \frac{58}{10^4} \left(1 + \frac{1}{10^2} + \frac{1}{10^4} + \dots\dots\dots\infty \right) = .32 + \frac{58}{10^4} \left(\frac{1}{1 - \frac{1}{100}} \right)$$

$$= \frac{32}{100} + \frac{58}{9900} = \frac{3168 + 58}{9900} = \frac{3226}{9900} = \frac{1613}{4950}$$

Do yourself - 3 :

- Find a three digit number whose consecutive digits form a G.P. If we subtract 792 from this number, we get a number consisting of the same digits written in the reverse order.
- If the third term of G.P. is 4, then find the product of first five terms.
- If a, b, c are respectively the p^{th} , q^{th} and r^{th} terms of the given G.P., then show that $(q - r) \log a + (r - p) \log b + (p - q) \log c = 0$, where a, b, c > 0.
- Find three numbers in G.P., whose sum is 52 and the sum of whose products in pairs is 624.
- The rational number which equals the number $2.\overline{357}$ with recurring decimal is -

(A) $\frac{2357}{999}$

(B) $\frac{2379}{997}$

(C) $\frac{785}{333}$

(D) $\frac{2355}{1001}$

Find the

6. sum $\frac{3}{4}, 1\frac{1}{2}, 3, \dots$ to 8 terms.

7. sum 1, 5, 25, ... to p terms.

8. sum $1, \sqrt{3}, 3, \dots$ to 12 terms.

9. sum $\frac{8}{5}, -1, \frac{5}{8}, \dots$ upto infinity

10. sum .45, .015, .0005, ... upto infinity

11. sum $3, \sqrt{3}, 1, \dots$ upto infinity

12. The sum of the first 6 terms of a G.P. is 9 times the sum of the first 3 terms; find the common ratio.

- ## 6. HARMONIC PROGRESSION (H.P.) :

If the sequence $a_1, a_2, a_3, \dots, a_n$ is an HP then $1/a_1, 1/a_2, \dots, 1/a_n$ is an AP. Here we do not have the formula for the sum of the n terms of an HP. The general form of a harmonic progression is

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots, \frac{1}{a+(n-1)d}$$

(i) If a, b, c are in HP, then $b = \frac{2ac}{a+c}$ or $\frac{a}{c} = \frac{a-b}{b-c}$

Solution : Three numbers are in H.P. can be taken as

$$\frac{1}{a-d}, \frac{1}{a}, \frac{1}{a+d}$$

$$\text{then } \frac{1}{a-d} + \frac{1}{a} + \frac{1}{a+d} = 37 \quad \dots\dots\dots(i)$$

$$\text{and } a - d + a + a + d = \frac{1}{4} \quad \Rightarrow \quad a = \frac{1}{12}$$

$$\text{from (i), } \frac{12}{1-12d} + 12 + \frac{12}{1+12d} = 37 \quad \Rightarrow \quad \frac{12}{1-12d} + \frac{12}{1+12d} = 25$$

$$\Rightarrow \frac{24}{1-144d^2} = 25$$

$$\Rightarrow 1-144d^2 = \frac{24}{25}$$

$$\Rightarrow d^2 = \frac{1}{25 \times 144}$$

$$\therefore d = \pm \frac{1}{60}$$

$$\therefore a-d, a, a+d \text{ are } \frac{1}{15}, \frac{1}{12}, \frac{1}{10} \text{ or } \frac{1}{10}, \frac{1}{12}, \frac{1}{15}$$

Hence, three numbers in H.P. are 15, 12, 10 or 10, 12, 15

Ans.

Illustration 12 : Suppose a is a fixed real number such that $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$

If p, q, r are in A.P., then prove that x, y, z are in H.P.

Solution :

$\therefore p, q, r$ are in A.P.

$$\therefore q-p = r-q \quad \dots\dots\dots (i)$$

$$\Rightarrow p-q = q-r = k \text{ (let)}$$

$$\text{given } \frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz} \quad \Rightarrow \quad \frac{\frac{a}{x}-1}{p} = \frac{\frac{a}{y}-1}{q} = \frac{\frac{a}{z}-1}{r}$$

$$\Rightarrow \frac{\left(\frac{a}{x}-1\right)-\left(\frac{a}{y}-1\right)}{p-q} = \frac{\left(\frac{a}{y}-1\right)-\left(\frac{a}{z}-1\right)}{q-r} \quad \text{(by law of proportion)}$$

$$\Rightarrow \frac{\frac{a}{x}-\frac{a}{y}}{k} = \frac{\frac{a}{y}-\frac{a}{z}}{k} \quad \{\text{from (i)}\}$$

$$\Rightarrow a\left(\frac{1}{x}-\frac{1}{y}\right) = a\left(\frac{1}{y}-\frac{1}{z}\right) \quad \Rightarrow \quad \frac{1}{x}-\frac{1}{y} = \frac{1}{y}-\frac{1}{z}$$

$$\therefore \frac{2}{y} = \frac{1}{x} + \frac{1}{z}$$

$$\therefore \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in A.P.}$$

Hence x, y, z are in H.P.

Do yourself - 4 :

1. If the 7th term of a H.P. is 8 and the 8th term is 7. Then find the 28th term.
2. In a H.P., if 5th term is 6 and 3rd term is 10. Find the 2nd term.
3. If the pth, qth and rth terms of a H.P. are a, b, c respectively, then prove that $\frac{q-r}{a} + \frac{r-p}{b} + \frac{p-q}{c} = 0$
4. Find the fourth term in each of the following Harmonic progressions :
(a) $2, 2\frac{1}{2}, 3\frac{1}{3}, \dots$ (b) $1\frac{1}{2}, 2, 3, \dots$ (c) $3\frac{3}{5}, 6, 18, \dots$
5. If a, b, c be in H.P., show that $a : a - b = a + c : a - c$.
6. If the mth term of a H.P. be equal to n, and the nth term be equal to m, prove that the
(m + n)th term is equal to $\frac{mn}{m+n}$.
7. The pth term T_p of H.P. is q(p + q) and qth term T_q is p(p + q) when p > 2, q > 2 (p ≠ q). Prove that
(a) T_{p+q} = pq; (b) T_{pq} = p + q; (c) T_{p+q} > T_{pq}
8. If the 10th term of a HP is 21 and 21st term of the same HP is 10, then find the 210th term.
9. If a, b, c, d, e be 5 numbers such that a, b, c are in AP ; b, c, d are in GP & c, d, e are in HP then :
(i) Prove that a, c, e are in GP.
(ii) Prove that e = (2b - a)²/a
(iii) If a = 2 & e = 18, find all possible values of b, c, d.

7. MEANS

(a) ARITHMETIC MEAN :

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if

a, b, c are in A.P., b is A.M. of a & c. So A.M. of a and c = $\frac{a+c}{2} = b$.

n-ARITHMETIC MEANS BETWEEN TWO NUMBERS :

If a, b be any two given numbers & A₁, A₂,, A_n, b are in AP, then A₁, A₂,, A_n are the 'n' A.M.'s between a & b then. A₁ = a + d, A₂ = a + 2d,, A_n = a + nd or b - d, where

$$d = \frac{b-a}{n+1}$$

$$\Rightarrow A_1 = a + \frac{b-a}{n+1}, A_2 = a + \frac{2(b-a)}{n+1}, \dots$$

Note: Sum of n A.M.'s inserted between a & b is equal to n times the single A.M. between a & b

$$\text{i.e. } \sum_{r=1}^n A_r = nA \text{ where } A \text{ is the single A.M. between } a \text{ \& } b.$$

(b) GEOMETRIC MEAN :

If a, b, c are in G.P., then b is the G.M. between a & c , $b^2 = ac$. So G.M. of a and $c = \sqrt{ac} = b$

n-GEOMETRIC MEANS BETWEEN TWO NUMBERS :

If a, b are two given positive numbers & $a, G_1, G_2, \dots, G_n, b$ are in G.P. Then $G_1, G_2, G_3, \dots, G_n$ are 'n' G.Ms between a & b . where $b = ar^{n+1} \Rightarrow r = (b/a)^{1/(n+1)}$

$$\begin{aligned} G_1 &= a(b/a)^{1/(n+1)}, & G_2 &= a(b/a)^{2/(n+1)}, & \dots, & & G_n &= a(b/a)^{n/(n+1)} \\ &= ar, & &= ar^2, & \dots, & & &= ar^n = b/r \end{aligned}$$

Note : The product of n G.Ms between a & b is equal to n^{th} power of the single G.M. between

a & b i.e. $\prod_{r=1}^n G_r = (G)^n$ where G is the single G.M. between a & b

(c) HARMONIC MEAN :

If a, b, c are in H.P., then b is H.M. between a & c . So H.M. of a and $c = \frac{2ac}{a+c} = b$.

Insertion of 'n' HM's between a and b :

$a, H_1, H_2, H_3, \dots, H_n, b \rightarrow$ H.P

$$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_n}, \frac{1}{b} \rightarrow \text{A.P.}$$

$$\frac{1}{b} = \frac{1}{a} + (n+1)D \Rightarrow D = \frac{\frac{1}{b} - \frac{1}{a}}{n+1}$$

$$\frac{1}{H_n} = \frac{1}{a} + n \left(\frac{\frac{1}{b} - \frac{1}{a}}{n+1} \right)$$

Important note :

- (i) If A, G, H , are respectively A.M., G.M., H.M. between two positive number a & b then
 (a) $G^2 = AH$ (A, G, H constitute a GP) (b) $A \geq G \geq H$ (c) $A = G = H \Leftrightarrow a = b$
 (ii) Let $a_1, a_2, \dots, a_n$ be n positive real numbers, then we define their arithmetic mean (A), geometric

mean (G) and harmonic mean (H) as $A = \frac{a_1 + a_2 + \dots + a_n}{n}$

$$G = (a_1 a_2 \dots a_n)^{1/n} \text{ and } H = \frac{n}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right)}$$

It can be shown that $A \geq G \geq H$. Moreover equality holds at either place if and only if $a_1 = a_2 = \dots = a_n$

Illustration 13 : If $2x^3 + ax^2 + bx + 4 = 0$ (a and b are positive real numbers) has 3 real roots, then prove that $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Solution : Let α, β, γ be the roots of $2x^3 + ax^2 + bx + 4 = 0$. Given that all the coefficients are positive, so all the roots will be negative.

$$\text{Let } \alpha_1 = -\alpha, \alpha_2 = -\beta, \alpha_3 = -\gamma \Rightarrow \alpha_1 + \alpha_2 + \alpha_3 = \frac{a}{2}$$

$$\alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_3\alpha_1 = \frac{b}{2}$$

$$\alpha_1\alpha_2\alpha_3 = 2$$

Applying AM $\geq$ GM, we have

$$\frac{\alpha_1 + \alpha_2 + \alpha_3}{3} \geq (\alpha_1\alpha_2\alpha_3)^{1/3} \Rightarrow a \geq 6 \times 2^{1/3}$$

$$\text{Also } \frac{\alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_1\alpha_3}{3} \geq (\alpha_1\alpha_2\alpha_3)^{2/3} \Rightarrow b \geq 6 \times 4^{1/3}$$

Therefore $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Illustration 14 : If $a_i > 0 \forall i \in N$ such that $\prod_{i=1}^n a_i = 1$,

then prove that $(1 + a_1)(1 + a_2)(1 + a_3) \dots (1 + a_n) \geq 2^n$

Solution : Using A.M. $\geq$ G.M.

$$1 + a_1 \geq 2\sqrt{a_1}$$

$$1 + a_2 \geq 2\sqrt{a_2}$$

$\vdots$

$$1 + a_n \geq 2\sqrt{a_n} \Rightarrow (1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n(a_1 a_2 a_3 \dots a_n)^{1/2}$$

$$\text{As } a_1 a_2 a_3 \dots a_n = 1$$

$$\text{Hence } (1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n.$$

Illustration 15 : If a, b, x, y are positive natural numbers such that $\frac{1}{x} + \frac{1}{y} = 1$ then prove that $\frac{a^x}{x} + \frac{b^y}{y} \geq ab$.

Solution : Consider the positive numbers $a^x, a^x, \dots, y$ times and $b^y, b^y, \dots, x$ times
For all these numbers,

$$\text{AM} = \frac{\{a^x + a^x + \dots, y \text{ times}\} + \{b^y + b^y + \dots, x \text{ times}\}}{x + y} = \frac{ya^x + xb^y}{(x + y)}$$

$$\text{GM} = \left\{ (a^x \cdot a^x \dots y \text{ times})(b^y \cdot b^y \dots x \text{ times}) \right\}^{\frac{1}{(x+y)}} = \left[(a^{xy}) \cdot (b^{xy}) \right]^{\frac{1}{(x+y)}} = (ab)^{\frac{xy}{(x+y)}}$$

$$\text{As } \frac{1}{x} + \frac{1}{y} = 1, \frac{x+y}{xy} = 1, \text{ i.e., } x + y = xy$$

$$\text{So using AM} \geq \text{GM } \frac{ya^x + xb^y}{x + y} \geq (ab)^{\frac{xy}{x+y}}$$

$$\therefore \frac{ya^x + xb^y}{xy} \geq ab \text{ or } \frac{a^x}{x} + \frac{b^y}{y} \geq ab.$$

(a) SUM OF N TERMS OF AN ARITHMETICO-GEOMETRIC SERIES :

$$\text{Let } S_n = a + (a+d)r + (a+2d)r^2 + \dots + [a+(n-1)d]r^{n-1}$$

$$\text{then } S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]r^n}{1-r}, \quad r \neq 1$$

(b) SUM TO INFINITY :

$$\text{If } 0 < |r| < 1 \text{ \& } n \rightarrow \infty, \text{ then } \lim_{n \rightarrow \infty} r^n = 0, S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

Illustration 16 : Find the sum of series $4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \dots \infty, |x| < 1$

Solution : Let $S = 4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \dots \infty$

$$-Sx = -4x + 9x^2 - 16x^3 + 25x^4 - 36x^5 + \dots \infty$$

On subtraction, we get

$$S(1+x) = 4 - 5x + 7x^2 - 9x^3 + 11x^4 - 13x^5 + \dots \infty$$

$$-S(1+x)x = -4x + 5x^2 - 7x^3 + 9x^4 - 11x^5 + \dots \infty$$

On subtraction, we get

$$S(1+x)^2 = 4 - x + 2x^2 - 2x^3 + 2x^4 - 2x^5 + \dots \infty$$

$$= 4 - x + 2x^2(1 - x + x^2 - \dots \infty) = 4 - x + \frac{2x^2}{1+x} = \frac{4+3x+x^2}{1+x}$$

$$S = \frac{4+3x+x^2}{(1+x)^3} \text{ Ans.}$$

Illustration 17 : Find the sum of series upto n terms $\left(\frac{2n+1}{2n-1}\right) + 3\left(\frac{2n+1}{2n-1}\right)^2 + 5\left(\frac{2n+1}{2n-1}\right)^3 + \dots$

Solution : For $x \neq 1$, let

$$S = x + 3x^2 + 5x^3 + \dots + (2n-3)x^{n-1} + (2n-1)x^n \dots \dots (i)$$

$$\Rightarrow xS = x^2 + 3x^3 + \dots + (2n-5)x^{n-1} + (2n-3)x^n + (2n-1)x^{n+1} \dots \dots (ii)$$

Subtracting (ii) from (i), we get

$$(1-x)S = x + 2x^2 + 2x^3 + \dots + 2x^{n-1} + 2x^n - (2n-1)x^{n+1}$$

$$= x + \frac{2x^2(1-x^{n-1})}{1-x} - (2n-1)x^{n+1}$$

$$= \frac{x}{1-x} [1-x+2x-2x^n-(2n-1)x^n+(2n-1)x^{n+1}]$$

$$\Rightarrow S = \frac{x}{(1-x)^2} [(2n-1)x^{n+1} - (2n+1)x^n + 1+x]$$

$$\text{Thus } \left(\frac{2n+1}{2n-1}\right) + 3\left(\frac{2n+1}{2n-1}\right)^2 + \dots + (2n-1)\left(\frac{2n+1}{2n-1}\right)^n$$

$$= \left(\frac{2n+1}{2n-1}\right)\left(\frac{2n-1}{2}\right)^2 \left[(2n-1)\left(\frac{2n+1}{2n-1}\right)^{n+1} - (2n+1)\left(\frac{2n+1}{2n-1}\right)^n + 1 + \frac{2n+1}{2n-1} \right]$$

$$= \frac{4n^2-1}{4} \cdot \frac{4n}{2n-1} = n(2n+1) \quad \text{Ans.}$$

Do yourself - 6 :

- Find sum to n terms of the series $3 + 5 \times \frac{1}{4} + 7 \times \frac{1}{4^2} + \dots$
- sum $1 + 2a + 3a^2 + 4a^3 + \dots$ to n terms.
- sum $1 + \frac{3}{4} + \frac{7}{16} + \frac{15}{64} + \frac{31}{256} + \dots$ to infinity.
- sum $\frac{2}{3} + \frac{3}{3^2} + \frac{2}{3^3} + \frac{3}{3^4} + \frac{2}{3^5} + \frac{3}{3^6} + \dots$ to infinity.
- Find the sum of n terms of the series the r^{th} term of which is $(2r+1)2^r$.
- Find the sum of the infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + \dots \infty$.

9. SIGMA NOTATIONS (Σ)

THEOREMS :

$$(a) \sum_{r=1}^n (a_r \pm b_r) = \sum_{r=1}^n a_r \pm \sum_{r=1}^n b_r \quad (b) \sum_{r=1}^n k a_r = k \sum_{r=1}^n a_r \quad (c) \sum_{r=1}^n k = nk ; \text{ where } k \text{ is a constant.}$$

10. RESULTS

- (a) $\sum_{r=1}^n r = \frac{n(n+1)}{2}$ (sum of the first n natural numbers)
- (b) $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ (sum of the squares of the first n natural numbers)
- (c) $\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4} = \left[\sum_{r=1}^n r \right]^2$ (sum of the cubes of the first n natural numbers)
- (d) $\sum_{r=1}^n r^4 = \frac{n}{30}(n+1)(2n+1)(3n^2+3n-1)$
- (e) $\sum_{r=1}^n (2r-1) = n^2$ (sum of first n odd natural numbers)
- (f) $\sum_{r=1}^n 2r = n(n+1)$ (sum of first n even natural numbers)

Note :

If n^{th} term of a sequence is given by $T_n = an^3 + bn^2 + cn + d$ where a, b, c, d are constants, then sum of n terms $S_n = \sum T_n = a\sum n^3 + b\sum n^2 + c\sum n + \sum d$

This can be evaluated using the above results.

Illustration 18 : Find the sum of the series to n terms whose n^{th} terms is $3n + 2$.

Solution : $S_n = \sum T_n = \sum (3n + 2) = 3\sum n + \sum 2 = \frac{3(n+1)n}{2} + 2n = \frac{n}{2}(3n + 7)$

Illustration 19 : If $T_k = k^3 + 3^k$, then find $\sum_{k=1}^n T_k$.

Solution : $\sum_{k=1}^n T_k = \sum_{k=1}^n k^3 + \sum_{k=1}^n 3^k = \left(\frac{n(n+1)}{2} \right)^2 + \frac{3(3^n - 1)}{3 - 1} = \left(\frac{n(n+1)}{2} \right)^2 + \frac{3}{2}(3^n - 1)$

Illustration 20 : Find the value of the expression $\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1$

Solution :
$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1 &= \sum_{i=1}^n \sum_{j=1}^i j = \sum_{i=1}^n \frac{i(i+1)}{2} = \frac{1}{2} \left[\sum_{i=1}^n i^2 + \sum_{i=1}^n i \right] = \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right] \\ &= \frac{n(n+1)}{12} [2n+1+3] = \frac{n(n+1)(n+2)}{6} \end{aligned}$$

Illustration 21 : Sum up to 16 terms of the series $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + \dots$ is

- (A) 450 (B) 456 (C) 446 (D) none of these

Solution :

$$t_n = \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{1 + 3 + 5 + \dots (2n-1)} = \frac{\left\{ \frac{n(n+1)}{2} \right\}^2}{\frac{n}{2} \{2 + 2(n-1)\}} = \frac{\frac{n^2(n+1)^2}{4}}{n^2} = \frac{(n+1)^2}{4} = \frac{n^2}{4} + \frac{n}{2} + \frac{1}{4}$$

$$\therefore S_n = \sum t_n = \frac{1}{4} \sum n^2 + \frac{1}{2} \sum n + \frac{1}{4} \sum 1 = \frac{1}{4} \cdot \frac{n(n+1)(2n+1)}{6} + \frac{1}{2} \cdot \frac{n(n+1)}{2} + \frac{1}{4} \cdot n$$

$$\therefore S_{16} = \frac{16 \cdot 17 \cdot 33}{24} + \frac{16 \cdot 17}{4} + \frac{16}{4} = 446$$

Ans. (C)

11. METHOD OF DIFFERENCE :

Some times the n^{th} term of a sequence or a series can not be determined by the method, we have discussed earlier. So we compute the difference between the successive terms of given sequence for obtained the n^{th} terms.

If $T_1, T_2, T_3, \dots, T_n$ are the terms of a sequence then some times the terms $T_2 - T_1, T_3 - T_2, \dots$ constitute an AP/GP. n^{th} term of the series is determined & the sum to n terms of the sequence can easily be obtained.

Case 1 :

- (a) If difference series are in A.P., then

Let $T_n = an^2 + bn + c$, where a, b, c are constant

- (b) If difference of difference series are in A.P.

Let $T_n = an^3 + bn^2 + cn + d$, where a, b, c, d are constant

Case 2 :

- (a) If difference are in G.P., then

Let $T_n = ar^n + b$, where r is common ratio & a, b are constant

- (b) If difference of difference are in G.P., then

Let $T_n = ar^n + bn + c$, where r is common ratio & a, b, c are constant

Determine constant by putting $n = 1, 2, 3, \dots, n$ and putting the value of $T_1, T_2, T_3, \dots$

and sum of series $(S_n) = \sum T_n$

Illustration 22 : Find the sum of n terms of the series $3 + 7 + 14 + 24 + 37 + \dots$

Solution : Clearly here the differences between the successive terms are

$7 - 3, 14 - 7, 24 - 14, \dots$ i.e. $4, 7, 10, 13, \dots$, which are in A.P.

Let $S = 3 + 7 + 14 + 24 + \dots + T_n$

$S = 3 + 7 + 14 + \dots + T_{n-1} + T_n$

Subtracting, we get

$$0 = 3 + [4 + 7 + 10 + 13 + \dots (n-1) \text{ terms}] - T_n$$

$\therefore T_n = 3 + S_{n-1}$ of an A.P. whose $a = 4$ and $d = 3$.

$$\therefore T_n = 3 + \left(\frac{n-1}{2} \right) (2 \cdot 4 + (n-2)3) = \frac{6 + (n-1)(3n+2)}{4} \quad \text{or, } T_n = \frac{1}{2}(3n^2 - n + 4)$$

Now putting $n = 1, 2, 3, \dots, n$ and adding

$$\therefore S_n = \frac{1}{2} [3 \sum n^2 - \sum n + 4n] = \frac{1}{2} \left[3 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n \right] = \frac{n}{2} (n^2 + n + 4) \text{ Ans.}$$

Aliter Method :

Let $T_n = an^2 + bn + c$

$$\text{Now, } T_1 = 3 = a + b + c \quad \dots(i)$$

$$T_2 = 7 = 4a + 2b + c \quad \dots(ii)$$

$$T_3 = 14 = 9a + 3b + c \quad \dots(iii)$$

Solving (i), (ii) & (iii) we get

$$a = \frac{3}{2}, b = -\frac{1}{2} \text{ \& } c = 2 \quad \therefore T_n = \frac{1}{2}(3n^2 - n + 4)$$

$$\Rightarrow S_n = \sum T_n = \frac{1}{2} [3 \sum n^2 - \sum n + 4n] = \frac{1}{2} \left[3 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n \right] = \frac{n}{2} (n^2 + n + 4)$$

Ans.

Illustration 23 : Find the sum of n-terms of the series $1 + 4 + 10 + 22 + \dots$

Solution : Let $S = 1 + 4 + 10 + 22 + \dots + T_n$ (i)

$$S = 1 + 4 + 10 + \dots + T_{n-1} + T_n \text{ (ii)}$$

$$(i) - (ii) \Rightarrow T_n = 1 + (3 + 6 + 12 + \dots + T_n - T_{n-1})$$

$$T_n = 1 + 3 \left(\frac{2^{n-1} - 1}{2 - 1} \right)$$

$$T_n = 3 \cdot 2^{n-1} - 2$$

$$\text{So } S_n = \sum T_n = 3 \sum 2^{n-1} - \sum 2$$

$$= 3 \left(\frac{2^n - 1}{2 - 1} \right) - 2n = 3 \cdot 2^n - 2n - 3$$

Ans.

Aliter Method :

Let $T_n = ar^n + b$, where $r = 2$

$$\text{Now } T_1 = 1 = ar + b \text{(i)}$$

$$T_2 = 4 = ar^2 + b \text{(ii)}$$

Solving (i) & (ii), we get

$$a = \frac{3}{2}, b = -2$$

$$\therefore T_n = 3 \cdot 2^{n-1} - 2$$

$$\Rightarrow S_n = \sum T_n = 3 \sum 2^{n-1} - \sum 2$$

$$= 3 \left(\frac{2^n - 1}{2 - 1} \right) - 2n = 3 \cdot 2^n - 2n - 3$$

Ans.

Illustration 24 : Find the general term and sum of n terms of the series

$$1 + 5 + 19 + 49 + 101 + 181 + 295 + \dots$$

Solution : The sequence of difference between successive term 4, 14, 30, 52, 80

The sequence of the second order difference is 10, 16, 22, 28, clearly it is an A.P.

So, let n^{th} term

$$T_n = an^3 + bn^2 + cn + d$$

$$a + b + c + d = 1 \quad \dots (i)$$

$$8a + 4b + 2c + d = 5 \quad \dots (ii)$$

$$27a + 9b + 3c + d = 19 \quad \dots (iii)$$

$$64a + 16b + 4c + d = 49 \quad \dots (iv)$$

from (i), (ii), (iii) & (iv)

$$a = 1, b = -1, c = 0, d = 1 \Rightarrow T_n = n^3 - n^2 + 1$$

$$\therefore S_n = \sum (n^3 - n^2 + 1) = \left(\frac{n(n+1)}{2} \right)^2 - \frac{n(n+1)(2n+1)}{6} + n = \frac{n(n^2-1)(3n+2)}{12} + n$$

Illustration 25 : If $\sum_{r=1}^n T_r = \frac{n}{8}(n+1)(n+2)(n+3)$, then find $\sum_{r=1}^n \frac{1}{T_r}$.

Solution :

$$\therefore T_n = S_n - S_{n-1}$$

$$= \sum_{r=1}^n T_r - \sum_{r=1}^{n-1} T_r = \frac{n(n+1)(n+2)(n+3)}{8} - \frac{(n-1)n(n+1)(n+2)}{8} = \frac{n(n+1)(n+2)}{8} [(n+3) - (n-1)]$$

$$T_n = \frac{n(n+1)(n+2)}{8} (4) = \frac{n(n+1)(n+2)}{2}$$

$$\Rightarrow \frac{1}{T_n} = \frac{2}{n(n+1)(n+2)} = \frac{(n+2) - n}{n(n+1)(n+2)} = \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \quad \dots (i)$$

$$\text{Let } V_n = \frac{1}{n(n+1)}$$

$$\therefore \frac{1}{T_n} = V_n - V_{n+1}$$

Putting $n = 1, 2, 3, \dots, n$

$$\Rightarrow \frac{1}{T_1} + \frac{1}{T_2} + \frac{1}{T_3} + \dots + \frac{1}{T_n} = (V_1 - V_{n+1}) \Rightarrow \sum_{r=1}^n \frac{1}{T_r} = \frac{n^2 + 3n}{2(n+1)(n+2)}$$

Illustration 26 : Find the sum of n terms of the series $1 \cdot 3 \cdot 5 + 3 \cdot 5 \cdot 7 + 5 \cdot 7 \cdot 9 + \dots$

Solution : The n^{th} term is $(2n-1)(2n+1)(2n+3)$

$$T_n = (2n-1)(2n+1)(2n+3)$$

$$T_n = \frac{1}{8}(2n-1)(2n+1)(2n+3)\{(2n+5)-(2n-3)\}$$

$$= \frac{1}{8}(V_n - V_{n-1}) \quad [\text{Let } V_n = (2n-1)(2n+1)(2n+3)(2n+5)]$$

$$S_n = \sum T_n = \frac{1}{8}[V_n - V_0]$$

$$\therefore S_n = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{8} + \frac{15}{8} = n(2n^3 + 8n^2 + 7n - 2) \quad \text{Ans.}$$

Illustration 27 : Find the natural number 'a' for which $\sum_{k=1}^n f(a+k) = 16(2^n - 1)$, where the function f satisfied

$$f(x+y) = f(x) \cdot f(y) \text{ for all natural number } x, y \text{ and further } f(1) = 2.$$

Solution : It is given that

$$f(x+y) = f(x) \cdot f(y) \text{ and } f(1) = 2$$

$$f(1+1) = f(1) \cdot f(1) \Rightarrow f(2) = 2^2, f(1+2) = f(1) \cdot f(2) \Rightarrow f(3) = 2^3, f(2+2) = f(2)$$

$$f(2) \Rightarrow f(4) = 2^4$$

$$\text{Similarly } f(k) = 2^k \text{ and } f(a) = 2^a$$

$$\text{Hence, } \sum_{k=1}^n f(a+k) = \sum_{k=1}^n f(a)f(k) = f(a) \sum_{k=1}^n f(k) = 2^a \sum_{k=1}^n 2^k = 2^a \{2^1 + 2^2 + \dots + 2^n\}$$

$$= 2^a \left\{ \frac{2(2^n - 1)}{2 - 1} \right\} = 2^{a+1}(2^n - 1)$$

$$\text{But } \sum_{k=1}^n f(a+k) = 16(2^n - 1)$$

$$2^{a+1}(2^n - 1) = 16(2^n - 1)$$

$$\therefore 2^{a+1} = 2^4$$

$$\therefore a+1 = 4 \Rightarrow a = 3$$

Ans.

Do yourself - 7 :

1. Find the sum of the series upto n terms $1 + \frac{1+2}{2} + \frac{1+2+3}{3} + \frac{1+2+3+4}{4} + \dots$
2. Find the sum of 'n' terms of the series whose n^{th} term is
 - (a) $3n^2 + 2n$
 - (b) $3n^2 - n$
 - (c) $n^3 + \frac{3}{2}n$
 - (d) $n(n+2)$
 - (e) $n^2(2n+3)$
 - (f) $3^n - 2^n$
 - (g) $3(4^n + 2n^2) - 4n^3$
3. Sum the following series to n terms :
 - (a) $1 \cdot 4 \cdot 7 + 2 \cdot 5 \cdot 8 + 3 \cdot 6 \cdot 9 + \dots$
 - (b) $1 \cdot 5 \cdot 9 + 2 \cdot 6 \cdot 10 + 3 \cdot 7 \cdot 11 + \dots$
4. Find the n^{th} term and the sum of n terms of the series :
 - (a) 4, 14, 30, 52, 80, 114,
 - (b) 9, 16, 29, 54, 103,
5. Find the n^{th} term and the sum to n terms of the sequence :
 - (i) $1 + 5 + 13 + 29 + 61 + \dots$
 - (ii) $6 + 13 + 22 + 33 + \dots$
6. Sum the following series to n terms and to infinity (where it is finite and defined) :
 - (i) $\frac{1}{1 \cdot 4 \cdot 7} + \frac{1}{4 \cdot 7 \cdot 10} + \frac{1}{7 \cdot 10 \cdot 13} + \dots$
 - (ii) $\sum_{r=1}^n r(r+1)(r+2)(r+3)$
 - (iii) $\sum_{r=1}^n \frac{1}{4r^2 - 1}$
7. The value of $1^2 + 3^2 + 5^2 + \dots + 25^2$ is :

(A) 1728	(B) 1456	(C) 2925	(D) 1469
----------	----------	----------	----------
8. The sum of the series : $(2)^2 + 2(4)^2 + 3(6)^2 + \dots$ upto 10 terms is :

(A) 11300	(B) 12100	(C) 12300	(D) 11200
-----------	-----------	-----------	-----------
9. The value of $(1 \cdot 1^2 + 3 \cdot 2^2 + 5 \cdot 3^2 + \dots + \text{upto 10 terms})$ is equal to :

(A) 6050	(B) 5965	(C) 5665	(D) 5385
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EXERCISE (O-1)

Straight Objective Type

1. If $a_1, a_2, a_3, \dots, a_n, \dots$ are in A.P. such that $a_4 - a_7 + a_{10} = m$, then the sum of first 13 terms of this A.P., is :
 (A) 15 m (B) 10 m (C) 12 m (D) 13 m SS0001
2. If a, b, c are in AP, then $(a - c)^2$ equals
 (A) $4(b^2 - ac)$ (B) $4(b^2 + ac)$ (C) $4b^2 - ac$ (D) $b^2 - 4ac$ SS0002
3. Let $a_1, a_2, a_3, \dots$ be an A.P. such that $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + a_3 + \dots + a_q} = \frac{p^2}{q^2}$; $p \neq q$. Then $\frac{a_6}{a_{21}}$ is equal to :
 (A) $\frac{121}{1861}$ (B) $\frac{11}{41}$ (C) $\frac{121}{1681}$ (D) $\frac{41}{11}$ SS0003
4. Given sum of the first n terms of an A. P. is $2n + 3n^2$. Another A. P. is formed with the same first term and double of the common difference, the sum of n terms of the new A. P. is :-
 (A) $n + 4n^2$ (B) $n^2 + 4n$ (C) $3n + 2n^2$ (D) $6n^2 - n$ SS0004
5. If for an A.P. $a_1, a_2, a_3, \dots, a_n, \dots$
 $a_1 + a_3 + a_5 = -12$ and $a_1 a_2 a_3 = 8$, then the value of $a_2 + a_4 + a_6$ equals
 (A) -12 (B) -16 (C) -18 (D) -21 SS0021
6. If the sum of the first 11 terms of an arithmetical progression equals that of the first 19 terms, then the sum of its first 30 terms, is
 (A) equal to 0 (B) equal to -1 (C) equal to 1 (D) non unique SS0022
7. Let $s_1, s_2, s_3, \dots$ and $t_1, t_2, t_3, \dots$ are two arithmetic sequences such that $s_1 = t_1 \neq 0$; $s_2 = 2t_2$ and
 $\sum_{i=1}^{10} s_i = \sum_{i=1}^{15} t_i$. Then the value of $\frac{s_2 - s_1}{t_2 - t_1}$ is
 (A) $8/3$ (B) $3/2$ (C) $19/8$ (D) 2 SS0023
8. Along a road lies an odd number of stones placed at intervals of 10 m. These stones have to be assembled around the middle stone. A person can carry only one stone at a time. A man carried out the job starting with the stone in the middle, carrying stones in succession, thereby covering a distance of 4.8 km. Then the number of stones is
 (A) 15 (B) 29 (C) 31 (D) 35 SS0025

9. In an A.P. with first term 'a' and the common difference d ($a, d \neq 0$), the ratio ' ρ ' of the sum of the first n terms to sum of n terms succeeding them does not depend on n. Then the ratio $\frac{a}{d}$ and the ratio ' ρ ', respectively are

(A) $\frac{1}{2}, \frac{1}{4}$ (B) $2, \frac{1}{3}$ (C) $\frac{1}{2}, \frac{1}{3}$ (D) $\frac{1}{2}, 2$

SS0026

10. Let $a_n, n \in \mathbb{N}$ is an A.P. with common difference 'd' and all whose terms are non-zero. If n approaches infinity, then the sum $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_n a_{n+1}}$ will approach

(A) $\frac{1}{a_1 d}$ (B) $\frac{2}{a_1 d}$ (C) $\frac{1}{2a_1 d}$ (D) $a_1 d$

SS0027

11. The first term of an infinite G.P. is 1 and every term is equals to the sum of the successive terms, then its fourth term will be-

(A) $\frac{1}{2}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{16}$

SS0006

12. If $a = \sum_{n=0}^{\infty} x^n, b = \sum_{n=0}^{\infty} y^n, c = \sum_{n=0}^{\infty} (xy)^n$ where $|x|, |y| < 1$; then-

(A) $abc = a + b + c$ (B) $ab + bc = ac + b$ (C) $ac + bc = ab + c$ (D) $ab + ac = bc + a$

SS0007

13. If $r > 1$ and $x = a + \frac{a}{r} + \frac{a}{r^2} + \dots$ to ∞ , $y = b - \frac{b}{r} + \frac{b}{r^2} - \dots$ to ∞ and $z = c + \frac{c}{r^2} + \frac{c}{r^4} + \dots$ to ∞ , then

$\frac{xy}{z} =$

(A) $\frac{ab}{c}$ (B) $\frac{ac}{b}$ (C) $\frac{bc}{a}$ (D) None of these

SS0008

14. In a GP, first term is 1. If $4T_2 + 5T_3$ is minimum, then its common ratio is

(A) $\frac{2}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $-\frac{3}{5}$

SS0009

15. Given a sequence of 4 numbers, first three of which are in G.P. and the last three are in A.P. with common difference six. If first and last terms of this sequence are equal, then the last term is :
 (A) 8 (B) 16 (C) 2 (D) 4

SS0010

16. If G be the GM between x and y, then the value of $\frac{1}{G^2 - x^2} + \frac{1}{G^2 - y^2}$ is equal to
 (A) G^2 (B) $\frac{2}{G^2}$ (C) $\frac{1}{G^2}$ (D) $3G^2$

SS0011

17. If $\frac{1+3+5+\dots\text{upto } n \text{ terms}}{4+7+10+\dots\text{upto } n \text{ terms}} = \frac{20}{7 \log_{10} x}$ and

$$n = \log_{10} x + \log_{10} x^{\frac{1}{2}} + \log_{10} x^{\frac{1}{4}} + \log_{10} x^{\frac{1}{8}} + \dots + \infty, \text{ then } x \text{ is equal to}$$

- (A) 10^3 (B) 10^5 (C) 10^6 (D) 10^7

SS0029

18. If $a \neq 1$ and $\ln a^2 + (\ln a^2)^2 + (\ln a^2)^3 + \dots = 3 (\ln a + (\ln a)^2 + (\ln a)^3 + (\ln a)^4 + \dots)$ then 'a' is equal to
 (A) $e^{1/5}$ (B) $\sqrt{e}$ (C) $\sqrt[3]{e}$ (D) $\sqrt[4]{e}$

SS0030

19. If a, b, c are in HP, then $\frac{a-b}{b-c}$ is equal to

- (A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\frac{a}{c}$ (D) $\frac{c}{b}$

SS0012

20. If a, b, c are distinct positive real in H.P., then the value of the expression, $\frac{b+a}{b-a} + \frac{b+c}{b-c}$ is equal to
 (A) 1 (B) 2 (C) 3 (D) 4

SS0031

21. An H.M. is inserted between the number $1/3$ and an unknown number. If we diminish the reciprocal of the inserted number by 6, it is the G.M. of the reciprocal of $1/3$ and that of the unknown number. If all the terms of the respective H.P. are distinct then
 (A) the unknown number is 27 (B) the unknown number is $1/27$
 (C) the H.M. is 15 (D) the G.M. is 21

SS0032

22. If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in A.P. and $|a| < 1$, $|b| < 1$, $|c| < 1$, then x, y, z are in-
 (A) HP (B) Arithmetic - Geometric Progression
 (C) AP (D) GP

SS0035

23. If a, b and c are positive real numbers then $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is greater than or equal to
 (A) 3 (B) 6 (C) 27 (D) 5 SS0013
24. If $a_1, a_2, a_3, \dots, a_n \in \mathbb{R}^+$ and $a_1 \cdot a_2 \cdot a_3 \dots a_n = 1$, then minimum value of $(1 + a_1 + a_1^2)(1 + a_2 + a_2^2) \dots (1 + a_n + a_n^2)$ is equal to :-
 (A) 3^{n+1} (B) 3^n (C) 3^{n-1} (D) none of these SS0014
25. If a, b, c are positive real numbers such that $ab^2c^3 = 64$ then minimum value of $\left(\frac{1}{a} + \frac{2}{b} + \frac{3}{c}\right)$ is equal to:-
 (A) 6 (B) 2 (C) 3 (D) None of these SS0015
26. If $abcd = 1$ where a, b, c, d are positive reals then the minimum value of $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$ is
 (A) 6 (B) 10 (C) 12 (D) 20 SS0033
27. If $x \in \mathbb{R}$, the numbers $(5^{1+x} + 5^{1-x}), a/2, (25^x + 25^{-x})$ form an A.P. then 'a' must lie in the interval
 (A) $[1, 5]$ (B) $[2, 5]$ (C) $[5, 12]$ (D) $[12, \infty)$ SS0024
28. $2 + 4 + 7 + 11 + 16 + \dots$ to n terms =
 (A) $\frac{1}{6}(n^2 + 3n + 8)$ (B) $\frac{n}{6}(n^2 + 3n + 8)$ (C) $\frac{1}{6}(n^2 - 3n + 8)$ (D) $\frac{n}{6}(n^2 - 3n + 8)$ SS0018
29. The sum $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots$ upto 11-terms is :-
 (A) $\frac{11}{4}$ (B) $\frac{60}{11}$ (C) $\frac{7}{2}$ (D) $\frac{11}{2}$ SS0019
30. The sum of the series : $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots$ up to 10 terms, is:
 (A) $\frac{22}{13}$ (B) $\frac{18}{11}$ (C) $\frac{20}{11}$ (D) $\frac{16}{9}$ SS0020

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $a_1, a_2, \dots, a_n$ are distinct terms of an A.P., then
 (A) $a_1 + 2a_2 + a_3 = 0$ (B) $a_1 - 2a_2 + a_3 = 0$
 (C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$ (D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$ **SS0126**
2. Let $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be arithmetic progressions such that $a_1 = 25, b_1 = 75$ and $a_{100} + b_{100} = 100$. Then
 (A) the difference between successive terms in progression 'a' is opposite of the difference in progression 'b'.
 (B) $a_n + b_n = 100$ for any n .
 (C) $(a_1 + b_1), (a_2 + b_2), (a_3 + b_3), \dots$ are in A.P.
 (D) $\sum_{r=1}^{100} (a_r + b_r) = 10000$ **SS0038**
3. If $1, \log_y x, \log_z y, -15 \log_x z$ are in A.P., then
 (A) $z^3 = x$ (B) $x = y^{-1}$ (C) $z^{-3} = y$ (D) $x = y^{-1} = z^3$ **SS0127**
4. Real number x_1, x_2, x_3 in A.P. satisfying the equation $x^3 - x^2 + \alpha x + \beta = 0$ then select correct alternative(s)
 (A) $\alpha \leq \frac{1}{3}$ (B) $\beta \geq -\frac{1}{27}$ (C) $\alpha > \frac{1}{3}$ (D) $\beta < -\frac{1}{27}$ **SS0128**
5. Let a, b, c are non zero real numbers such that $3(a^2 + b^2 + c^2 + 1) = 2(a + b + c + ab + bc + ca)$ then
 (A) a, b, c are in AP (B) a, b, c are in GP
 (C) a, b, c are in HP (D) a^2, b^2, c^2 are in GP **SS0129**
6. If a satisfies the equation $a^{2017} - 2a + 1 = 0$ and $S = 1 + a + a^2 + \dots + a^{2016}$ then possible value(s) of S is/are
 (A) 2016 (B) 2018 (C) 2017 (D) 2 **SS0130**
7. If $a_1, a_2, a_3, \dots$ are positive numbers in G.P., then which of the following is/are true -
 (A) $a_1^3, a_2^3, a_3^3, \dots$ are in G.P. (B) $\frac{a_2}{a_1}, \left(\frac{a_3}{a_2}\right)^2, \left(\frac{a_4}{a_3}\right)^3, \dots$ are in G.P.
 (C) $\ln a_1, \ln a_2, \ln a_3, \dots$ are in A.P. (D) $a_1 a_2, a_2 a_3, a_3 a_4, \dots$ are in G.P. **SS0131**
8. The increasing geometric sequence of natural numbers $x_1, x_2, x_3, \dots$ consist entirely of integral powers of 2. Given $\sum_{n=1}^8 \log_2(x_n) = 196$ then which of the following is/are correct?
 (A) Total number of all possible such sequences is 4
 (B) Total number of all possible such sequences is more than four
 (C) Maximum value of x_4 is 2^{24}
 (D) Maximum value of x_4 is 2^{23} **SS0132**

9. If equation $16x^4 - mx^3 + (2m + 17)x^2 - mx + 16 = 0$ has four distinct roots forming a geometric progression, and one of the root is $\frac{1}{2}$, then :

(A) common ratio of G.P. is 4

(B) $m = 170$

(C) sum of the roots = $\frac{85}{8}$

(D) Product of root = 3

SS0133

10. Three numbers a, b, c between 2 and 18 are such that

(i) their sum is 25

(ii) the numbers 2, a, b , are in A.P.

(iii) the number $b, c, 18$ are in G.P.

then which of the following options are correct.

(A) $a = 5$

(B) $b = 8$

(C) $b + c = 20$

(D) $a + b + c = 25$ SS0134

11. Let $\{a_n\}$ be a sequence such that $a_0 = 1, a_1 = 0, a_n = 3a_{n-1} - 2a_{n-2}$. The correct statement is: SS0135

(A) $\sum_{i=1}^{50} a_i = 2(51 - 2^{50})$ (B) $a_{51} = 2^{25} - 2$ (C) $a_{48} = 2(1 - 2^{47})$ (D) $\sum_{i=1}^{49} a_i = (2^{49} - 50)$

12. If a, b, c are positive integers forming first three terms of increasing geometric progression with common ratio 'r', where $b - a$ is perfect cube and $\log_6 a + \log_6 b + \log_6 c = 6$. Let S denotes sum of 10 terms of

another G.P. whose first term is 'a' and common ratio is $\frac{5r}{2}$, then

(A) $a + b + c = 189$

(B) $a + b + c = 273$

(C) sum of digits of S is 90.

(D) $S = 10^{10} - 1$

SS0136

13. The roots of the equation $x^4 - 8x^3 + ax^2 - bx + 16 = 0$, are positive, if

(A) $a = 24$

(B) $a = 12$

(C) $b = 8$

(D) $b = 32$

SS0137

14. Choose the correct statement(s)

(A) Sum of ten arithmetic means between $\sin^2 18^\circ$ and $\cos^2 18^\circ$ is 5

(B) Product of ten geometric means between $4\tan 15^\circ$ and $\cot 15^\circ$ is 1024

(C) If $27abc \geq (a + b + c)^3$ and $3a + 4b + 5c = 12$ then $\frac{1}{a^2} + \frac{1}{b^3} + \frac{1}{c^5} = 10$; where

a, b, c are positive real numbers.

(D) For positive real numbers A.M. $\geq$ G.M.

SS0138

15. Let $x, y, z \in \mathbb{R}^+$ and $x + y + z = 2$, then choose the correct option(s)-

(A) maximum value of $x^2 y^3 z^2$ is $\frac{2^{11} \cdot 3^3}{7^7}$

(B) maximum value of $x^2 y^3 z^2$ is 1.

(C) minimum value of $\frac{4}{x} + \frac{1}{y} + \frac{9}{z}$ is 18

(D) maximum value of xyz is $\frac{8}{27}$

SS0139

16. Let $0 < a < b$ such that $2, 2^x - \frac{7}{2}, 2^x - 5$ are harmonic mean, arithmetic mean and geometric mean of a, b respectively then which of the following option(s) is/are correct
 (A) number of possible values of x is 2 (B) number of value of x is 1
 (C) $a^2 + b^2$ can be 63 (D) $a^2 + b^2$ can be 3 **SS0140**

17. The value of $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{5^n}$ is $\frac{a}{b}$, where a & b are relatively prime then
 (A) $a > b$ (B) $a < b$ (C) $a > 3$ (D) $b > 10$ **SS0141**

18. Let $S_r = 2 + 4 + 6 + \dots + r$ terms then -

- (A) $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{S_r} = 1$ (B) $\sum_{r=1}^n S_r = \frac{n(n+1)(n+2)}{6}$
 (C) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sum_{r=1}^k S_r} = \frac{3}{4}$ (D) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sum_{r=1}^k S_r} = \frac{1}{4}$ **SS0142**

19. The value of $\sum_{r=2}^{100} \frac{2^r(2-r)}{r(r+1)(r+2)}$ is equal to

- (A) $\frac{2}{3} - \frac{2^{100}}{(51)(101)}$ (B) $\frac{2}{3} - \frac{2^{101}}{(101)(102)}$ (C) $\frac{1}{2} - \frac{2^{100}}{100.101}$ (D) $\frac{1}{2} - \frac{2^{100}}{(101)(51)}$ **SS0143**

20. Consider the arrangement

1
 2 3 4
 5 6 7 8 9
 10 11 12 13 14 15 16

and so on

then correct option's is/are

- (A) first term of 15th row is 197
 (B) first term of 15th row is 201

- (C) let t_i be the number of terms in i^{th} row then $\sum_{i=1}^n t_i = n^2$

- (D) let t_i be the number of terms in i^{th} row then $\sum_{i=1}^n t_i = 2n - 1$ **SS0144**

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 2

A geometric series has first term 'a' and common ratio 'r'. The second term of the series is 4 and the sum upto infinite terms of the series is 25

1. The sum of all possible values of a is-
(A) 5 (B) 15 (C) 20 (D) 25 **SS0145**
2. If 'r' is taken the larger of its two possible values then the smallest value of n for which S_n exceeds 15 is-
(A) 4 (B) 5 (C) 6 (D) 7 **SS0146**

Paragraph for Question 3 to 4

From first 'n' natural numbers, 2 consecutive numbers are removed the arithmetic mean of the remaining number is $\frac{105}{4}$ then answer the following

3. n is divisible by
(A) 15 (B) 25 (C) 5 (D) 10 **SS0147**
4. The removed number is(are)
(A) 6 (B) 7 (C) 8 (D) 9 **SS0148**

Paragraph for Question 5 to 6

In a G.P. the sum of the first and last term is 66 and the product of second and the second last term is 128. Sum of all terms is 126

On the basis of above information, answer the following questions :

5. If an increasing G.P. is considered then number of terms **SS0149**
6. Difference of greatest and least term is **SS0150**

Matrix Match Type

7. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

(P) If p, q, r are in GP ; $a^p = b^q = c^r$, then ℓna , ℓnb , ℓnc are always in

(Q) If $\frac{a+be^x}{a-be^x} = \frac{b+ce^x}{b-ce^x} = \frac{c+de^x}{c-de^x}$, then a, b, c, d are in

(R) If a, b, c are in AP; a, p, b are in GP; b, q, c are in GP, then p^2 , b^2 , q^2 are in

(S) If a, b, c are in GP, then $\log_a 10$, $\log_b 10$, $\log_c 10$ are in

List-II

(1) AP

(2) GP

(3) HP

(4) None

Codes :

	P	Q	R	S
(A)	1	4	3	1
(B)	4	3	1	2
(C)	3	4	2	2
(D)	2	2	1	3

SS0151

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I**List-II**

- (P) Number of common integers in two arithmetic progression 1, 3, 5, 7, ... 157 and 2, 5, 8, 11 ... 161 is (1) 25
- (Q) If a and b are roots of $11x^2 - 4x - 2 = 0$, then value of $10(1 + a + a^2 + \dots) (1 + b + b^2 + \dots)$ is (2) 26
- (R) If the sum of first 20 terms is same as the sum of first 10 terms in an arithmetic progression then the sum of first 30 terms is less than (3) 22
- (S) If sum of $\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} \dots + \frac{199}{99^2 \cdot 100^2}$ is $\frac{a}{b}$ (4) 20
(where a and b are coprime) the value of $20|b - a|$ is (5) 10
- (A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 1, 2, 3; S \rightarrow 3$ (B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1, 2, 3, 4, 5; S \rightarrow 4$
- (C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1, 2; S \rightarrow 2$ (D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3, 4; S \rightarrow 1$ **SS0152**

Match the column

9. **Column - I**

Column - II

- (A) The coefficient of x^{49} in the product $(x - 1)(x - 3)(x - 5)(x - 7) \dots (x - 99)$ (p) -2500
- (B) Let S_n denote sum of first n terms of an A.P. If $S_{2n} = 3S_n$, then $\frac{S_{3n}}{S_n}$ is (q) 9

- (C) The sum $\sum_{r=2}^{\infty} \frac{1}{r^2 - 1}$ is equal to (r) $\frac{3}{4}$

- (D) The length, breadth, height of a rectangular box are in G.P. (length > breadth > height) The volume is 27, the total surface area is 78. Then the length is (s) 6

SS0153

10. **Column - I**

Column - II

- (A) The value of xyz is $\frac{15}{2}$ or $\frac{18}{5}$ according as the series a, x, y, z, b are in an A.P. or H.P. then 'a + b' equals where a, b are positive integers. (p) 2
- (B) The value of $2^{\frac{1}{4}} 4^{\frac{1}{8}} 8^{\frac{1}{16}} \dots \infty$ is equal to (q) 1
- (C) If x, y, z are in A.P., then $(x + 2y - z)(2y + z - x)(z + x - y) = kxyz$, where $k \in \mathbb{N}$, then k is equal to (r) 3
- (D) There are m A.M. between 1 and 31. If the ratio of the 7^{th} and $(m - 1)^{\text{th}}$ means is 5 : 9, then $\frac{m}{7}$ is equal to (s) 4

SS0154

EXERCISE (O-4)

Numerical Grid Type

1. The interior angles of a convex polygon form an arithmetic progression with a common difference of 4° . Determine the number of sides of the polygon if its largest interior angle is 172° .

SS0040

2. The sequence $a_1, a_2, a_3, \dots, a_{98}$ satisfies the relation $a_{n+1} = a_n + 1$ for $n = 1, 2, 3, \dots, 97$ and has the sum equal to 4949. Evaluate $\sum_{k=1}^{49} a_{2k}$.

SS0042

3. The first term of an arithmetic progression is 1 and the sum of the first nine terms equal to 369. The first and the ninth term of a geometric progression with real common ratio coincide with the first and the ninth term of the arithmetic progression. Find the seventh term of the geometric progression.

SS0044

4. For an increasing G.P. $a_1, a_2, a_3, \dots, a_n$, if $a_6 = 4a_4$, $a_9 - a_7 = 192$, then the value of $\sum_{i=1}^{\infty} \frac{1}{a_i}$ is

SS0045

5. If $a > 0$, then minimum value of $a + 2a^2 + a^3 + 15 + a^{-1} + a^{-3} + a^{-4}$ is

SS0041

6. If $a, b, c, d > 0$ such that $a + 2b + 3c + 4d = 50$, then find the maximum value of $\left(\frac{a^2 b^4 c^3 d}{16} \right)^{1/10}$

SS0051

7. If $x > 0$, then find greatest value of the expression $\frac{x^{100}}{1 + x + x^2 + x^3 + \dots + x^{200}}$.

SS0155

8. If number of coins earned in n^{th} game is $n2^{n+2} - 2^n$ and total number of coins earned in first 10 games is $10(B \cdot 2^{10} + 1)$, where $B \in \mathbb{N}$, then the value of B is

SS0052

9. Find the sum of the sequence $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots \infty$

SS0067

10. Let $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$. Find $[S]$.

Where $[y]$ denotes largest integer less than or equal to y .

SS0069

EXERCISE (JM)

1. If m is the A.M. of two distinct real numbers l and n ($l, n > 1$) and G_1, G_2 and G_3 are three geometric means between l and n , then $G_1^4 + 2G_2^4 + G_3^4$ equals - [JEE(Main)-2015]
 (1) $4lmn^2$ (2) $4l^2m^2n^2$ (3) $4l^2mn$ (4) $4lm^2n$ SS0070
2. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is:- [JEE(Main)-2016]
 (1) $\frac{7}{4}$ (2) $\frac{8}{5}$ (3) $\frac{4}{3}$ (4) 1 SS0071
3. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$ is $\frac{16}{5}m$, then m is equal to :- [JEE(Main)-2016]
 (1) 99 (2) 102 (3) 101 (4) 100 SS0072
4. If, for a positive integer n , the quadratic equation, $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$ has two consecutive integral solutions, then n is equal to : [JEE(Main)-2017]
 (1) 11 (2) 12 (3) 9 (4) 10 SS0073
5. For any three positive real numbers a, b and c , $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then : [JEE(Main)-2017]
 (1) a, b and c are in G.P. (2) b, c and a are in G.P.
 (3) b, c and a are in A.P. (4) a, b and c are in A.P. SS0074
6. Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$. If $a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$, then m is equal to- [JEE(Main)-2018]
 (1) 68 (2) 34 (3) 33 (4) 66 SS0075
7. Let A be the sum of the first 20 terms and B be the sum of the first 40 terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$. If $B - 2A = 100\lambda$, then λ is equal to : [JEE(Main)-2018]
 (1) 248 (2) 464 (3) 496 (4) 232 SS0076
8. If a, b and c be three distinct real numbers in G. P. and $a + b + c = xb$, then x cannot be : [JEE(Main)-2019]
 (1) 4 (2) -3 (3) -2 (4) 2 SS0077

9. Let $a_1, a_2, \dots, a_{30}$ be an A. P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{(2i-1)}$. If $a_5 = 27$ and $S - 2T = 75$, then

a_{10} is equal to :

[JEE(Main)-2019]

- (1) 57 (2) 47 (3) 42 (4) 52

SS0078

10. The sum of the following series

$$1 + 6 + \frac{9(1^2 + 2^2 + 3^2)}{7} + \frac{12(1^2 + 2^2 + 3^2 + 4^2)}{9} + \frac{15(1^2 + 2^2 + \dots + 5^2)}{11} + \dots \text{ up to 15 terms, is :}$$

[JEE(Main)-2019]

- (1) 7820 (2) 7830
(3) 7520 (4) 7510

SS0079

11. Let x, y be positive real numbers and m, n positive integers. The maximum value of the expression

$$\frac{x^m y^n}{(1 + x^{2m})(1 + y^{2n})} \text{ is :-}$$

[JEE(Main)-2019]

- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$
(3) $\frac{m+n}{6mn}$ (4) 1

SS0080

12. If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to :

[JEE(Main)-2019]

- (1) $\beta\gamma$ (2) 0 (3) $\alpha\gamma$ (4) $\alpha\beta$

SS0081

13. The sum of all natural numbers 'n' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is :

[JEE(Main)- 2019]

- (1) 3221 (2) 3121 (3) 3203 (4) 3303

SS0082

14. If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct?

[JEE(Main)-2019]

- (1) d, e, f are in A.P. (2) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.
(3) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (4) d, e, f are in G.P.

SS0083

15. Let the sum of the first n terms of a non-constant A.P., $a_1, a_2, a_3, \dots$ be $50n + \frac{n(n-7)}{2}A$, where A is a constant. If d is the common difference of this A.P., then the ordered pair (d, a_{50}) is equal to

[JEE(Main)- 2019]

- (1) $(A, 50+46A)$ (2) $(A, 50+45A)$ (3) $(50, 50+46A)$ (4) $(50, 50+45A)$

SS0084

16. The sum $\frac{3 \times 1^3}{1^2} + \frac{5 \times (1^3 + 2^3)}{1^2 + 2^2} + \frac{7 \times (1^3 + 2^3 + 3^3)}{1^2 + 2^2 + 3^2} + \dots$ upto 10th term, is :

[JEE(Main)- 2019]

- (1) 660 (2) 620 (3) 680 (4) 600

SS0085

17. If $a_1, a_2, a_3, \dots, a_n$ are in A.P. and $a_1 + a_4 + a_7 + \dots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to :

[JEE(Main)- 2019]

- (1) 38 (2) 98 (3) 76 (4) 64

SS0086

18. Let a, b and c be in G. P. with common ratio r , where $a \neq 0$ and $0 < r \leq \frac{1}{2}$. If $3a, 7b$ and $15c$ are the first three terms of an A. P., then the 4th term of this A. P. is :

[JEE(Main)- 2019]

- (1) $\frac{7}{3}a$ (2) a (3) $\frac{2}{3}a$ (4) $5a$

SS0087

19. If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r + \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$ is equal to :

[JEE(Main)- 2019]

- (1) $\frac{21}{346}$ (2) $\frac{29}{358}$ (3) $\frac{1}{12}$ (4) $\frac{7}{116}$

SS0088

20. If the sum of the first 40 terms of the series, $3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + \dots$ is $(102)m$, then m is equal to :

[JEE(Main)- 2020]

- (1) 20 (2) 5 (3) 10 (4) 25

SS0089

21. Let $a_1, a_2, a_3, \dots$ be a G.P. such that $a_1 < 0$, $a_1 + a_2 = 4$ and $a_3 + a_4 = 16$. If $\sum_{i=1}^9 a_i = 4\lambda$, then λ is equal to :

[JEE(Main)- 2020]

- (1) -171 (2) 171 (3) $\frac{511}{3}$ (4) -513

SS0090

22. Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $-\frac{1}{2}$, then the greatest number amongst them is : [JEE(Main)- 2020]

(1) $\frac{21}{2}$ (2) 27 (3) 16 (4) 7

SS0091

23. The greatest positive integer k, for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + \dots + 49^2 + 49 + 1$, is : [JEE(Main)- 2020]

(1) 32 (2) 60 (3) 63 (4) 65

SS0092

24. The sum, $\sum_{n=1}^7 \frac{n(n+1)(2n+1)}{4}$ is equal to _____. [JEE(Main)- 2020]

SS0093

25. The sum $\sum_{k=1}^{20} (1 + 2 + 3 + \dots + k)$ is [JEE(Main)- 2020]

SS0094

26. The number of terms common to the two A.P.'s 3, 7, 11,, 407 and 2, 9, 16,, 709 is _____. [JEE(Main)- 2020]

SS0095

27. The product $2^{\frac{1}{4}} \cdot 4^{\frac{1}{16}} \cdot 8^{\frac{1}{48}} \cdot 16^{\frac{1}{128}} \cdot \dots$ to ∞ is equal to : [JEE(Main)- 2020]

(1) $2^{\frac{1}{2}}$ (2) $2^{\frac{1}{4}}$ (3) 2 (4) 1

SS0096

28. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of

$47 \sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ is equal to _____. [JEE(Main)- 2021]

SS0156

EXERCISE (JA)

1. If the sum of first n terms of an A.P. is cn^2 , then the sum of squares of these n terms is

(A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$ (C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$

[JEE 2009, 3 (-1)]

SS0097

2. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$.

If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to [JEE 2010, 3+3]

SS0098

3. The minimum value of the sum of real numbers a^{-5} , a^{-4} , $3a^{-3}$, 1 , a^8 and a^{10} with $a > 0$ is

[JEE 2011, 4]

SS0099

4. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i$, $1 \leq p \leq 100$. For any

integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n , then a_2 is [JEE 2011, 4]

SS0100

5. Let $a_1, a_2, a_3, \dots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is [JEE 2012, 3 (-1)]

(A) 22 (B) 23 (C) 24 (D) 25

SS0101

6. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) [JEE-Advanced 2013, 4, (-1)]

(A) 1056 (B) 1088 (C) 1120 (D) 1332

SS0102

7. A pack contains n cards numbered from 1 to n . Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller to the numbers on the removed cards is k , then $k - 20 =$ [JEE-Advanced 2013, 4, (-1)]

SS0103

8. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is [JEE(Advanced)-2014, 3]

SS0104

9. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is $6 : 11$ and the seventh term lies in between 130 and 140, then the common difference of this A.P. is **[JEE 2015, 4M, -0M]**
SS0105
10. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + \dots + b_{51}$ and $s = a_1 + a_2 + \dots + a_{51}$ then **[JEE(Advanced)-2016, 3(-1)]**
(A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$
(C) $s < t$ and $a_{101} > b_{101}$ (D) $s < t$ and $a_{101} < b_{101}$
SS0106
11. The sides of the right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side ? **[JEE(Advanced)-2017, 3]**
SS0107
12. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, $\dots$, and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, $\dots$. Then, the number of elements in the set $X \cup Y$ is _____. **[JEE(Advanced)-2018, 3]**
SS0108
13. Let m be the minimum possible value of $\log_3(3^{y_1} + 3^{y_2} + 3^{y_3})$, where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2(m^3) + \log_3(M^2)$ is _____. **[JEE(Advanced)-2020]**
SS0124
14. Let $a_1, a_2, a_3, \dots$ be a sequence of positive integers in arithmetic progression with common difference 2. Also, let $b_1, b_2, b_3, \dots$ be a sequence of positive integers in geometric progression with common ratio 2. If $a_1 = b_1 = c$, then the number of all possible values of c , for which the equality $2(a_1 + a_2 + \dots + a_n) = b_1 + b_2 + \dots + b_n$ holds for some positive integer n , is _____. **[JEE(Advanced)-2020]**
SS0125
15. Let $\ell_1, \ell_2, \dots, \ell_{100}$ be consecutive terms of an arithmetic progression with common difference d_1 , and let $w_1, w_2, \dots, w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2 , where $d_1 d_2 = 10$. For each $i = 1, 2, \dots, 100$, let R_i be a rectangle with length ℓ_i , width w_i and area A_i . If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is _____. **[JEE(Advanced)-2022]**

ANSWER KEY**Do yourself - 1 :**

1. (a) $\frac{2}{1}, \frac{4}{2}, \frac{8}{3}, \frac{16}{4}, \dots$, (b) $\frac{2}{3}, \frac{4}{9}, \frac{2}{27}, \frac{4}{81}, \dots$;
 3. $-(p+q)$ 4. 153 5. 0 6. -185
 7. $820a - 1680b$

Do yourself - 2 :

1. 900 2. 14 3. 3 4. 5 5. 612
 6. 4, 9, 14 7. 1, 4, 7 8. 495 9. n^2 10. 3, 5, 7, 9
 11. $6n - 1$ 13. $(14n - 6)/(8n + 23)$ 14. $2Q$

Do yourself - 3 :

1. 931 2. 4^5 4. 4, 12, 36 5. (C) 6. $191\frac{1}{4}$
 7. $\frac{1}{4}(5^p - 1)$ 8. $364(\sqrt{3} + 1)$ 9. $\frac{64}{65}$ 10. $\frac{27}{58}$
 11. $\frac{3(3 + \sqrt{3})}{2}$ 12. 2 13. 16, 24, 36, ... 14. 8, 12, 18
 15. 2, 6, 18 16. $6, -3, 1\frac{1}{2}, \dots$ 17. $4, 1, \frac{1}{4}, \dots$
 18. $\frac{x(x^n - 1)}{x - 1} + \frac{(n(n+1))a}{2}$ 19. $n = 38$

Do yourself - 4 :

1. 2 2. 15 4. (a) 5 (b) 6 (c) -18
 8. 1 9. (iii) $b = 4, c = 6, d = 9$ or $b = -2, c = -6, d = -18$

Do yourself - 5 :

1. $\frac{1}{2}$ 3. $-\frac{1}{4}, -\frac{3}{4}, \dots, -9\frac{1}{4}$ 4. $1, -1\frac{1}{2}, \dots, -39$.
 5. $\frac{3}{2}, 1, \frac{2}{3}$ 6. $\frac{16}{3}, 8, \dots, 27$ 7. $6\frac{1}{9}, 7\frac{6}{7}$
 8. $\frac{2}{5}, \frac{2}{7}, \frac{2}{9}, \frac{2}{11}$ 10. (a) 6, 3 (b) 120, 30 12. 28 13. 102

Do yourself - 6 :

1. $4 + \frac{8}{9} \left(1 - \frac{1}{4^{n-1}} \right) - \left(\frac{2n+1}{3 \times 4^{n-1}} \right)$
2. $\frac{1-a^n}{(1-a)^2} - \frac{na^n}{1-a}$
3. $\frac{8}{3}$
4. $\frac{9}{8}$
5. $n \cdot 2^{n+2} - 2^{n+1} + 2$
6. 23

Do yourself - 7 :

1. $\frac{n(n+3)}{4}$
2. (a) $\frac{n(n+1)(2n+3)}{2}$ (b) $n^2(n+1)$
- (c) $\frac{1}{4}n(n+1)(n^2+n+3)$ (d) $\frac{1}{6}n(n+1)(2n+7)$
- (e) $\frac{1}{2}n(n+1)(n^2+3n+1)$ (f) $\frac{1}{2}(3^{n+1}+1)-2^{n+1}$
- (g) $4^{n+1} - 4 - n(n+1)(n^2 - n - 1)$
3. (a) $\frac{n}{4}(n+1)(n+6)(n+7)$ (b) $\frac{n}{4}(n+1)(n+8)(n+9)$
4. (a) $3n^2 + n; n(n+1)^2$
- (b) $3 \cdot 2^n + n + 2; 6(2^n - 1) + \frac{n(n+5)}{2}$
5. (i) $2^{n+1} - 3; 2^{n+2} - 4 - 3n$ (ii) $n^2 + 4n + 1; (1/6)n(n+1)(2n+13) + n$
6. (i) $s_n = (1/24) - [1/\{6(3n+1)(3n+4)\}] ; s_\infty = 1/24$
- (ii) $(1/5)n(n+1)(n+2)(n+3)(n+4)$
- (iii) $n/(2n+1)$
7. C
8. B
9. C

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	D	D	A	C	C	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	A	B	A	C	B	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	A	B	C	B	D	B	D	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	A,B,C,D	A,B,C,D	A,B	A,B,C,D	C,D	A,B,C,D	A,C	A,B,C	A,B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	A,C,D	A,D	A,B,D	A,C,D	B,C	B,C,D	A,C	A,B	A,C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8		
Ans.	D	B	B,C,D	B,C	6.00	62.00	D	B		
Que.	Q.9	A	B	C	D	Q.10	A	B	C	D
Ans.		p	s	r	q		s	p	s	p

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	12	2499	27	2	22	5	$\frac{1}{201}$	7	0.50	49

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	3	1	3	2	1	4	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	3	1	1	3	2	3	1
Que.	21	22	23	24	25	26	27	28		
Ans.	1	3	3	504	1540.00	14	1	7		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	0	8	9 or 3	D	A,D	5	4	9	B
Que.	11	12	13	14	15					
Ans.	6	3748	8.00	1.00	18900.00					

PRINCIPLE OF MATHEMATICAL INDUCTION

1. THEOREM-I

If $P(n)$ is a statement depending upon n , then to prove $P(n)$ by induction, we proceed as follows :

- (i) Verify the validity of $P(n)$ for $n = 1$
- (ii) Assume that $P(n)$ is true for any positive integer m and then using it establish the validity of $P(n)$ for $n = m + 1$.

Then $P(n)$ is true for each $n \in \mathbb{N}$

2. THEOREM-II

If $P(n)$ is a statement depending upon n but beginning with any positive integer k , then to prove $P(n)$ by Induction, we proceed as follows :

- (i) Verify the validity of $P(n)$ for $n = k$.
- (ii) Assume that the $P(n)$ is true for $n = m \geq k$.

Then using it establish the validity of $P(n)$ for $n = m + 1$.

Then $P(n)$ is true for each $n \geq k$

3. SOME USEFUL RESULT BASED ON PRINCIPLE OF MATHEMATICAL INDUCTION :

For any natural number n

- (i) $1 + 2 + 3 + \dots + n = \Sigma n = \frac{n(n+1)}{2}$
- (ii) $1^2 + 2^2 + 3^2 + \dots + n^2 = \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}$
- (iii) $1^3 + 2^3 + 3^3 + \dots + n^3 = \Sigma n^3 = (\Sigma n)^2 = \left\{ \frac{n(n+1)}{2} \right\}^2$
- (iv) $2 + 4 + 6 + \dots + 2n = \Sigma 2n = n(n+1)$
- (v) $1 + 3 + 5 + \dots + (2n-1) = \Sigma (2n-1) = n^2$
- (vi) $x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \dots + xy^{n-2} + y^{n-1})$
- (vii) $x^n + y^n = (x+y)(x^{n-1} - x^{n-2}y + x^{n-3}y^2 + \dots - xy^{n-2} + y^{n-1})$

when n is odd positive integer

4. IMPORTANT TIPS :

- (i) Product of r successive integers is divisible by $r!$
- (ii) For $x \neq y$, $x^n - y^n$ is divisible by
 - (a) $x + y$, if n is even
 - (b) $x - y$, if n is even or odd
- (iii) $x^n + y^n$ is divisible by
 - (a) $x + y$, If n is odd
- (iv) For solving objective question related to natural numbers we find out the correct alternative by negative examination of this principle. If the given statement is $P(n)$, then by putting $n = 1, 2, 3, \dots$ in $P(n)$ we decide the correct answer. We also use the above formulae established by this principle to find the sum of n terms of a given series. For this we first express T_n as a polynomial in n and then for finding S_n , we put Σ before each term of this polynomial and then use above results of $\Sigma n, \Sigma n^2, \Sigma n^3$ etc.

SOLVED EXAMPLES

Ex.1 Use the principle of mathematical induction to show that $5^{2n+1} + 3^{n+2} \cdot 2^{n-1}$ divisible by 19 for all natural numbers n .

Sol. Let $P(n) = 5^{2n+1} + 3^{n+2} \cdot 2^{n-1}$

Step I : For $n = 1$

$$P(1) = 5^{2+1} + 3^{1+2} \cdot 2^{1-1}$$

$$= 125 + 27$$

$$= 152, \text{ which is divisible by 19.}$$

Therefore, the result is true for $n = 1$.

Step II : Assume that the result is true for $n = k$, i.e. $P(k) = 5^{2k+1} + 3^{k+2} \cdot 2^{k-1}$ is divisible by 19.

$\Rightarrow P(k) = 19r$, where r is an integer.

Step III : For $n = k + 1$

$$P(k+1) = 5^{2(k+1)+1} + 3^{k+1+2} \cdot 2^{k+1-1}$$

$$= 5^{2k+3} + 3^{k+3} \cdot 2^k$$

$$= 25 \cdot 5^{2k+1} + 3 \cdot 3^{k+2} \cdot 2 \cdot 2^{k-1}$$

$$= 25 \cdot 5^{2k+1} + 6 \cdot 3^{k+2} \cdot 2^{k-1}$$

$$\text{Now } 25 \cdot 5^{2k+1} + 6 \cdot 3^{k+2} \cdot 2^{k-1} = 25 \cdot (5^{2k+1} + 3^{k+2} \cdot 2^{k-1}) - 19 \cdot 3^{k+2} \cdot 2^{k-1}$$

$$\text{i.e. } P(k+1) = 25 P(k) - 19 \cdot 3^{k+2} \cdot 2^{k-1}$$

But we know that $P(k)$ is divisible by 19. Also $19 \cdot 3^{k+2} \cdot 2^{k-1}$ is clearly divisible by 19.

Hence $P(k+1)$ is divisible by 19. This shows that the result is true for $n = k + 1$. Hence by the principle of mathematical induction, the result is true for all $n \in \mathbb{N}$.

Ex.2. Use the principle of mathematical induction to show that $1 \cdot 3 + 2 \cdot 4 + \dots + n \cdot (n+2) = \frac{1}{6} n(n+1)(2n+7)$.

Sol. Let $P(n) : 1 \cdot 3 + 2 \cdot 4 + \dots + n \cdot (n+2) = \frac{1}{6} n(n+1)(2n+7)$

Step I : For $n = 1$

$$\text{LHS of } P(1) = 1 \cdot 3 = 3 = \frac{1}{6} \cdot 1 \cdot 2 \cdot 9 = \frac{1}{6} \cdot 1 \cdot (1+1) \cdot (2 \cdot 1 + 7) = \text{RHS of } P(1)$$

So $P(1)$ is true

Step II : Now assume $P(k)$ is true, for some natural number k , i.e.

$$1 \cdot 3 + 2 \cdot 4 + \dots + k \cdot (k+2) = \frac{1}{6} k(k+1)(2k+7).$$

Now deduce $P(k+1)$.

$$\text{LHS of } P(k+1) = 1 \cdot 3 + 2 \cdot 4 + \dots + k \cdot (k+2) + (k+1) \cdot (k+1+2)$$

$$= (\text{LHS of } P(k)) + (k+1)(k+3)$$

$$= (\text{RHS of } P(k)) + (k+1)(k+3), \text{ (by inductive assumption)}$$

$$= \frac{1}{6} k(k+1)(2k+7) + (k+1)(k+3)$$

$$= \frac{1}{6} (k+1)(k(2k+7) + 6(k+3))$$

$$= \frac{1}{6} (k+1)(2k^2 + 13k + 18)$$

$$= \frac{1}{6}(k+1)(k+2)(2k+9)$$

$$= \frac{1}{6}(k+1)(k+1+1)(2(k+1)+7)$$

$$= \text{RHS of } P(k+1).$$

So $P(k+1)$ is true, if $P(k)$ is true.

Hence by induction $P(n)$ is true for all natural numbers n .

Ex.3 Use the principle of mathematical induction to show that for any positive integer number n , $n^3 + 2n$, is divisible by 3.

Sol. Statement $P(n)$ is defined by $n^3 + 2n$ is divisible 3

Step 1 : We first show that $P(1)$ is true. Let $n = 1$ and calculate $n^3 + 2n$

$$1^3 + 2(1) = 3$$

Hence $P(1)$ is true.

Step 2 : We now assume that $P(k)$ is true $k^3 + 2k$ is divisible by 3. is equivalent to

$$k^3 + 2k = 3M, \text{ where } M \text{ is a positive integer.}$$

We now consider the algebraic expression $(k+1)^3 + 2(k+1)$; expand it and group like terms.

$$(k+1)^3 + 2(k+1) = k^3 + 3k^2 + 5k + 3$$

$$= [k^3 + 2k] + [3k^2 + 3k + 3]$$

$$= 3M + 3[k^2 + k + 1] = 3[M + k^2 + k + 1]$$

Hence $(k+1)^3 + 2(k+1)$ is also divisible by 3 and therefore statement $P(k+1)$ is true.

Ex.4 Prove that $3^n > n^2$ for $n = 1, n = 2$ and use the mathematical induction to prove that $3^n > n^2$ for n , a positive integer greater than 2.

Sol Statement $P(n)$ is defined by

$$3^n > n^2$$

Step 1 : We first show that $P(1)$ is true. Let $n = 1$ and calculate 3^1 and 1^2 and compare them

$$3^1 = 3$$

$$1^2 = 1$$

3 is greater than 1 and hence $P(1)$ is true.

Let us also show that $P(2)$ is true.

$$3^2 = 9$$

$$2^2 = 4$$

Hence $P(2)$ is also true.

Step 2 : We now assume that $P(k)$ is true

$$3^k > k^2$$

Multiply both sides of the above inequality by 3.

$$3 * 3^k > 3 * k^2$$

The left side is equal to 3^{k+1} . For $k > 2$, we can write

$$k^2 > 2k \text{ and } k^2 > 1$$

We now combine the above inequalities by adding the left hand sides and the right hand sides of the two inequalities.

$$2k^2 > 2k + 1$$

We now add k^2 to both sides of the above inequality to obtain the inequality

$$3k^2 > k^2 + 2k + 1$$

Factor the right side we can write

$$3 * k^2 > (k+1)^2$$

If $3 \cdot 3^k > 3 \cdot k^2$ and $3 \cdot k^2 > (k+1)^2$ then

$$3 \cdot 3^k > (k+1)^2$$

Rewrite the left side as 3^{k+1}

$$3^{k+1} > (k+1)^2$$

* Which proves that $P(k+1)$ is true.

Ex.5. Use mathematical induction to prove De Moivre's theorem

$[R(\cos t + i \sin t)]^n = R^n(\cos nt + i \sin nt)$ for n a positive integer.

Sol. Step 1 : For $n = 1$

$$[R(\cos t + i \sin t)]^1 = R^1(\cos 1 \cdot t + i \sin 1 \cdot t)$$

It can be easily be seen that the two sides are equal.

Step 2 : We now assume that the theorem is true for $n = k$, hence $[R(\cos t + i \sin t)]^k = R^k(\cos kt + i \sin kt)$

Multiply both sides of the above equation by $R(\cos t + i \sin t)$

$$[R(\cos t + i \sin t)]^k R(\cos t + i \sin t) = R^k(\cos kt + i \sin kt) R(\cos t + i \sin t)$$

Rewrite the above as follows

$$[R(\cos t + i \sin t)]^{k+1} = R^{k+1}[(\cos kt \cos t - \sin kt \sin t) + i(\sin kt \cos t + \cos kt \sin t)]$$

Trigonometric identities can be used to write the trigonometric expressions $(\cos kt \cos t - \sin kt \sin t)$

and $(\sin kt \cos t + \cos kt \sin t)$ as follows

$$(\cos kt \cos t - \sin kt \sin t) = \cos(kt + t) = \cos(k+1)t$$

$$(\sin kt \cos t + \cos kt \sin t) = \sin(kt + t) = \sin(k+1)t$$

Substitute the above into the last equation to obtain.

$$[R(\cos t + i \sin t)]^{k+1} = R^{k+1}[\cos(k+1)t + i \sin(k+1)t]$$

It has been established that the theorem is true for $n = 1$ and that if it assumed true for $n = k$ it is true for $n = k+1$.

CHECK YOUR GRASP PRINCIPLE OF MATHEMATICAL INDUCTION EXERCISE-I

1. Let $P(n) : n^2 + n ; n \in \mathbb{N}$ is an odd integer. It is seen that truth of $P(n) \Rightarrow$ the truth of $P(n+1)$. Therefore, $P(n)$ is true for all-

- (1) $n > 1$ (2) n
(3) $n > 2$ (4) None of these

MI0001

2. If $n \in \mathbb{N}$, then $x^{2n-1} + y^{2n-1}$ is divisible by-

- (1) $x + y$ (2) $x - y$ (3) $x^2 + y^2$ (4) $x^2 + xy$

MI0002

3. If $n \in \mathbb{N}$, then $11^{n+2} + 12^{2n+1}$ is divisible by-

- (1) 113 (2) 123
(3) 133 (4) None of these

MI0003

4. If $n \in \mathbb{N}$, then $3^{4n+2} + 5^{2n+1}$ is a multiple of-

- (1) 14 (2) 16 (3) 18 (4) 20

MI0004

5. For every positive integer

$$n, \frac{n^7}{7} + \frac{n^5}{5} + \frac{2n^3}{3} - \frac{n}{105} \text{ is-}$$

- (1) an integer
(2) a rational number which is not an integer
(3) a negative real number
(4) an odd integer

MI0005

6. Sum of the infinite series $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$ equals-

- (1) $1/3$ (2) 3 (3) $1/4$ (4) 2

MI0006

7. If $a_k = \frac{1}{k(k+1)}$; then $\left(\sum_{k=1}^n a_k \right)^2$ is equal to-

- (1) $\frac{n^2}{(n+1)^2}$ (2) $\frac{n^4}{(n+1)^4}$ (3) $\frac{n^2}{n^2+1}$ (4) $\frac{n}{n+1}$

MI0007

8. The sum of n terms of the series

$$\frac{1}{1^3} + \frac{2}{1^3+2^3} + \frac{3}{1^3+2^3+3^3} + \dots \text{ is-}$$

- (1) $\frac{1}{n(n+1)}$ (2) $\frac{n}{n+1}$ (3) $\frac{n+1}{n}$ (4) $\frac{n+1}{n+2}$

MI0008

9. For all $n \in \mathbb{N}$, $7^{2n} - 48n - 1$ is divisible by-

- (1) 25 (2) 26 (3) 1234 (4) 2304

MI0009

10. For all positive integral values of n , $3^{2n} - 2n + 1$ is divisible by-

- (1) 2 (2) 4 (3) 8 (4) 12

MI0010

11. The smallest positive integer n for which the statement $3^{n+1} < 4^n$ holds is-

- (1) 1 (2) 2 (3) 3 (4) 4

MI0011

12. For positive integer n , $10^{n-2} > 81n$ when-

- (1) $n < 5$ (2) $n > 5$ (3) $n \geq 5$ (4) $n > 6$

MI0012

13. If P is a prime number then $n^p - n ; n \in \mathbb{N}$ is divisible by p when n is a

- (1) All natural number greater than or equal to 1
(2) odd number only
(3) even number only
(4) None of these

MI0013

14. A student was asked to prove a statement by induction. He proved

- (i) $P(5)$ is true and
(ii) Truth of $P(n) \Rightarrow$ truth of $p(n+1), n \in \mathbb{N}$

On the basis of this, he could conclude that $P(n)$ is true for

- (1) no $n \in \mathbb{N}$ (2) all $n \in \mathbb{N}$
(3) all $n \geq 5$ (4) None of these

MI0014

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	4	1	3	1	1	1	1	2	4	1	4	3	1	3	

BRAIN TEASERS

PRINCIPLE OF MATHEMATICAL INDUCTION

EXERCISE-II

1. If $x > -1$, then the statement $P(n) : (1+x)^n > 1+nx$ is true for-
 (1) all $n \in \mathbb{N}$ (2) all $n > 1; n \in \mathbb{N}$
 (3) all $n > 1; n \in \mathbb{N}$ and $x \neq 0$ (4) None of these
MI0015
2. For every positive integral value of n , $3^n > n^3$ when-
 (1) $n > 2$ (2) $n \geq 3$ (3) $n \geq 4$ (4) $n < 4$
MI0016
3. $P(n) : 3^{2n+2} - 8n - 9$ is divisible by 64, is true for-
 (1) all $n \in \mathbb{N} \cup \{0\}$ (2) only for $n \geq 2, n \in \mathbb{N}$
 (3) $n \in \mathbb{N}, n > 2$ (4) None of these
MI0017
4. If only for m, n are any two odd positive integer with $n < m$, then the largest positive integers which divides all the numbers of the type $m^2 - n^2$ is-
 (1) 4 (2) 6 (3) 8 (4) 9
MI0018
5. For all $n \in \mathbb{N}$, $\cos \theta \cos 2\theta \cos 4\theta \dots \cos 2^{n-1}\theta$ equals to
 (1) $\frac{\sin 2^n \theta}{2^n \sin \theta}$ (2) $\frac{\sin 2^n \theta}{\sin \theta}$
 (3) $\frac{\cos 2^n \theta}{2^n \cos 2\theta}$ (4) $\frac{\cos 2^n \theta}{2^n \sin \theta}$ **MI0019**
6. $x(x^{n-1} - na^{n-1}) + a^n(n-1)$ is divisible by $(x-a)^2$ for-
 (1) $n > 1$ (2) $n > 2$
 (3) all $n \in \mathbb{N}$ (4) None of these
MI0020
7. For any odd integer $n \geq 1$
 $n^3 - (n-1)^3 + \dots + (-1)^{n-1} \cdot 1^3$ is equal to-
 (1) $\frac{1}{4}(n+1)^2(2n-1)$ (2) $\frac{1}{4}(n-1)^2(2n-1)$
 (3) $\frac{1}{2}(n-1)^2(2n-1)$ (4) $\frac{1}{2}(n+1)^2(2n-1)$
MI0021
8. If p and q are respectively. The sum and the sum of squares of n successive integers beginning with a , then $nq - p^2$ is-
 (1) independent of a (2) independent of n
 (3) dependent on a (4) None of these
MI0022
9. The sum of first n terms of the given series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$ is $\frac{n(n+1)^2}{2}$, when n is even. When n is odd, then sum will be-
[AIEEE-2004]
 (1) $\frac{n(n+1)^2}{2}$ (2) $\frac{1}{2}n^2(n+1)$
 (3) $n(n+1)^2$ (4) None
MI0023
10. Let $S(k) = 1 + 3 + 5 + \dots + (2k-1) = 3 + k^2$, then which of the following is true? **[AIEEE-2004]**
 (1) $S(1)$ is true
 (2) $S(k) \Rightarrow S(k+1)$
 (3) $S(k) \not\Rightarrow S(k+1)$
 (4) Principle of mathematical Induction can be used to prove that formula **MI0024**
11. The sum of n terms of the series $1 + (1+a) + (1+a+a^2) + (1+a+a^2+a^3) + \dots$, is-
 (1) $\frac{n}{1-a} - \frac{a(1-a^n)}{(1-a)^2}$ (2) $\frac{n}{1-a} + \frac{a(1-a^n)}{(1-a)^2}$
 (3) $\frac{n}{1-a} + \frac{a(1+a^n)}{(1-a)^2}$ (4) $-\frac{n}{1-a} + \frac{a(1-a^n)}{(1-a)^2}$
MI0025
12. **Statement -1** : For every natural number $n \geq 2$
 $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$
Statement -2 : For every natural number $n \geq 2$, $\sqrt{n(n+1)} < n+1$. **[AIEEE-2008]**
 (1) Statement -1 is false, Statement -2 is true
 (2) Statement -1 is true, Statement -2 is false
 (3) Statement -1 is true, Statement -2 is true; Statement -2 is a correct explanation for Statement -1
 (4) Statement -1 is true, Statement -2 is true; Statement -2 is not a correct explanation for Statement -1 **MI0026**
13. **Statement - 1**: For each natural number n , $(n+1)^7 - n^7 - 1$ is divisible by 7.
Statement - 2: For each natural number n , $n^7 - n$ is divisible by 7. **[AIEEE-2011]**
 (1) Statement-1 is false, statement-2 is true.
 (2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement-1.
 (3) Statement-1 is true, statement-2 is true; Statement-2 is not a correct explanation for statement-1.
 (4) Statement-1 is true, statement-2 is false. **MI0027**
14. Consider the statement : " $P(n): n^2 - n + 41$ is prime." Then which one of the following is true?
[JEE(Main) Jan-19]
 (1) $P(5)$ is false but $P(3)$ is true
 (2) Both $P(3)$ and $P(5)$ are false
 (3) $P(3)$ is false but $P(5)$ is true
 (4) Both $P(3)$ and $P(5)$ are true **MI0028**

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	3	3	1	3	1	3	1	1	2	2	1	3	2	4	

Important Notes

QUADRATIC EQUATION

1. INTRODUCTION :

The algebraic expression of the form $ax^2 + bx + c$, $a \neq 0$ is called a quadratic expression, because the highest order term in it is of second degree. Quadratic equation means, $ax^2 + bx + c = 0$. In general whenever one says zeroes of the expression $ax^2 + bx + c$, it implies roots of the equation $ax^2 + bx + c = 0$, unless specified otherwise.

A quadratic equation has exactly two roots which may be real (equal or unequal) or imaginary.

2. SOLUTION OF QUADRATIC EQUATION & RELATION BETWEEN ROOTS & CO-EFFICIENTS :

(a) The general form of quadratic equation is $ax^2 + bx + c = 0$, $a \neq 0$.

The roots can be found in following manner :

$$a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right) = 0 \Rightarrow \left(x + \frac{b}{2a} \right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0$$

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

This expression can be directly used to find the two roots of a quadratic equation.

(b) The expression $b^2 - 4ac \equiv D$ is called the discriminant of the quadratic equation.

(c) If α & β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then :

$$(i) \alpha + \beta = -b/a \quad (ii) \alpha\beta = c/a \quad (iii) |\alpha - \beta| = \frac{\sqrt{D}}{|a|}$$

(d) A quadratic equation whose roots are α & β is $(x - \alpha)(x - \beta) = 0$ i.e.

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \text{ i.e. } x^2 - (\text{sum of roots})x + \text{product of roots} = 0.$$

Illustration 1 : If α, β are the roots of a quadratic equation $x^2 - 3x + 5 = 0$, then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is -

$$(A) x^2 + 4x + 1 = 0 \quad (B) x^2 - 4x + 4 = 0 \quad (C) x^2 - 4x - 1 = 0 \quad (D) x^2 + 2x + 3 = 0$$

Solution : Since α, β are the roots of equation $x^2 - 3x + 5 = 0$

$$\text{So } \alpha^2 - 3\alpha + 5 = 0$$

$$\beta^2 - 3\beta + 5 = 0$$

$$\therefore \alpha^2 - 3\alpha = -5$$

$$\beta^2 - 3\beta = -5$$

$$\text{Putting in } (\alpha^2 - 3\alpha + 7) \text{ \& } (\beta^2 - 3\beta + 7) \quad \dots\dots\dots(i)$$

$$-5 + 7, -5 + 7$$

$$\therefore 2 \text{ and } 2 \text{ are the roots.}$$

$$\therefore \text{ The required equation is } x^2 - 4x + 4 = 0.$$

Ans. (B)

Illustration 2 : If α and β are the roots of $ax^2 + bx + c = 0$, find the value of $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$.

Solution : We know that $\alpha + \beta = -\frac{b}{a}$ & $\alpha\beta = \frac{c}{a}$

$$\begin{aligned}(a\alpha + b)^{-2} + (a\beta + b)^{-2} &= \frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} \\&= \frac{a^2\beta^2 + b^2 + 2ab\beta + a^2\alpha^2 + b^2 + 2ab\alpha}{(a^2\alpha\beta + ba\beta + ba\alpha + b^2)^2} = \frac{a^2(\alpha^2 + \beta^2) + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} \\&(\alpha^2 + \beta^2 \text{ can always be written as } (\alpha + \beta)^2 - 2\alpha\beta) \\&= \frac{a^2[(\alpha + \beta)^2 - 2\alpha\beta] + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} = \frac{a^2\left[\frac{b^2 - 2ac}{a^2}\right] + 2ab\left(-\frac{b}{a}\right) + 2b^2}{\left(a^2\frac{c}{a} + ab\left(-\frac{b}{a}\right) + b^2\right)^2} = \frac{b^2 - 2ac}{a^2c^2}\end{aligned}$$

Alternatively :

If α & β are roots of $ax^2 + bx + c = 0$

then, $a\alpha^2 + b\alpha + c = 0$

$$\Rightarrow a\alpha + b = -\frac{c}{\alpha}$$

$$\text{same as } a\beta + b = -\frac{c}{\beta}$$

$$\begin{aligned}\therefore (a\alpha + b)^{-2} &= (\alpha\beta + b)^{-2} = \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2} \\&= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{c^2} \\&= \frac{(-b/a)^2 - 2(c/a)}{c^2} \\&= \frac{b^2 - 2ac}{a^2c^2}\end{aligned}$$

Do yourself - 1 :

1. Find the roots of following equations :

$$(a) \quad x^2 + 3x + 2 = 0 \quad (b) \quad x^2 - 8x + 16 = 0 \quad (c) \quad x^2 - 2x - 1 = 0$$

2. Find the roots of the equation $a(x^2 + 1) - (a^2 + 1)x = 0$, where $a \neq 0$.

3. Solve : $\frac{6-x}{x^2-4} = 2 + \frac{x}{x+2}$

4. If the roots of $4x^2 + 5k = (5k + 1)x$ differ by unity, then find the values of k .

5. Form a quadratic equation whose roots are the numbers $\frac{1}{10-\sqrt{72}}$ and $\frac{1}{10+6\sqrt{2}}$.
6. For what values of a is the sum of the roots of the equation $x^2 + (2 - a - a^2)x - a^2 = 0$ equal to zero?
7. For what values of a is the ratio of the roots of the equation $ax^2 - (a + 3)x + 3 = 0$ equal to 1.5?
8. The roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$. Find p .
9. Find k in the equation $5x^2 - kx + 1 = 0$ such that the difference between the roots of the equation is unity.
10. Find p in the equation $x^2 - 4x + p = 0$ if it is known that the sum of the squares of its roots is equal to 16.
11. For what values of a is the difference between the roots of the equation $2x^2 - (a + 1)x + (a - 1) = 0$ equal to their product?
12. Express $x_1^3 + x_2^3$ in terms of the coefficients of the equation $x^2 + px + q = 0$, where x_1 and x_2 are the roots of the equation.
13. Assume that x_1 and x_2 are roots of the equation $3x^2 - ax + 2a - 1 = 0$. Calculate $x_1^3 + x_2^3$.
14. Without solving the equation $3x^2 - 5x - 2 = 0$, find the sum of the cubes of its roots.

3. NATURE OF ROOTS :

- (a) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{R}$ & $a \neq 0$ then ;

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

- (i) $D > 0 \Leftrightarrow$ roots are real & distinct (unequal).
- (ii) $D = 0 \Leftrightarrow$ roots are real & coincident (equal)
- (iii) $D < 0 \Leftrightarrow$ roots are non real and complex conjugate of each other.
- (iv) If $p + iq$ is one root of a quadratic equation, then the other root must be the conjugate $p - iq$ & vice versa. ($p, q \in \mathbb{R}$ & $i = \sqrt{-1}$).

- (b) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{Q}$ & $a \neq 0$ then ;

- (i) If D is a perfect square of some rational number, then roots are rational.
- (ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is an irrational) then other root will be $p - \sqrt{q}$.

Illustration 6 : Find all the integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots.

Solution : Here the equation is $x^2 - (a + 10)x + 10a + 1 = 0$. Since integral roots will always be rational it means D should be a perfect square.

$$\text{From (i) } D = a^2 - 20a + 96.$$

$$\Rightarrow D = (a - 10)^2 - 4 \Rightarrow 4 = (a - 10)^2 - D$$

If D is a perfect square it means we want difference of two perfect square as 4 which is possible only when $(a - 10)^2 = 4$ and $D = 0$.

$$\Rightarrow (a - 10) = \pm 2 \Rightarrow a = 12, 8$$

Ans.

Illustration 7 : If D_1 and D_2 are the discriminant of two Quadratic Equations with real coefficients respectively then comment upon the nature of the roots of the Quadratic Equations under the following conditions

$$(i) D_1 + D_2 \geq 0 \quad (ii) D_1 + D_2 < 0 \quad (iii) D_1 D_2 < 0 \quad (iv) D_1 D_2 > 0 \quad (v) D_1 D_2 = 0$$

Solution : (i) If D_1 and D_2 are discriminant of two quadratic equations and $D_1 + D_2 \geq 0$, then at least one of D_1 and $D_2 \geq 0$

$\Rightarrow$ At least one of the equation has real roots.

$$(ii) D_1 + D_2 < 0 \Rightarrow \text{At least one of } D_1 \text{ and } D_2 < 0$$

$\Rightarrow$ At least one of the equation has imaginary roots.

$$(iii) \text{ If } D_1 D_2 < 0. \Rightarrow (D_1 > 0 \text{ and } D_2 < 0) \text{ or } (D_1 < 0 \text{ and } D_2 > 0).$$

Then one of the equations has real root and other equation has imaginary roots.

$$(iv) D_1 D_2 > 0$$

Case I : If $D_1 > 0$ and $D_2 > 0$, then both the equations has real roots

Case II : If $D_1 < 0$ and $D_2 < 0$, then both the equations has non real roots

$$(v) \text{ If } D_1 D_2 = 0, \text{ then } D_1 = 0 \text{ or } D_2 = 0$$

$\Rightarrow$ at least of equation has equal roots.

Illustration 8 : Let $a > 0$, $b > 0$ and $c > 0$. Then both the roots of the equation $ax^2 + bx + c = 0$ can

(a) be real and negative

(b) have negative real parts

(c) be rational numbers

(d) none of these

Solution : (abc)

$$\text{We have } D = b^2 - 4ac.$$

$$\text{If } D \geq 0, \text{ then the roots of the equation are given by } x = \frac{-b \pm \sqrt{D}}{2a}$$

As $D = b^2 - 4ac < b^2$ ($\because a > 0, c > 0$), it follows that the roots of the quadratic equation are negative.

$$\text{In case } D < 0, \text{ then the roots of the equation are given by } x = \frac{-b \pm i\sqrt{-D}}{2a}$$

which have negative real parts. ($\because$ Both a and b are positive)

Roots will be rational numbers when $\frac{b}{a}, \frac{c}{a} \in \mathbb{Q}$ and $\frac{D}{4a^2}$ is perfect square of some rational number.

4. SYMMETRIC EXPRESSIONS OF α AND β .

Let a and b be the roots of the equation $ax^2 + bx + c = 0$ such that

$f(\alpha, \beta) = f(\beta, \alpha)$ then $f(\alpha, \beta)$ denotes symmetric expression of the roots.

e.g. $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$

It is to be noted that every symmetric expression in α, β can be expressed in terms of two symmetric expression $\alpha + \beta$ and $\alpha\beta$

Do yourself - 2 :

- For the following equations, find the nature of the roots (real & distinct, real & coincident or imaginary).
 - $x^2 - 6x + 10 = 0$
 - $x^2 - (7 + \sqrt{3})x + 6(1 + \sqrt{3}) = 0$
 - $4x^2 + 28x + 49 = 0$
- Prove that roots of $(x - a)(x - b) = h^2$ are always real.
- For what values of a does the equation $9x^2 - 2x + a = 6 - ax$ possess equal roots ?
- Find the value of k for which the equation $(k - 1)x^2 + (k + 4)x + k + 7 = 0$ has equal roots.
- Find the values of a for which the roots of the equation $(2a - 5)x^2 - 2(a - 1)x + 3 = 0$ are equal.
- For what values of ' a ' does the quadratic equation $x^2 + (2a\sqrt{a^2 - 3})x + 4 = 0$ possess equal roots ?
- Find the least integral value of k for which the equation $x^2 - 2(k + 2)x + 12 + k^2 = 0$ has two different real roots.
- If ℓ, m are real and $\ell \neq m$, then show that the roots of $(\ell - m)x^2 - 5(\ell + m)x - 2(\ell - m) = 0$ are real and unequal.
- For what values of m does the equation $x^2 - x + m = 0$ possess no real roots ?
- For what values of c does the equation $(c - 2)x^2 + 2(c - 2)x + 2 = 0$ possess no real roots ?
- Find integral values of k for which the quadratic equation $(k - 12)x^2 + 2(k - 12)x + 2 = 0$ possess no real roots ?
- For what values of m does the equation $x^2 - x + m^2 = 0$ possess no real roots ?
- For what values of m does the equation $mx^2 - (m + 1)x + 2m - 1 = 0$ possess no real roots ?
- If $2 + \sqrt{3}$ is a root of the equation $x^2 + bx + c = 0$, where $b, c \in \mathbb{Q}$, find b, c .
- Find the value of $x^3 + x^2 - x + 22$ when $x = 1 + 2i$.
- Find the value of $x^3 - 3x^2 - 8x + 15$ when $x = 3 + i$.

17. Consider $f(x) = x^2 + bx + c$.
- (a) Find c if $x = 0$ is a root of $f(x) = 0$.
- (b) Find c if $\alpha, \frac{1}{\alpha}$ are roots of $f(x) = 0$.
- (c) Comment on sign of b & c , if $\alpha < 0 < \beta$ & $|\beta| > |\alpha|$, where α, β are roots of $f(x) = 0$.
18. Which of the following expressions in α, β will denote the symmetric functions in α, β . Give proper reasoning.
- (i) $f(\alpha, \beta) = \alpha^2 - \beta$ (ii) $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$
- (iii) $f(\alpha, \beta) = \alpha^3 + \beta^3$ (iv) $f(\alpha, \beta) = \frac{\alpha}{\beta} + \frac{\beta}{\alpha}$
19. If α, β are the roots of the equation $ax^2 + bx + c = 0$, find the value of
- (A) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ (B) $\alpha^4\beta^7 + \alpha^7\beta^4$ (C) $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$
20. Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.
21. For what value of a is the difference between the roots of the equation $(a-2)x^2 - (a-4)x - 2 = 0$ equal to 3?
22. Find the value of a for which one root of the equation $x^2 + (2a-1)x + a^2 + 2 = 0$ is twice as large as the other.
23. For what values of a is the ratio of the roots of the equation $x^2 + ax + a + 2 = 0$ equal to 2?
24. For what values of a do the roots x_1 and x_2 of the equation $x^2 - (3a+2)x + a^2 = 0$ satisfy the relation $x_1 = 9x_2$? Find the roots.
25. Find a such that one of the roots of the equation $x^2 - \frac{15}{4}x + a = 0$ is the square of the other.
26. Find all the values of a for which the sum of the roots of the equation $x^2 - 2a(x-1) - 1 = 0$ is equal to the sum of the squares of its roots.
27. Find the coefficients of the equation $x^2 + px + q = 0$ such that its roots are equal to p and q .

5. TRANSFORMATION OF THE EQUATION :

Let $ax^2 + bx + c = 0$ be a quadratic equation with two roots α and β . If we have to find an equation whose roots are $f(\alpha)$ and $f(\beta)$, i.e. some expression in α & β , then this equation can be found by finding α in terms of y . Now as α satisfies given equation, put this α in terms of y directly in the equation.

$$y = f(\alpha)$$

By transformation, $\alpha = g(y)$

$$a(g(y))^2 + b(g(y)) + c = 0$$

This is the required equation in y .

Illustration 9 : If the roots of $ax^2 + bx + c = 0$ are α and β , then find the equation whose roots are :

(a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$ (b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$ (c) α^2, β^2

Solution : (a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$

$$\text{put, } y = \frac{-2}{\alpha} \Rightarrow \alpha = \frac{-2}{y}$$

$$a\left(-\frac{2}{y}\right)^2 + b\left(\frac{-2}{y}\right) + c = 0 \Rightarrow cy^2 - 2by + 4a = 0$$

Required equation is $cx^2 - 2bx + 4a = 0$

(b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

$$\text{put, } y = \frac{\alpha}{\alpha+1} \Rightarrow \alpha = \frac{y}{1-y}$$

$$\Rightarrow a\left(\frac{y}{1-y}\right)^2 + b\left(\frac{y}{1-y}\right) + c = 0 \Rightarrow (a+c-b)y^2 + (-2c+b)y + c = 0$$

Required equation is $(a+c-b)x^2 + (b-2c)x + c = 0$

(c) α^2, β^2

$$\text{put } y = \alpha^2 \Rightarrow \alpha = \sqrt{y}$$

$$ay + b\sqrt{y} + c = 0$$

$$b^2y = a^2y^2 + c^2 + 2acy$$

$$\Rightarrow a^2y^2 + (2ac - b^2)y + c^2 = 0$$

Required equation is $a^2x^2 + (2ac - b^2)x + c^2 = 0$

Illustration 10 : If the roots of $ax^3 + bx^2 + cx + d = 0$ are α, β, γ then find equation whose roots are

$$\frac{1}{\alpha\beta}, \frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}.$$

Solution : Put $y = \frac{1}{\alpha\beta} = \frac{\gamma}{\alpha\beta\gamma} = -\frac{a\gamma}{d}$ ($\because \alpha\beta\gamma = -\frac{d}{a}$)

Put $x = -\frac{dy}{a}$

$$\Rightarrow a\left(-\frac{dy}{a}\right)^3 + b\left(-\frac{dy}{a}\right)^2 + c\left(-\frac{dy}{a}\right) + d = 0$$

Required equation is $d^2x^3 - bdx^2 + acx - a^2 = 0$

Do yourself - 3 :

- If α, β are the roots of $ax^2 + bx + c = 0$, then find the equation whose roots are
 (a) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$ (b) $\frac{1}{a\alpha + b}, \frac{1}{a\beta + b}$ (c) $\alpha + \frac{1}{\beta}, \beta + \frac{1}{\alpha}$
- If α and β are the roots of $x^2 + px + q = 0$, form the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$.
- If α, β are roots of $x^2 - px + q = 0$, then find the quadratic equation whose root are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ and $\alpha^2\beta^3 + \alpha^3\beta^2$.

6. EQUATION VS IDENTITY :

An equation which is true for every value of the variable within the domain is called an identity, for example : $5(a - 3) = 5a - 15$, $(a + b)^2 = a^2 + b^2 + 2ab$ for all $a, b \in \mathbb{R}$.

Note : An equation of degree ≤ 2 cannot have three or more roots & if it has, it becomes an identity.

If $ax^2 + bx + c = 0$ is an identity $\Leftrightarrow a = b = c = 0$

Illustration 11 : If the equation $(\lambda^2 - 5\lambda + 6)x^2 + (\lambda^2 - 3\lambda + 2)x + (\lambda^2 - 4) = 0$ has more than two roots, then find the value of λ ?

Solution : As the equation has more than two roots so it becomes an identity. Hence

$$\lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda = 2, 3$$

$$\text{and } \lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = 1, 2$$

$$\text{and } \lambda^2 - 4 = 0 \Rightarrow \lambda = 2, -2$$

$$\text{So } \lambda = 2$$

Ans. $\lambda = 2$

7. COMMON ROOTS OF TWO QUADRATIC EQUATIONS :

(a) Atleast one common root.

Let α be the common root of $ax^2 + bx + c = 0$ & $a'x^2 + b'x + c' = 0$ then

$$a\alpha^2 + b\alpha + c = 0 \text{ \& \> } a'\alpha^2 + b'\alpha + c' = 0. \text{ By Cramer's Rule } \frac{\alpha^2}{bc' - b'c} = \frac{\alpha}{a'c - ac'} = \frac{1}{ab' - a'b}$$

$$\text{Therefore, } \alpha = \frac{ca' - c'a}{ab' - a'b} = \frac{bc' - b'c}{a'c - ac'}$$

So the condition for a common root is $(ca' - c'a)^2 = (ab' - a'b)(bc' - b'c)$.

(b) If both roots are same then $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$.

Illustration 12 : Find p and q such that $px^2 + 5x + 2 = 0$ and $3x^2 + 10x + q = 0$ have both roots in common.

Solution : $a_1 = p, b_1 = 5, c_1 = 2$ and $a_2 = 3, b_2 = 10, c_2 = q$

We know that :

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{p}{3} = \frac{5}{10} = \frac{2}{q} \Rightarrow p = \frac{3}{2}; q = 4$$

Illustration 13 : Find the possible value(s) of a for which the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have atleast one common root.

Solution : Let α is a common root

$$\text{then } \alpha^2 + a\alpha + 1 = 0$$

$$\& \quad \alpha^2 + \alpha + a = 0$$

by cramer's rule

$$\frac{\alpha^2}{a^2 - 1} = \frac{\alpha}{1 - a} = \frac{1}{1 - a}$$

$$\Rightarrow (1 - a)^2 = (a^2 - 1)(1 - a)$$

$$\Rightarrow a = 1, -2$$

Illustration 14 : If the equations $x^2 - ax + b = 0$ and $x^2 - cx + d = 0$ have one root in common and second equation has equal roots, prove that $ac = 2(b + d)$

Solution : The equation $x^2 - cx + d = 0$ has equal roots.

$$\Rightarrow D = 0 \Rightarrow D = c^2 - 4d = 0 \quad \dots(i)$$

Also, the equal roots are $x = -\frac{b}{2a}$ (for $ax^2 + bx + c = 0$ having equal roots)

$$\Rightarrow x = \frac{c}{2} \text{ is the equal root of this equation.}$$

Now this should be the common root.

$$\therefore x = \frac{c}{2} \text{ will satisfy the first equation}$$

$$\Rightarrow \frac{c^2}{4} - a\left(\frac{c}{2}\right) + b = 0 \Rightarrow c^2 + 4b = 2ac$$

$$\Rightarrow 4d + 4b = 2ac \quad [\text{Using (i)}]$$

$$\Rightarrow 2(d + b) = ac$$

$$\text{Hence } ac = 2(b + d)$$

Illustration 15 : If $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a root in common, find the relation between a , b and c .

Solution : Solve the two equations :

$$ax^2 + bx + c = 0 \text{ and } bx^2 + cx + a = 0$$

$$\frac{x^2}{ba - c^2} = \frac{-x}{a^2 - bc} = \frac{1}{ac - b^2}$$

$$\Rightarrow (a^2 - bc)^2 = (ba - c^2)(ac - b^2)$$

$$\text{Simplify to get : } a(a^3 + b^3 + c^3 - 3abc) = 0$$

$$\Rightarrow a = 0 \text{ or } a^3 + b^3 + c^3 = 3abc$$

This is the relation between a , b and c .

Illustration 16 : Find the conditions on a, b, c and d such that equations

$$2ax^3 + bx^2 + cx + d = 0 \text{ and } 2ax^2 + 3bx + 4c = 0 \text{ have a common root.}$$

Solution :

Let ' α ' be a common root of the given equations,

$$\text{then } 2a\alpha^3 + b\alpha^2 + c\alpha + d = 0 \quad \dots(1)$$

$$\text{and } 2a\alpha^2 + 3b\alpha + 4c = 0 \quad \dots(2)$$

Multiply (2) by α and then subtract (1) from it, to get

$$2b\alpha^2 + 3c\alpha - d = 0 \quad \dots(3)$$

Now (2) and (3) are quadratic having a common root α , so

$$\frac{\alpha^2}{3bd - 12c^2} = \frac{\alpha}{8bc + 2ad} = \frac{1}{6ac - 6b^2}, \quad \alpha^2 = \frac{bd + 4c^2}{2b^2 - 2ac}, \quad \alpha = \frac{4bc + ad}{3ac - 3b^2}$$

Eliminating α from these two equations, we get

$$(4bc + ad)^2 = \frac{9}{2} (bd + 4c^2)(b^2 - ac), \text{ which is the required condition.}$$

Illustration 17 : The values of 'a' for which the equations $x^3 - 6x^2 + (6 + a)x - 6 = 0$ and $x^2 - ax + 4 = 0$ have a common root.

Solution :

Let ' α ' is the common root.

$$\alpha^3 - 6\alpha^2 + (6 + a)\alpha - 6 = 0$$

$$\alpha^2 - a\alpha + 4 = 0$$

Adding

$$\alpha^3 - 5\alpha^2 + 6\alpha - 2 = 0$$

$\Rightarrow \alpha = 1$ is one solution of above equation.

$$\begin{array}{r} \alpha^2 - 4\alpha + 2 \\ \alpha^3 - 5\alpha^2 + 6\alpha - 2 \\ \hline \alpha^3 - \alpha^2 \\ \hline -4\alpha^2 + 6\alpha - 2 \\ -4\alpha^2 + 6\alpha \\ \hline 2\alpha - 2 \\ 2\alpha - 2 \\ \hline 0 \end{array}$$

$$\Rightarrow (\alpha - 1)(\alpha^2 - 4\alpha + 2) = 0$$

$$\Rightarrow (\alpha - 1)\{\alpha - (2 - \sqrt{2})\}\{\alpha - (2 + \sqrt{2})\} = 0$$

$$\alpha = 1, \alpha = 2 - \sqrt{2} \text{ or } \alpha = 2 + \sqrt{2}$$

This common root can be 1, $2 - \sqrt{2}$, $2 + \sqrt{2}$.

We can obtain values of 'a'.

$$\text{For } \alpha = 1, 1 - a + 4 = 0 \Rightarrow a = 5$$

$$\text{For } \alpha = 2 - \sqrt{2}, (6 - 4\sqrt{2}) - a(2 - \sqrt{2}) + 4 = 0$$

$$\Rightarrow a(2 - \sqrt{2}) = 10 - 4\sqrt{2}$$

$$\Rightarrow a = \frac{10 - 4\sqrt{2}}{2 - \sqrt{2}} = \frac{(10 - 4\sqrt{2})(2 + \sqrt{2})}{(4 - 2)}$$

$$\Rightarrow a = (5 - 2\sqrt{2})(2 + \sqrt{2})$$

$$\Rightarrow a = 10 + 5\sqrt{2} - 4\sqrt{2} - 4$$

$$\Rightarrow a = 6 + \sqrt{2}$$

$$\text{For } \alpha = 2 + \sqrt{2} \Rightarrow (6 + 4\sqrt{2}) - a(2 + \sqrt{2}) + 4 = 0$$

$$\Rightarrow a = \frac{10 + 4\sqrt{2}}{2 + \sqrt{2}} = \frac{(10 + 4\sqrt{2})(2 - \sqrt{2})}{2}$$

$$\Rightarrow a = (5 + 2\sqrt{2})(2 - \sqrt{2})$$

$$\Rightarrow a = 10 - 5\sqrt{2} + 4\sqrt{2} - 4$$

$$\Rightarrow a = 6 - \sqrt{2}$$

Thus possible values of 'a' are $5, 6 \pm \sqrt{2}$.

Illustration 18 : If each pair of the following three equations $x^2 + ax + b = 0$, $x^2 + cx + d = 0$, $x^2 + ex + f = 0$ has exactly one root in common and no roots common to all three equations, then show that $(a + c + e)^2 = 4(ac + ce + ea - b - d - f)$.

Solution : Given equations are

$$x^2 + ax + b = 0 \quad \dots(1)$$

$$x^2 + cx + d = 0 \quad \dots(2)$$

$$x^2 + ex + f = 0 \quad \dots(3)$$

Let α, β be the roots of (1), β, γ be the roots of (2) and γ, α be the roots of (3).

$$\therefore \alpha + \beta = -a, \alpha\beta = b \quad \dots(4)$$

$$\beta + \gamma = -c, \beta\gamma = d \quad \dots(5)$$

$$\gamma + \alpha = -e, \gamma\alpha = f \quad \dots(6)$$

$$\begin{aligned} \text{LHS} &= (a + c + e)^2 \\ &= (-\alpha - \beta - \beta - \gamma - \gamma - \alpha)^2 \quad \{\text{from (4), (5), (6)}\} \\ &= 4(\alpha + \beta + \gamma)^2 \quad \dots(7) \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 4(ac + ce + ea - b - d - f) \\ &= 4\{(\alpha + \beta)(\beta + \gamma) + (\beta + \gamma)(\gamma + \alpha) + (\alpha + \beta)(\gamma + \alpha) - \alpha\beta - \beta\gamma - \gamma\alpha\} \quad \{\text{from (4), (5), (6)}\} \\ &= 4(\alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\beta\gamma + 2\gamma\alpha) \\ &= 4(\alpha + \beta + \gamma)^2 \quad \dots(8) \end{aligned}$$

From (7) and (8),

$$(a + c + e)^2 = 4(ac + ce + ea - b - d - f)$$

Illustrations 19 : If α, β are the roots of $x^2 + px + q = 0$, and γ, δ are the roots of $x^2 + rx + s = 0$, evaluate $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ in terms of p, q, r and s . Deduce the condition that the equations have a common root.

Solution :

α, β are the roots of $x^2 + px + q = 0$

$$\therefore \alpha + \beta = -p, \alpha\beta = q \quad \dots\dots(1)$$

and γ, δ are the roots of $x^2 + rx + s = 0$

$$\therefore \gamma + \delta = -r, \gamma\delta = s \quad \dots\dots(2)$$

Now, $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$

$$\begin{aligned} &= [\alpha^2 - \alpha(\gamma + \delta) + \gamma\delta] [\beta^2 - \beta(\gamma + \delta) + \gamma\delta] \\ &= (\alpha^2 + r\alpha + s) (\beta^2 + r\beta + s) \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s(\alpha^2 + \beta^2) + sr(\alpha + \beta) + s^2 \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s((\alpha + \beta)^2 - 2\alpha\beta) + sr(\alpha + \beta) + s^2 \\ &= q^2 - pqr + r^2q + s(p^2 - 2q) + sr(-p) + s^2 \\ &= (q - s)^2 - rpq + r^2q + sp^2 - prs \\ &= (q - s)^2 - rq(p - r) + sp(p - r) \\ &= (q - s)^2 + (p - r)(sp - rq) \end{aligned}$$

$$\text{For a common root (Let } \alpha = \gamma \text{ or } \beta = \delta) \quad \dots\dots(3)$$

$$\text{then } (\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta) = 0 \quad \dots\dots(4)$$

from (3) and (4), we get

$$(q - s)^2 + (p - r)(sp - rq) = 0$$

$$\Rightarrow (q - s)^2 = (p - r)(rq - sp), \text{ which is the required condition.}$$

Do yourself - 4 :

- For what values of a , the equation

$$(a + 3)(a - 1)x^2 + (a - 1)(4a + 3)x + (a^2 - 3a + 2) = 0 \text{ has more than two roots.}$$

- If $a. \frac{(x-b)(x-c)}{(a-b)(a-c)} + b. \frac{(x-c)(x-a)}{(b-c)(b-a)} + c. \frac{(x-a)(x-b)}{(c-a)(c-b)} = x$ then prove that it is an identity.

- If $x^2 + bx + c = 0$ & $2x^2 + 9x + 10 = 0$ have both roots in common, find b & c .

- If $x^2 - 7x + 10 = 0$ & $x^2 - 5x + c = 0$ have a common root, find c .

- Show that $x^2 + (a^2 - 2)x - 2a^2 = 0$ and $x^2 - 3x + 2 = 0$ have exactly one common root for all $a \in \mathbb{R}$.

- For what values of a do the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have exactly one root in common ?

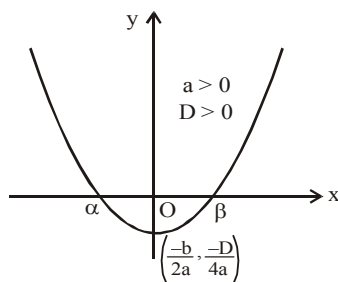
- Given two quadratic equations $x^2 - x + m = 0$ and $x^2 - x + 3m = 0, m \neq 0$. Find the value of m for which one of the roots of the second equation is equal to double the root of the first equation.

8. QUADRATIC EXPRESSION AND ITS GRAPHS :

Consider the quadratic expression, $y = ax^2 + bx + c$, $a \neq 0$ & $a, b, c \in \mathbb{R}$ then ;

- (a) The graph between x, y is always a parabola. If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.
- (b) The graph of $y = ax^2 + bx + c$ can be divided in 6 broad categories which are as follows :
(Let the real roots of quadratic equation $ax^2 + bx + c = 0$ be α & β where $\alpha \leq \beta$).

Fig. 1

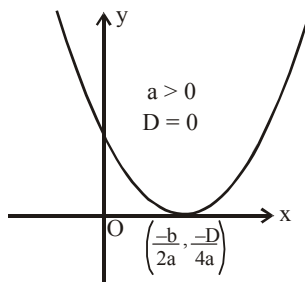


Roots are real & distinct

$$ax^2 + bx + c > 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

$$ax^2 + bx + c < 0 \quad \forall x \in (\alpha, \beta)$$

Fig. 2

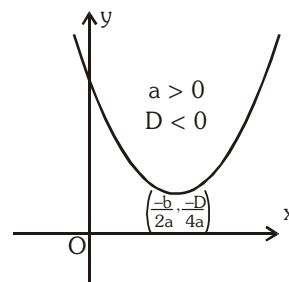


Roots are coincident

$$ax^2 + bx + c > 0 \quad \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \quad \text{for } x = \alpha = \beta$$

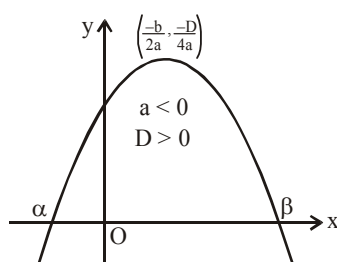
Fig. 3



Roots are complex conjugate

$$ax^2 + bx + c > 0 \quad \forall x \in \mathbb{R}$$

Fig. 4

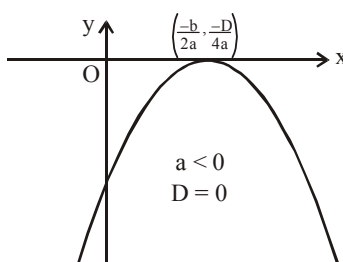


Roots are real & distinct

$$ax^2 + bx + c > 0 \quad \forall x \in (\alpha, \beta)$$

$$ax^2 + bx + c < 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

Fig. 5

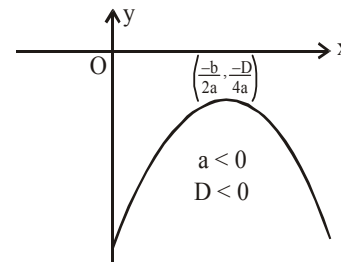


Roots are coincident

$$ax^2 + bx + c < 0 \quad \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \quad \text{for } x = \alpha = \beta$$

Fig. 6



Roots are complex conjugate

$$ax^2 + bx + c < 0 \quad \forall x \in \mathbb{R}$$

Important Note :

- (i) The quadratic expression $ax^2 + bx + c > 0$ for each $x \in \mathbb{R} \Rightarrow a > 0, D < 0$ & vice-versa (Fig. 3)
- (ii) The quadratic expression $ax^2 + bx + c < 0$ for each $x \in \mathbb{R} \Rightarrow a < 0, D < 0$ & vice-versa (Fig. 6)

9. INEQUALITIES (rational, irrational and modulus)

9.1 Rational inequalities (When numerator or denominator contains irreducible quadratic factor)

Illustration 20 : Find x such that $3x^2 - 7x + 6 < 0$

Solution :

$$f(x) = 3x^2 - 7x + 6$$

$$D = 49 - 72 < 0 \text{ and } a > 0$$

$$\Rightarrow 3x^2 - 7x + 6 > 0 \quad \forall x \in \mathbb{R} \Rightarrow x \in \phi$$

Illustration 21: Find x such that $2x^2 + 4x + 9 > 0$

Solution : $f(x) = 2x^2 + 4x + 9$
 $a > 0$ and $D < 0$
 $\Rightarrow 2x^2 + 4x + 9 > 0 \forall x \in \mathbb{R}$
 $\Rightarrow x \in (-\infty, \infty)$

Illustration-22 : Solve for x : $\frac{(x^2 - 1)^2 + 1}{x^2 - 1} \geq 2$.

Solution : Recall :

The sum of any real number except zero and its reciprocal should not lie in $(-2, 2)$.

Case I :

When $t + \frac{1}{t} \geq 2$. This holds $\forall t > 0$ and equality holds only when $t = 1$.

Case II :

When $t + \frac{1}{t} \leq -2$. This holds $\forall t < 0$ and equality holds if $t = -1$.

Note : $x^2 - 1 \neq 0, x^2 \neq 1 \Rightarrow x \neq \pm 1$

Rewrite the given inequality $\frac{(x^2 - 1)^2 + 1}{x^2 - 1} \geq 2$

as $(x^2 - 1) + \frac{1}{(x^2 - 1)} \geq 2$

$\Rightarrow x^2 - 1 > 0 \Rightarrow (x + 1)(x - 1) > 0$



$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$

Do yourself- 5 :

Solve for $x \in \mathbb{R}$

1. $x^2 - 3x + 4 > 0$

2. $\frac{x}{x^2 + 2} < 1$

3. $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} < 3$

4. $\frac{(2x - 1)(x + 3)(2 - x)(1 - x)^2}{x^4(x + 6)(x - 9)(2x^2 + 4x + 9)} < 0$

5. $\frac{7x - 17}{x^2 - 3x + 4} \geq 1$

6. $\frac{x^2 + 6x - 7}{x^2 + 1} \leq 2$

7. $\frac{2x}{x^2 - 9} \leq \frac{1}{x + 2}$

8. $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} > 3$

9. $\frac{x^2 - 5x + 6}{x^2 + x + 1} < 0$

10. $\frac{x^2 + 4x + 4}{2x^2 - x - 1} > 0$

9.2. IRRATIONAL INEQUALITIES :

While solving Irrational Inequality, first we have to find largest interval of x for which left hand side and right hand side both make sense. This interval is called domain of the Inequality.

Note :

- (i) For $\sqrt{f(x)}$ to be well defined $f(x)$ must be greater than or equal to 0.
- (ii) $\sqrt{f(x)} \geq 0$ whenever defined.

Type 1 : $\sqrt{f(x)} \geq g(x)$

Case-I : $f(x) \geq 0$ and $g(x) > 0$... (1)

Above inequality become

$$f(x) \geq g^2(x) \quad \dots (2)$$

Answer in this case is intersection of solutions of (1) and (2)

Case-II : $f(x) \geq 0$ and $g(x) \leq 0$.

In this type final answer of the given Inequality is union of case I and case II.

Type 2 : $g(x) \geq \sqrt{f(x)}$

Since $\sqrt{f(x)} \geq 0$ for $\forall f(x) \geq 0$

So $g(x)$ must be non-negative

$$g(x) \geq 0 \quad \dots (1)$$

$$f(x) \geq 0 \quad \dots (2)$$

$$g^2(x) \geq f(x) \quad \dots (3)$$

Answer of the inequality is intersection of (1), (2) and (3).

Illustration-23 : Solve : $x + 1 \geq \sqrt{5 - x}$

Solution : $5 - x \geq 0$ and $x + 1 \geq 0$

$$\Rightarrow x \in [-1, 5] \quad \dots (1)$$

For $x \in [-1, 5]$

$$x + 1 \geq \sqrt{5 - x}$$

$$\Rightarrow (x + 1)^2 \geq 5 - x \Rightarrow x^2 + 3x - 4 \geq 0$$

$$\Rightarrow (x - 1)(x + 4) \geq 0 \Rightarrow x \in (-\infty, -4] \cup [1, \infty)$$

$$\Rightarrow x \in [1, 5]$$

Illustration-24 : Solve : $\sqrt{6+x} \geq 6-x$

Solution :

Case-I $6+x \geq 0$ and $6-x > 0$

$$x \in [-6, 6)$$

$$\sqrt{6+x} \geq 6-x \Rightarrow 6+x \geq 36-12x+x^2$$

$$\Rightarrow 0 \geq x^2-13x+30 \Rightarrow (x-10)(x-3) \leq 0 \Rightarrow x \in [3, 10]$$

$$x \in [3, 10] \cap [-6, 6) = [3, 6)$$

Case-II $6+x \geq 0$ and $6-x \leq 0 \Rightarrow x \in [6, \infty)$

In this case given Inequality is always true.

From case-I and case-II,

$$x \in [3, \infty).$$

Illustration 25 : Solve for x, if $\sqrt{x^2-3x+2} > x-2$

Solution :

$$\left[\begin{array}{l} \left\{ \begin{array}{l} x^2-3x+2 \geq 0 \\ x-2 \geq 0 \\ (x^2-3x+2) > (x-2)^2 \end{array} \right. \\ \text{or} \\ \left\{ \begin{array}{l} x^2-3x+2 \geq 0 \\ x-2 < 0 \end{array} \right. \end{array} \right] \Rightarrow \left[\begin{array}{l} \left\{ \begin{array}{l} (x-1)(x-2) \geq 0 \\ (x-2) \geq 0 \\ x-2 > 0 \end{array} \right. \Rightarrow x > 2 \\ \text{or} \\ \left\{ \begin{array}{l} (x-1)(x-2) \geq 0 \\ x-2 < 0 \end{array} \right. \Rightarrow x \leq 1 \end{array} \right]$$

Hence, solution set of the original inequation is $x \in \mathbb{R} - (1, 2]$

Do yourself- 6 :

Solve for x if

1. $\sqrt{x^2-x} > (x-1)$

2. $\sqrt{\frac{x-2}{1-2x}} > -1$

3. $\frac{\sqrt{x-3}}{x-2} > 0$

4. $2\sqrt{x-1} < x$

5. $\sqrt{9x-20} < x$

6. $\sqrt{5-2x} < 6x-1$

7. $\sqrt{2x-x^2} < 5-x$

8. $\sqrt{x+18} < 2-x$

9.3 Modulus inequalities

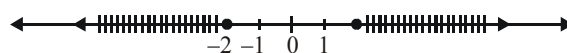
- $|x| \leq a$, if $a \geq 0$
 $\Rightarrow -a \leq x \leq a$
 if $a < 0$, then $x \in \phi$
- $|x| \geq a$
 if $a \geq 0$, then $x \in (-\infty, -a] \cup [a, \infty)$
 if $a < 0$, then $x \in (-\infty, \infty)$
- $|xy| = |x| |y|$
- $||x| - |y|| \leq |x + y| \leq |x| + |y|$
- $|x| + |y| = |x + y| \Leftrightarrow xy \geq 0$
- $|x| + |y| = |x - y| \Leftrightarrow xy \leq 0$

Illustration-26 : Represent x on the number line when

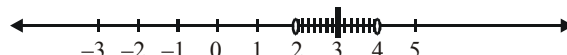
(i) $|x| \geq 2$ (ii) $|x - 3| \leq 1$ (iii) $|5 - x| < 2$

(iv) $1 \leq |x + 2| < 5$ (v) $|2x - 5| < 1$

Solution : (i) $|x| \geq 2 \Rightarrow$ distance of x from O on the number line is greater than or equal to 2.



(ii) $|x - 3| \leq 1 \Rightarrow$ distance of x from 3 is less than or equal to 1.



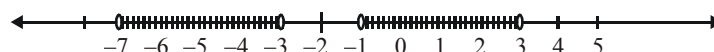
(iii) $|5 - x| < 2 \Rightarrow$ distance of x from 2 is less than 2.



(iv) $1 \leq |x + 2| < 5$

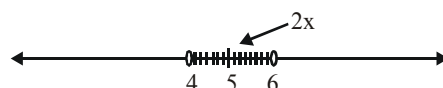
$$\Rightarrow 1 \leq |x - (-2)| < 5$$

$\Rightarrow$ distance of x from -2 is greater than or equal to 1 and less than 5.



(v) $|2x - 5| < 1$

$\Rightarrow$ distance of $2x$ from 5 on the number line is less than or equal to 1.



$$\Rightarrow 4 < 2x < 6 \Rightarrow 2 < x < 3$$

Illustration-27 : Solve following

- (i) $|x^2 - 4x| = x + 1$
 (ii) $|x^2 + 1| = |x - 2|$
 (iii) $|x^2 - 16| - 8|x - 2| = x(8 - x)$
 (iv) $a^2|x + a| + |a^2x + 1| = 1 - a^3$

where a is a real constant.

Solution :

(i) Case-1

$$x^2 - 4x \geq 0$$

$$\Rightarrow x \in (-\infty, 0] \cup [4, \infty) \quad \dots(1)$$

$$|x^2 - 4x| = x + 1$$

$$\Rightarrow x^2 - 4x = x + 1 \Rightarrow x^2 - 5x - 1 = 0$$

$$\Rightarrow x = \frac{5 \pm \sqrt{29}}{2}$$

from (1) and (2)

$$x = \frac{5 \pm \sqrt{29}}{2}$$

Case -2

where $x^2 - 4x < 0$

$$\Rightarrow x \in (0, 4) \quad \dots(3)$$

$$|x^2 - 4x| = x + 1$$

$$\Rightarrow 4x - x^2 = x + 1 \Rightarrow x^2 - 3x + 1 = 0$$

$$\Rightarrow x = \frac{3 \pm \sqrt{5}}{2} \quad \dots(4)$$

From (3) and (4)

Set of all solutions of the given equation is

$$\left\{ \frac{5 - \sqrt{29}}{2}, \frac{3 - \sqrt{5}}{2}, \frac{3 + \sqrt{5}}{2}, \frac{5 + \sqrt{29}}{2} \right\}$$

(ii) $|x^2 + 1| = |x - 2|$

$$\text{As } x^2 + 1 > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow |x^2 + 1| = x^2 + 1, \quad \forall x \in \mathbb{R}$$

Case-1

$$x - 2 \geq 0 \Rightarrow x \in [2, \infty)$$

$$|x^2 + 1| = |x - 2|$$

$$\Rightarrow x^2 + 1 = x - 2 \Rightarrow x^2 - x + 3 = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{11}i}{2}$$

Case-2

$$x - 2 < 0 \Rightarrow x \in (-\infty, 2)$$

$$|x^2 + 1| = |x - 2|$$

$$\Rightarrow x^2 + 1 = -x + 2 \Rightarrow x^2 + x - 1 = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{5}}{2}$$

Both x belongs to $(-\infty, 2)$

$$\text{Hence set of solutions is } \left\{ \frac{-1 \pm \sqrt{5}}{2} \right\}$$

$$(iii) |x^2 - 16| - 8|x - 2| = x(8 - x)$$

Case 1 : $x \leq -4$

$$x^2 - 16 + 8x - 16 = 8x - x^2$$

$$2x^2 = 32$$

Case 2 : $x \in [-4, 2]$

$$-x^2 + 16 + 8x - 16 = 8x - x^2$$

$$x = x \text{ always true}$$

$$x \in [-4, 2]$$

Case 3 : $x \in [2, 4]$

$$-x^2 + 76 - 8x + 16 = 8x - x^2$$

$$16x = 32$$

$$x = 2$$

Case 4 : $x \geq 4$

$$x^2 - 16 - 8x + 16 = 8x - x^2$$

$$2x^2 = 16x$$

$$x = 8, 0 \text{ (rejected)}$$

$$x \in [-4, 2] \cup \{8\}$$

$$(iv) a^2|x + a| + |a^2x + 1| = 1 - a^3 \Rightarrow |a^2x + a^3| + |a^2x + 1| = 1 - a^3$$

$$\text{Let } A = a^2x + a^3, B = a^2x + 1 \text{ and } C = 1 - a^3$$

Given equation can be written as

$$|A| + |B| = C = B - A$$

$$\Rightarrow B \geq 0 \text{ and } A \leq 0$$

$$a^2x + 1 \geq 0 \text{ and } a^2x + a^3 \leq 0$$

$$\text{when } a = 0, \quad x \in \mathbb{R}$$

$$\text{when } a \neq 0, \quad x \geq \frac{-1}{a^2} \text{ and } x \leq -a \Rightarrow x \in \left[-\frac{1}{a^2}, \infty \right) \cap (-\infty, -a]$$

Case-1

$$\text{when } -\frac{1}{a^2} > -a \Rightarrow \frac{1}{a^2} < a \Rightarrow 1 < a^3 \Rightarrow a \in (1, \infty)$$

$$x \in \left[-\frac{1}{a^2}, \infty\right) \cap (-\infty, -a] = \phi$$

Case-2

$$\text{when } -a \geq -\frac{1}{a^2} \Rightarrow a \leq \frac{1}{a^2} \text{ and } a \neq 0$$

$$\Rightarrow a^3 \leq 1 \text{ and } a \neq 0 \Rightarrow a < 1 \text{ and } a \neq 0 \Rightarrow a \in (-\infty, 1] - \{0\}$$

$$x \in \left[-\frac{1}{a^2}, \infty\right) \cap (-\infty, -a]$$

$$x \in \left[-\frac{1}{a^2}, -a\right]$$

Solutions of the given equation is

$$x \in \mathbb{R} \text{ when } a = 0$$

$$x \in \phi \text{ when } a > 1$$

$$x \in \left[-\frac{1}{a^2}, -a\right] \text{ when } a \in (-\infty, 1] - \{0\}$$

Illustrations-28 : Let $A = \{x : x \in \mathbb{R}, |x| < 1\}$; $B = \{x : x \in \mathbb{R}, |x - 1| \geq 1\}$ and $A \cup B = \mathbb{R} - D$, then the set D is-

- (1) $[x : 1 < x \leq 2]$ (2) $[x : 1 \leq x < 2]$ (3) $[x : 1 \leq x \leq 2]$ (4) none of these

Solution :

$$A = \{x : x \in \mathbb{R}, -1 < x < 1\}$$

$$B = \{x : x \in \mathbb{R} : x - 1 \leq -1 \text{ or } x - 1 \geq 1\}$$

$$= \{x : x \in \mathbb{R} : x \leq 0 \text{ or } x \geq 2\}$$

$$\therefore A \cup B = \mathbb{R} - D$$

$$\text{where } D = \{x : x \in \mathbb{R}, 1 \leq x < 2\}$$

Thus (2) is the correct answer.

Illustration-29 : If $|x - 1||x - 2| = -(x^2 - 3x + 2)$, then find the interval in which x lies ?

Solution :

$$|x - 1||x - 2| = -(x - 2)(x - 1)$$

$$\Rightarrow (x - 1)(x - 2) \leq 0$$

$$\Rightarrow 1 \leq x \leq 2$$



Illustration-30 : Solve : (i) $||x - 2| - 5| \geq 1$

Solution : $||x - 2| - 5| \geq 1$
 $\Rightarrow |x - 2| - 5 \geq 1$ or $|x - 2| - 5 \leq -1$
 $\Rightarrow |x - 2| \geq 6$ or $|x - 2| \leq 4$

Case-1

$$|x - 2| \geq 6$$

$$\Rightarrow x - 2 \geq 6 \text{ or } x - 2 \leq -6 \Rightarrow x \geq 8 \text{ or } x \leq -4$$

$$\Rightarrow x \in (-\infty, -4] \cup [8, \infty) \quad \dots(1)$$

Case-2

$$|x - 2| \leq 4$$

$$\Rightarrow -4 \leq x - 2 \leq 4 \Rightarrow -2 \leq x \leq 6$$

$$\Rightarrow x \in [-2, 6] \quad \dots(2)$$

From (1) and (2) $x \in (-\infty, -4] \cup [-2, 6] \cup [8, \infty)$

Illustration 31 : If x satisfies $|x - 1| + |x - 2| + |x - 3| \geq 6$, then

- (A) $0 \leq x \leq 4$ (B) $x \leq -2$ or $x \geq 4$
 (C) $x \leq 0$ or $x \geq 4$ (D) none of these

Solution :

Case I : $x \leq 1$, then
 $1 - x + 2 - x + 3 - x \geq 6 \Rightarrow x \leq 0$
 Hence $x \leq 0$...(i)

Case II : $1 < x \leq 2$, then
 $x - 1 + 2 - x + 3 - x \geq 6 \Rightarrow x \leq -2$
 But $1 < x \leq 2 \Rightarrow$ No solution. ...(ii)

Case III : $2 < x \leq 3$, then
 $x - 1 + x - 2 + 3 - x \geq 6 \Rightarrow x \geq 6$
 But $2 < x \leq 3 \Rightarrow$ No solution. ...(iii)

Case IV : $x > 3$, then
 $x - 1 + x - 2 + x - 3 \geq 6 \Rightarrow x \geq 4$
 Hence $x \geq 4$...(iv)

From (i), (ii), (iii) and (iv) the given inequality holds for $x \leq 0$ or $x \geq 4$.

Illustration 32 : Solve for x : (a) $||x - 1| + 2| \leq 4$. (b) $\frac{x-4}{x+2} \leq \left| \frac{x-2}{x-1} \right|$

Solution :

(a) $||x - 1| + 2| \leq 4 \Rightarrow -4 \leq |x - 1| + 2 \leq 4$
 $\Rightarrow -6 \leq |x - 1| \leq 2$
 $\Rightarrow |x - 1| \leq 2 \Rightarrow -2 \leq x - 1 \leq 2$
 $\Rightarrow -1 \leq x \leq 3 \Rightarrow x \in [-1, 3]$

(b) **Case 1 :** Given inequation will be satisfied for all x such that

$$\frac{x-4}{x+2} \leq 0 \Rightarrow x \in (-2, 4] - \{1\} \quad \dots\dots\dots(i)$$

(Note : $\{1\}$ is not in domain of RHS)

Case 2 : $\frac{x-4}{x+2} > 0 \Rightarrow x \in (-\infty, -2) \cup (4, \infty)$ (ii)

Given inequation becomes

$$\frac{x-2}{x-1} \geq \frac{x-4}{x+2} \quad \text{or} \quad \frac{x-2}{x-1} \leq -\frac{x-4}{x+2}$$

on solving we get

$$x \in (-2, 4/5) \cup (1, \infty)$$

taking intersection with (ii) we get

$$x \in (4, \infty)$$

.....(iii)

on solving we get

$$x \in (-2, 0] \cup (1, 5/2]$$

taking intersection with (ii) we get

$$x \in \phi$$

Hence, solution of the original inequation : $x \in (-2, \infty) - \{1\}$ (taking union of (i) & (iii))

Illustration 33: The equation $|x| + \left| \frac{x}{x-1} \right| = \frac{x^2}{|x-1|}$ is always true for x belongs to

(A) $\{0\}$

(B) $(1, \infty)$

(C) $(-1, 1)$

(D) $(-\infty, \infty)$

Solution :

$$\frac{x^2}{|x-1|} = \left| x + \frac{x}{x-1} \right|$$

$$\therefore |x| + \left| \frac{x}{x-1} \right| = \left| x + \frac{x}{x-1} \right| \text{ is true only if } \left(x \cdot \frac{x}{x-1} \right) \geq 0 \Rightarrow x \in \{0\} \cup (1, \infty). \text{ Ans (A,B)}$$

Do yourself-7

Solve for $x \in \mathbb{R}$

1. $|x| \geq 10$

2. $|x-5| \leq 5$

3. $|2-x| \leq 4$

4. $1 \leq |2x-3| \leq 10$

5. $|x| < \sqrt{2}$

6. $|x| < 0$

7. $|2x| < 3$

8. $|3x| + 5 < 0$

9. $|3x-5| < 2$

10. $|1-x| \leq 2$

11. $|x| > 2$

12. $|2-7x| < 8$

13. $|2x| + 4 > 0$

14. $|2x| - 7 > 0$

10. MAXIMUM & MINIMUM VALUES OF QUADRATIC EXPRESSIONS : $y = ax^2 + bx + c$:

We know that $y = ax^2 + bx + c$ takes following form : $y = a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{(b^2 - 4ac)}{4a^2} \right]$,

which is a parabola.

$$\therefore \text{vertex} = \left(\frac{-b}{2a}, \frac{-D}{4a} \right)$$

When $a > 0$, y will take a minimum value at vertex ; $x = \frac{-b}{2a}$; $y_{\min} = \frac{-D}{4a}$

When $a < 0$, y will take a maximum value at vertex ; $x = \frac{-b}{2a}$; $y_{\max} = \frac{-D}{4a}$.

If quadratic expression $ax^2 + bx + c$ is a perfect square, then $a > 0$ and $D = 0$

Illustration 34 : The value of the expression $x^2 + 2bx + c$ will be positive for all real x if -

- (A) $b^2 - 4c > 0$ (B) $b^2 - 4c < 0$ (C) $c^2 < b$ (D) $b^2 < c$

Solution : As $a > 0$, so this expression will be positive if $D < 0$

$$\text{so } 4b^2 - 4c < 0$$

$$b^2 < c$$



Ans. (D)

Illustration 35 : The minimum value of the expression $4x^2 + 2x + 1$ is -

- (A) $1/4$ (B) $1/2$ (C) $3/4$ (D) 1

Solution : Since $a = 4 > 0$

$$\text{therefore its minimum value} = -\frac{D}{4a} = \frac{4(4)(1) - (2)^2}{4(4)} = \frac{16 - 4}{16} = \frac{12}{16} = \frac{3}{4} \quad \text{Ans. (C)}$$

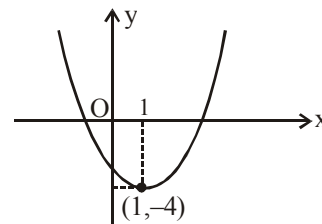
Illustration 36 : If $y = x^2 - 2x - 3$, then find the range of y when :

- (i) $x \in \mathbb{R}$ (ii) $x \in [0, 3]$ (iii) $x \in [-2, 0]$

Solution : We know that minimum value of y will occur at

$$x = -\frac{b}{2a} = -\frac{(-2)}{2 \times 1} = 1$$

$$y_{\min} = -\frac{D}{4a} = \frac{-(4 + 3 \times 4)}{4} = -4$$



- (i) $x \in \mathbb{R};$

$$y \in [-4, \infty)$$

Ans.

- (ii) $x \in [0, 3]$

$$f(0) = -3, f(1) = -4, \quad f(3) = 0$$

$$\therefore f(3) > f(0)$$

$\therefore y$ will take all the values from minimum to $f(3)$.

$$y \in [-4, 0]$$

Ans.

- (iii) $x \in [-2, 0]$

This interval does not contain the minimum value of y for $x \in \mathbb{R}$.

y will take values from $f(0)$ to $f(-2)$

$$f(0) = -3$$

$$f(-2) = 5$$

$$y \in [-3, 5]$$

Ans.

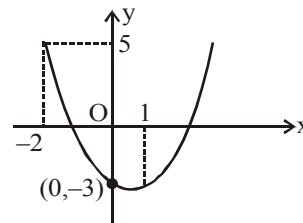


Illustration 37 : If $ax^2 + bx + 10 = 0$ does not have real & distinct roots, find the minimum value of $5a - b$.

Solution : Either $f(x) \geq 0 \forall x \in \mathbb{R}$ or $f(x) \leq 0 \forall x \in \mathbb{R}$

$$\therefore f(0) = 10 > 0 \Rightarrow f(x) \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow f(-5) = 25a - 5b + 10 \geq 0$$

$$\Rightarrow 5a - b \geq -2$$

Ans.

Illustration-38 : Find possible values (range) of x^2 if

$$(i) x \in [1, 3] \quad (ii) x \in [-2, 0] \quad (iii) x \in [-1, 2] \quad (iv) x \in [-2, 1] \quad (v) x \in [-2, 2]$$

Solution : (i) $1 \leq x \leq 3 \Rightarrow 1 \leq x^2 \leq 9 \Rightarrow x^2 \in [1, 9]$

(ii) $-2 \leq x \leq 0 \Rightarrow 4 \geq x^2 \geq 0 \Rightarrow x^2 \in [0, 4]$

(iii) $-1 \leq x < 0$ or $0 \leq x \leq 2$

$$\Rightarrow 1 \geq x^2 > 0 \text{ or } 0 \leq x^2 \leq 4 \Rightarrow 0 \leq x^2 \leq 4 \Rightarrow x^2 \in [0, 4]$$

(iv) $-2 \leq x < 0$ or $0 \leq x \leq 1$

$$\Rightarrow 4 \geq x^2 > 0 \text{ or } 0 \leq x^2 \leq 1 \Rightarrow x^2 \in [0, 4]$$

(v) $-2 \leq x < 0$ or $0 \leq x \leq 2$

$$\Rightarrow 4 \geq x^2 > 0 \text{ or } 0 \leq x^2 \leq 4 \Rightarrow 0 \leq x^2 \leq 4 \Rightarrow x^2 \in [0, 4]$$

Illustration-39 : If $x \in \left[-\frac{3}{2}, 1\right]$ then find range of (i) $(2x - 1)^2 - 3$ (ii) $4x^3 + 1$

Solution : (i) $[-3, 13]$ (ii) $\left[-\frac{25}{2}, 5\right]$

Do yourself - 8

1. Find the minimum value of :

(a) $y = x^2 + 2x + 2$

(b) $y = 4x^2 - 16x + 15$

2. For following graphs of $y = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, comment on the sign of :

(i) a

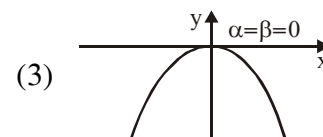
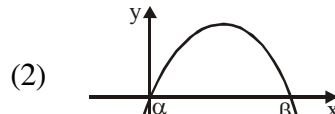
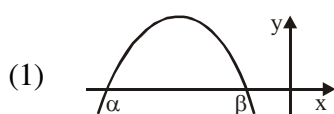
(ii) b

(iii) c

(iv) D

(v) $\alpha + \beta$

(vi) $\alpha\beta$



3. Given the roots of equation $ax^2 + bx + c = 0$ are real & distinct, where $a, b, c \in \mathbb{R}^+$, then the vertex of the graph will lie in which quadrant.

4. Find the range of 'a' for which :
 (a) $ax^2 + 3x + 4 > 0 \quad \forall x \in \mathbb{R}$ (b) $ax^2 + 4x - 2 < 0 \quad \forall x \in \mathbb{R}$
5. The trinomial $ax^2 + bx + c$ has no real roots, $a + b + c < 0$. Find the sign of the number c.
6. For what values of a do the graphs of the functions $y = 2ax + 1$ and $y = (a - 6)x^2 - 2$ not intersect?
7. For what values of k is the inequality $x^2 - (k - 3)x - k + 6 > 0$ valid for all real x?
8. For what integral k is the inequality $x^2 - 2(4k - 1)x + 15k^2 - 2k - 7 > 0$ valid for any real x?
9. For what values of a is the inequality $ax^2 + 2ax + 0.5 > 0$ valid for all real x?
10. For what least integral k is the expression $(k - 2)x^2 + 8x + k + 4 > 0$ for all values of x?
11. Find all values of a for which the inequality $(a - 3)x^2 - 2ax + 3a - 6 > 0$ is true for all $x \in \mathbb{R}$.

11. MAXIMUM & MINIMUM VALUES OF RATIONAL ALGEBRAIC EXPRESSIONS :

$$y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}, \frac{1}{ax^2 + bx + c}, \frac{a_1x + b_1}{a_2x^2 + b_2x + c_2}, \frac{a_1x^2 + b_1x + c_1}{a_2x + b_2} :$$

Sometime we have to find range of expression of form $\frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$.

The following procedure is used :

Step 1 : Equate the given expression to y i.e. $y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$

Step 2 : By cross multiplying and simplifying, obtain a quadratic equation in x.

$$(a_1 - a_2y)x^2 + (b_1 - b_2y)x + (c_1 - c_2y) = 0$$

Step 3 : Put Discriminant ≥ 0 and solve the inequality for possible set of values of y.

Illustration 40 : If $x \in [-2, 3]$, find all possible values (Range) of

(i) $\frac{1}{x+1}$

(ii) $\frac{x+2}{x+4}$

(iii) $\frac{2x}{x-1}$

Solution : (i) $-2 \leq x \leq 3 \Rightarrow -1 \leq x + 1 \leq 4$

(For $x + 1 = 0$, $\frac{1}{x+1}$ is undefined)

so, $-1 \leq x + 1 < 0$ or $0 < x + 1 \leq 4$

$$\Rightarrow -1 \geq \frac{1}{x+1} > -\infty \text{ or } \infty > x+1 \geq \frac{1}{4}$$

So range of $\frac{1}{x+1}$ is $(-\infty, -1] \cup \left[\frac{1}{4}, \infty\right)$

$$(ii) \quad \frac{x+2}{x+4} = 1 - \frac{2}{x+4}$$

$$\text{Given } -2 \leq x \leq 3$$

$$\Rightarrow 2 \leq x+4 \leq 7 \Rightarrow \frac{1}{2} \geq \frac{1}{x+4} \geq \frac{1}{7}$$

$$\Rightarrow \frac{2}{2} \geq \frac{2}{x+4} \geq \frac{2}{7} \Rightarrow -1 \leq \frac{-2}{x+4} \leq \frac{-2}{7}$$

$$\Rightarrow 0 \leq 1 - \frac{2}{x+4} \leq \frac{5}{7} \Rightarrow \text{Range} = \left[0, \frac{5}{7}\right]$$

$$(iii) \quad \frac{2x}{x-1} = 2 + \frac{2}{x-1}$$

$$\text{for } -2 \leq x \leq 3 \Rightarrow -3 \leq x-1 \leq 2$$

$$\Rightarrow -3 \leq x-1 < 0 \text{ or } 0 < x-1 \leq 2$$

$$\Rightarrow \frac{-2}{3} \geq \frac{2}{x-1} > -\infty \text{ or } \infty > \frac{2}{x-1} \geq \frac{2}{2}$$

$$\Rightarrow \frac{4}{3} \geq 2 + \frac{2}{x-1} \text{ or } 2 + \frac{2}{x-1} \geq 3$$

$$\text{so range} = \left(-\infty, \frac{4}{3}\right] \cup [3, \infty)$$

Illustration 41 : For $x \in \mathbb{R}$, find the set of values attainable by $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$.

Solution : Let $y = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$

$$x^2(y-1) + 3x(y+1) + 4(y-1) = 0$$

Case- I : $y \neq 1$

For $y \neq 1$ above equation is a quadratic equation.

So for $x \in \mathbb{R}$, $D \geq 0$

$$\Rightarrow 9(y+1)^2 - 16(y-1)^2 \geq 0 \Rightarrow 7y^2 - 50y + 7 \leq 0$$

$$\Rightarrow (7y-1)(y-7) \leq 0 \Rightarrow y \in \left[\frac{1}{7}, 7\right] - \{1\}$$

Case II : when $y = 1$

$$\Rightarrow 1 = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$$

$$\Rightarrow x^2 + 3x + 4 = x^2 - 3x + 4$$

$$\Rightarrow x = 0$$

Hence $y = 1$ for real value of x .

$$\text{so range of } y \text{ is } \left[\frac{1}{7}, 7\right]$$

Illustration 42 : Find the values of a for which the expression $\frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$ assumes all real values for real values of x .

Solution :

$$\text{Let } y = \frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$$

$$x^2(a + 4y) + 3(1 - y)x - (4 + ay) = 0$$

$$\text{If } x \in \mathbb{R}, D \geq 0$$

$$\Rightarrow 9(1 - y)^2 + 4(a + 4y)(4 + ay) \geq 0 \Rightarrow (9 + 16a)y^2 + (4a^2 + 46)y + (9 + 16a) \geq 0$$

$$\text{for all } y \in \mathbb{R}, (9 + 16a) > 0 \text{ \& } D \leq 0$$

$$\Rightarrow (4a^2 + 46)^2 - 4(9 + 16a)(9 + 16a) \leq 0 \Rightarrow 4(a^2 - 8a + 7)(a^2 + 8a + 16) \leq 0$$

$$\Rightarrow a^2 - 8a + 7 \leq 0 \Rightarrow 1 \leq a \leq 7$$

$$9 + 16a > 0 \text{ \& } 1 \leq a \leq 7$$

$$\text{Taking intersection, } a \in [1, 7]$$

Now, checking the boundary values of a

$$\text{For } a = 1$$

$$y = \frac{x^2 + 3x - 4}{3x - 4x^2 + 1} = -\frac{(x - 1)(x + 4)}{(x - 1)(4x + 1)}$$

$$\therefore x \neq 1 \Rightarrow y \neq -1$$

$$\Rightarrow a = 1 \text{ is not possible.}$$

$$\text{for } a = 7,$$

$$y = \frac{7x^2 + 3x - 4}{3x - 4x^2 + 7} = \frac{(7x - 4)(x + 1)}{(7 - 4x)(x + 1)} \quad \therefore x \neq -1 \Rightarrow y \neq -1$$

So y will assume all real values for some real values of x .

$$\text{So } a \in (1, 7)$$

Illustration 43 : Find the values of m so that the inequality $\left| \frac{x^2 + mx + 1}{x^2 + x + 1} \right| < 3$ holds for all $x \in \mathbb{R}$.

Solution :

We know that

$$|a| < b \Rightarrow -b < a < b \quad (\text{for } b > 0)$$

$$\text{Hence } \left| \frac{x^2 + mx + 1}{x^2 + x + 1} \right| < 3 \Rightarrow -3 < \frac{x^2 + mx + 1}{x^2 + x + 1} < 3$$

Case I :

$$\frac{x^2 + mx + 1}{x^2 + x + 1} < 3$$

$$\Rightarrow \frac{(x^2 + mx + 1) - 3(x^2 + x + 1)}{x^2 + x + 1} < 0 \Rightarrow \frac{-2x^2 + (m - 3)x - 2}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} < 0$$

Multiplying both sides by denominator, we get :

$$\Rightarrow -2x^2 + (m-3)x - 2 < 0 \text{ (because denominator is always positive)}$$

$$\Rightarrow 2x^2 - (m-3)x + 2 > 0$$

A quadratic expression in x is always positive if coefficient of $x^2 > 0$ and $D < 0$.

$$\Rightarrow (m-3)^2 - 4(2)(2) < 0 \Rightarrow (m-3)^2 - 4^2 < 0 \Rightarrow (m-7)(m+1) < 0 \Rightarrow m \in (-1, 7) \dots(i)$$

Case II :

$$-3 < \frac{x^2 + mx + 1}{x^2 + x + 1}$$

$$\Rightarrow \frac{(x^2 + mx + 1) + 3(x^2 + x + 1)}{x^2 + x + 1} > 0 \Rightarrow 4x^2 + (m+3)x + 4 > 0$$

For this to be true for all $x \in \mathbb{R}$, coefficient of $x^2 > 0$ and $D < 0$

$$\Rightarrow (m+3)^2 - 4(4)(4) < 0$$

$$\Rightarrow (m+3-8)(m+3+8) < 0$$

$$\Rightarrow (m-5)(m+11) < 0$$

$$\Rightarrow m \in (-11, 5) \dots(ii)$$

We will combine (i) and (ii) because both must be satisfied.

$$\Rightarrow \text{The common solution is } m \in (-1, 5).$$

Illustration 44 : Find the values of m for which the expression : $\frac{2x^2 - 5x + 3}{4x - m}$ can take all real values for

$$x \in \mathbb{R} - \left\{ \frac{m}{4} \right\}$$

Solution : Let $\frac{2x^2 - 5x + 3}{4x - m} = y \Rightarrow 2x^2 - (4y + 5)x + 3 + my = 0$

$$\Rightarrow \text{As } x \in \mathbb{R}, \text{ discriminant } \geq 0$$

$$\Rightarrow (4y + 5)^2 - 8(3 + my) \geq 0$$

$$\Rightarrow 16y^2 + (40 - 8m)y + 1 \geq 0 \dots(i)$$

A quadratic in y is non-negative for all values of y if coefficient of y^2 is positive and discriminant ≤ 0 .

$$\Rightarrow (40 - 8m)^2 - 4(16)(1) \leq 0 \Rightarrow (5 - m)^2 - 1 \leq 0$$

$$\Rightarrow (m - 5 - 1)(m - 5 + 1) \leq 0 \Rightarrow (m - 6)(m - 4) \leq 0$$

$$\Rightarrow m \in [4, 6] \text{ but } m = 4, 6 \text{ will be rejected as shown in Illustration 41}$$

So for the given expression to take all real values, m should take values : $m \in (4, 6)$.

Illustrations45: If $(y^2 - 5y + 3)(x^2 + x + 1) < 2x$ for all $x \in \mathbb{R}$, then find the interval in which y lies.

Solution : $(y^2 - 5y + 3)(x^2 + x + 1) < 2x, \forall x \in \mathbb{R}$

$$\Rightarrow y^2 - 5y + 3 < \frac{2x}{x^2 + x + 1}$$

$$\text{Let } \frac{2x}{x^2 + x + 1} = P$$

$$\Rightarrow px^2 + (p - 2)x + p = 0$$

$$(1) \text{ Since } x \text{ is real, } (p - 2)^2 - 4p^2 \geq 0$$

$$\Rightarrow -2 \leq p \leq \frac{2}{3}$$

$$(2) \text{ The minimum value of } 2x/(x^2 + x + 1) \text{ is } -2.$$

$$\text{So, } y^2 - 5y + 3 < -2 \Rightarrow y^2 - 5y + 5 < 0$$

$$\Rightarrow y \in \left(\frac{5 - \sqrt{5}}{2}, \frac{5 + \sqrt{5}}{2} \right)$$

Do yourself - 9 :

1. If x is real prove that the expression $\frac{8x - 4}{x^2 + 2x - 1}$ where x is real cannot have values between 2 and 4, in its range.
2. Find the range of $\frac{x^2 + 2x + 1}{x^2 + 2x + 7}$, where x is real
3. If x is real, then prove that $\frac{x^2 - x + 1}{x^2 + x + 1}$ lies from $\frac{1}{3}$ to 3.
4. If x be real, prove that $\frac{x}{x^2 - 5x + 9}$ must lie between 1 and $-\frac{1}{11}$ (both inclusive).
5. If x be real, prove that $\frac{x^2 + 34x - 71}{x^2 + 2x - 7}$ can have no value between 5 and 9.
6. Find the greatest value of $\frac{x + 2}{2x^2 + 3x + 6}$ for real values of x .

12. LOCATION OF ROOTS :

This article deals with an elegant approach of solving problems on quadratic equations when the roots are located / specified on the number line with variety of constraints :

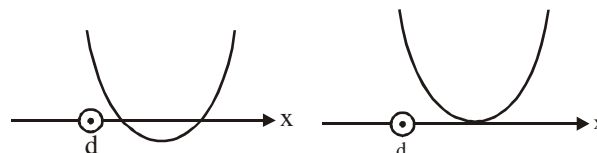
Consider the quadratic equation $ax^2 + bx + c = 0$ with $a > 0$ and let $f(x) = ax^2 + bx + c$

Type-1 :

Both roots of the quadratic equation are greater than a specific number (say d).

The necessary and sufficient condition for this are :

$$(i) D \geq 0; \quad (ii) f(d) > 0; \quad (iii) -\frac{b}{2a} > d$$



Note : When both roots of the quadratic equation are less than a specific number d than the necessary and sufficient condition will be :

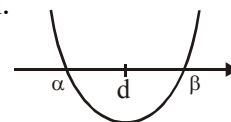
$$(i) D \geq 0; \quad (ii) f(d) > 0; \quad (iii) -\frac{b}{2a} < d$$

Type-2 :

Both roots lie on either side of a fixed number say (d). Alternatively one root is greater than ' d ' and other root less than ' d ' or ' d ' lies between the roots of the given equation.

The necessary and sufficient condition for this are : $f(d) < 0$

Note : Consideration of discriminant is not needed.

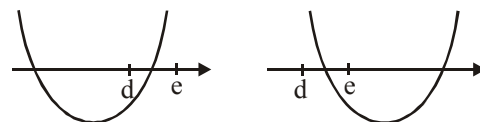


Type-3 :

Exactly one root lies in the interval (d, e).

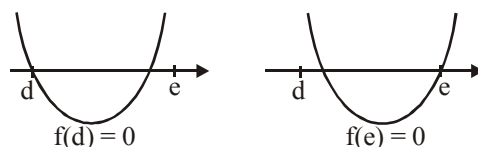
The necessary and sufficient condition for this are :

$$f(d) \cdot f(e) < 0$$



Note : The extremes of the intervals found by given conditions give ' d ' or ' e ' as the root of the equation.

Hence in this case also check for end points.



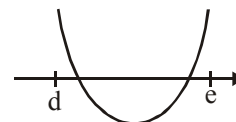
Type-4 :

When both roots are confined between the number d and e ($d < e$).

The necessary and sufficient condition for this are :

$$(i) D \geq 0; \quad (ii) f(d) > 0; \quad (iii) f(e) > 0$$

$$(iv) d < -\frac{b}{2a} < e$$



Type-5 :

One root is greater than e and the other roots is less than d ($d < e$).

The necessary and sufficient condition for this are : $f(d) < 0$ and $f(e) < 0$

Note : If $a < 0$ in the quadratic equation $ax^2 + bx + c = 0$ then we divide the whole equation by ' a '.

Now assume $x^2 + \frac{b}{a}x + \frac{c}{a}$ as $f(x)$. This makes the coefficient of x^2 positive and hence above cases are applicable.

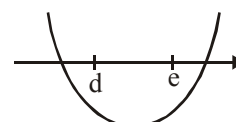


Illustration 46 : Let the quadratic equation $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$ has real roots then find the conditions such that

- (a) roots are equal in magnitude but opposite in sign
- (b) one root is zero other is $-b/a$
- (c) roots are reciprocal to each other
- (d) roots are of opposite signs
- (e) both roots are negative.
- (f) both roots are positive.
- (g) Greater root in magnitude is negative.
- (h) Greater root in magnitude is positive.
- (i) one root is 1 and second root is c/a or $(-b-a)/a$.

Solution

- (a) $b = 0, D > 0$ (b) $c = 0$ (c) $a = c, D \geq 0$
- (d) $a \cdot c < 0$ (e) $D \geq 0, \frac{b}{a} > 0, \frac{c}{a} > 0$ (f) $D \geq 0, \frac{b}{a} < 0, \frac{c}{a} > 0$
- (g) $\frac{b}{a} > 0, D \geq 0$ (h) $\frac{b}{a} < 0, D \geq 0$ (i) $a + b + c = 0$ and $a \neq 0$

Illustration 47 : Find the values of the parameter 'a' for which the roots of the quadratic equation $x^2 + 2(a-1)x + a + 5 = 0$ are

- (i) real and distinct (ii) equal
- (iii) opposite in sign (iv) equal in magnitude but opposite in sign
- (v) positive (vi) negative
- (vii) greater than 3 (viii) smaller than 3
- (ix) such that both the roots lie in the interval (1, 3)

Solution :

Let $f(x) = x^2 + 2(a-1)x + a + 5 = Ax^2 + Bx + C$ (say)

$\Rightarrow A = 1, B = 2(a-1), C = a + 5.$

Also $D = B^2 - 4AC = 4(a-1)^2 - 4(a+5) = 4(a+1)(a-4)$

(i) $D > 0$

$$\Rightarrow (a+1)(a-4) > 0 \Rightarrow a \in (-\infty, -1) \cup (4, \infty).$$

(ii) $D = 0$

$$\Rightarrow (a+1)(a-4) = 0 \Rightarrow a = -1, 4.$$

(iii) This means that 0 lies between the roots of the given equation.

$$\Rightarrow f(0) < 0$$

$$\Rightarrow a + 5 < 0 \Rightarrow a < -5 \Rightarrow a \in (-\infty, -5).$$

(iv) This means that the sum of the roots is zero

$$\Rightarrow -2(a-1) = 0 \text{ and } D > 0 \text{ i.e. } a \in (-\infty, -1) \cup (4, \infty) \Rightarrow a = 1$$

which does not belong to $(-\infty, -1) \cup (4, \infty)$

$$\Rightarrow a \in \phi.$$

(v) This implies that both the roots are greater than zero

$$\Rightarrow -\frac{B}{A} > 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) > 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a < 1, -5 < a, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-5, -1].$$

(vi) This implies that both the roots are less than zero

$$\Rightarrow -\frac{B}{A} < 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) < 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a > 1, a > -5, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in [4, \infty).$$

(vii) In this case

$$-\frac{B}{2a} > 3, A.f(3) > 0 \text{ and } D \geq 0.$$

$$\Rightarrow -(a-1) > 3, 7a+8 > 0 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a < -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

Since no value of 'a' can satisfy these conditions simultaneously, there can be no value of a for which both the roots will be greater than 3.

(viii) In this case

$$-\frac{B}{2a} < 3, A.f(3) > 0 \text{ and } D \geq 0.$$

$$\Rightarrow a > -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-8/7, -1] \cup [4, \infty)$$

(ix) In this case

$$1 < -\frac{B}{2A} < 3, A.f(1) > 0, A.f(3) > 0, D \geq 0.$$

$$\Rightarrow 1 < -1(a-1) < 3, 3a+4 > 0, 7a+8 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow -2 < a < 0, a > -4/3, a > -8/7, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in \left(-\frac{8}{7}, -1\right]$$

Illustration 48 : Find value of k for which one root of equation $x^2 - (k+1)x + k^2 + k - 8 = 0$ exceeds 2 & other is less than 2.

Solution : $4 - 2(k+1) + k^2 + k - 8 < 0 \Rightarrow k^2 - k - 6 < 0$

$$(k-3)(k+2) < 0 \Rightarrow -2 < k < 3$$

Taking intersection, $k \in (-2, 3)$.

Illustration 49 : Find all possible values of a for which exactly one root of $x^2 - (a+1)x + 2a = 0$ lies in interval (0, 3).

Solution : $f(0) \cdot f(3) < 0$

$$\Rightarrow 2a(9 - 3(a+1) + 2a) < 0 \Rightarrow 2a(-a+6) < 0$$

$$\Rightarrow a(a-6) > 0 \Rightarrow a < 0 \text{ or } a > 6$$

Checking the extremes.

If $a = 0$, $x^2 - x = 0$

$$x = 0, 1$$

$$1 \in (0, 3)$$

If $a = 6$, $x^2 - 7x + 12 = 0$

$$x = 3, 4 \quad \text{But } 4 \notin (0, 3)$$

Hence solution set is

$$a \in (-\infty, 0] \cup (6, \infty)$$

Illustration 50 : Let $x^2 - (m-3)x + m = 0$ ($m \in \mathbb{R}$) be a quadratic equation. Find the value of m for which the roots of the equation are

- (i) real and distinct
- (ii) equal
- (iii) not real
- (iv) opposite in sign
- (v) equal in magnitude but opposite in sign
- (vi) positive
- (vii) negative
- (viii) such that at least one is positive
- (ix) one root is smaller than 2 and the other root is greater than 2
- (x) both the roots are greater than 2
- (xi) both the roots are smaller than 2
- (xii) exactly one root lies in the interval $(1, 2)$
- (xiii) both the roots lie in the interval $(1, 2)$
- (xiv) such that at least one root lie in the interval $(1, 2)$
- (xv) one root is greater than 2 and the other root is smaller than 1

Solution :

Let $f(x) = x^2 - (m-3)x + m$

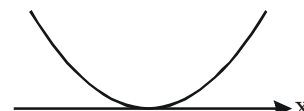
- (i) Both the roots are real and distinct $\Rightarrow D > 0$

$$\Rightarrow (m-3)^2 - 4m > 0 \Rightarrow m^2 - 10m + 9 > 0$$

$$\Rightarrow (m-1)(m-9) > 0 \Rightarrow m \in (-\infty, 1) \cup (9, \infty)$$

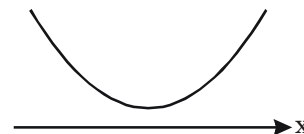


- (ii) Both the roots are equal $\Rightarrow D = 0 \Rightarrow m = 9$ or $m = 1$



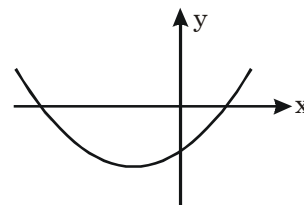
- (iii) Both the roots are imaginary $\Rightarrow D < 0$

$$(m-1)(m-9) < 0 \Rightarrow m \in (1, 9)$$



- (iv) Both the roots are opposite in sign

$$\Rightarrow f(0) < 0 \Rightarrow m < 0 \Rightarrow m \in (-\infty, 0)$$

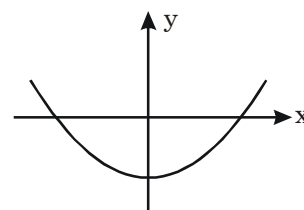


- (v) Roots are equal in magnitude but opposite in sign

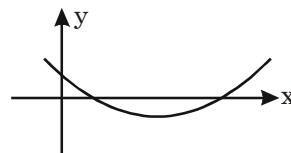
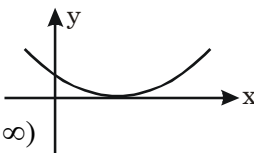
$$\Rightarrow \text{sum of roots is zero as well as } D \geq 0$$

$$\Rightarrow m \in (-\infty, 1] \cup [9, \infty) \text{ and } m-3 = 0 \text{ i.e., } m = 3$$

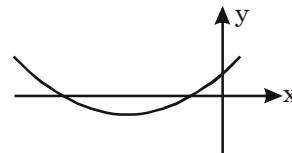
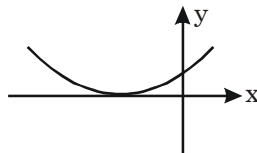
$$\Rightarrow \text{no such } m \text{ exists}$$



- (vi) Both the roots are positive $\Rightarrow D \geq 0$,
both sum and product of roots are positive
 $\Rightarrow m - 3 > 0, m > 0$ and $m \in (-\infty, 1] \cup [9, \infty)$
 $\Rightarrow m \in [9, \infty)$

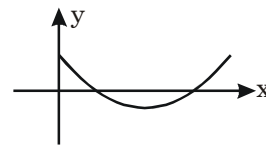
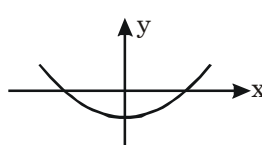
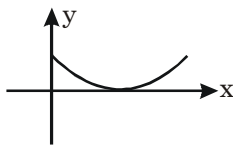


- (vii) Both the roots are negative



$\Rightarrow D \geq 0$, sum of roots is negative but product of roots is positive.
 $\Rightarrow m - 3 < 0, m > 0, m \in (-\infty, 1] \cup [9, \infty) \Rightarrow m \in (0, 1]$

- (viii) at least one root is positive

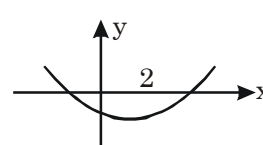
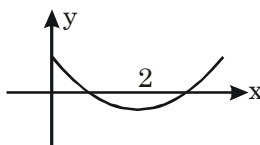


$\Rightarrow$ either one root is positive or both the roots are positive

$\Rightarrow$ union of (iv) and (vi) with $m = 0$

(i.e., one root is zero and in this case other root becomes negative) $\Rightarrow m \in (-\infty, 0) \cup [9, \infty)$

- (ix) one root is smaller than 2 and



other root is greater than 2 $\Rightarrow 2$ lies between the roots

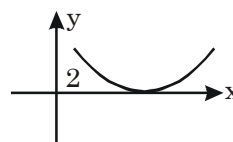
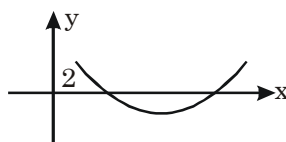
$\Rightarrow f(2) < 0 \Rightarrow 4 - 2(m - 3) + m < 0 \Rightarrow m > 10$

- (x) Both roots are greater than 2

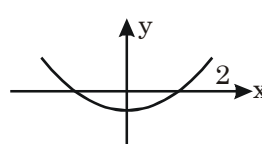
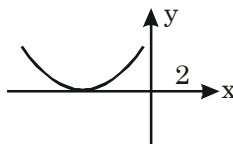
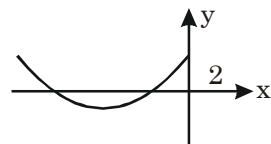
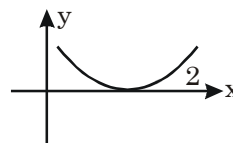
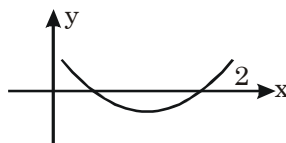
$\Rightarrow f(2) > 0, D \geq 0, \frac{-b}{2a} > 2$

$\Rightarrow m < 10$ and $m \in (-\infty, 1] \cup [9, \infty)$ and $m - 3 > 4$

$\Rightarrow m \in [9, 10)$

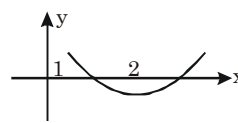
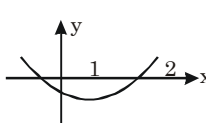
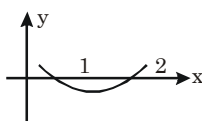


- (xi) Both the roots are smaller than 2



$\Rightarrow f(2) > 0, D \geq 0, \frac{-b}{2a} < 2 \Rightarrow m \in (-\infty, 1]$

- (xii) Exactly one root lies between (1, 2)



$\Rightarrow f(1) \cdot f(2) < 0 \Rightarrow 4(10 - m) < 0 \Rightarrow m \in (10, \infty)$

(xiii) Both roots lie in the interval (1, 2). Then

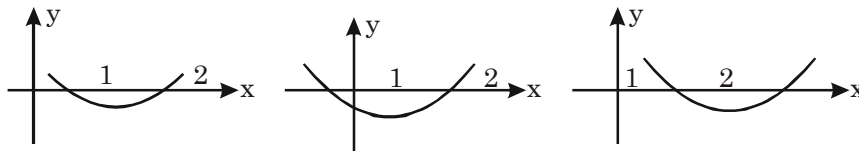
$$f(1) > 0, 1 - m + 3 + m > 0 \Rightarrow 4 > 0$$

$$f(2) > 0, 4 - 2m + 6 + m > 0 \Rightarrow m < 10$$

$$1 < \frac{m-3}{2} < 2 \Rightarrow 5 < m < 7$$

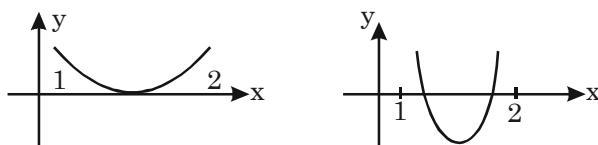
$$D \geq 0, m \in (-\infty, 1] \cup [9, \infty) \Rightarrow \text{no solution}$$

(xiv) **Case I :** Exactly one root lies in (1, 2)



$$f(1) \cdot f(2) < 0 \Rightarrow m > 10$$

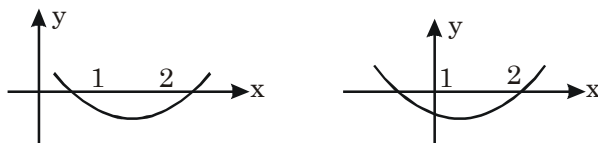
Case II : Both roots lie in (1, 2)



From (xiii) $m \in \phi$

At least one root lies in (1, 2) $\Rightarrow m \in (10, \infty)$

(xv) For one root greater than 2 and other is smaller than 1, conditions are



$$f(1) < 0 \quad \dots(1)$$

$$\text{and } f(2) < 0 \quad \dots(2)$$

From (1), $f(1) < 0$, but $f(1) = 4$ which is not possible

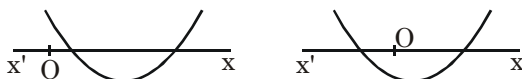
Thus no such 'm' exists.

Illustration 51 : Find the value(s) of 'a' for which $ax^2 + (a-3)x + 1 < 0$ for at least one positive x.

Solution : Let $f(x) = ax^2 + (a-3)x + 1$

Case I :

If $a > 0$, then $f(x)$ will be negative only for those values of x, which lie between the roots. From the graph we can see that, $f(x)$ will be less than zero for at least one positive real x, when $f(x) = 0$ has distinct roots and at least one of these roots is positive real root.



For this $D > 0$, i.e., $(a-3)^2 - 4a > 0$

$$\Rightarrow a < 1 \text{ or } a > 9 \quad \dots(1)$$

It is easy to see that for figure 1, $a < 3$ and figure 2 is not possible as $f(0) = 1$.

Hence for case-I $0 < a < 1$.

Case II :

If $a < 0$, then $f(x)$ will certainly be negative for infinitely many positive x . Thus $a \in (-\infty, 0)$.

Case III :

if $a = 0$, $f(x) = -3x + 1 \Rightarrow f(x) < 0 \forall x > 1/3$

Hence the required set of values of 'a' is $(-\infty, 1)$

Illustration 52 : Find the values of 'a' for which $4^t - (a - 4)2^t + \frac{9}{4}a < 0 \forall t \in (1, 2)$.

Solution : Let $2^t = x$ and $f(x) = x^2 - (a - 4)x + \frac{9}{4}a$

We want

$$f(x) < 0 \forall x \in (2^1, 2^2) \text{ i.e., } \forall x \in (2, 4)$$

Since we want $f(x) < 0 \forall x \in (2, 4)$, one of the roots of $f(x)$ should be less than or equal to 2 and the other must be greater than or equal to 4

i.e., $f(2) \leq 0$ and $f(4) \leq 0 \Rightarrow a \leq -48$ and $a \geq 128/7$, which is not possible.

Hence no such 'a' exists.

Illustration 53 : Find the value(s) of 'a' for which the inequality $\tan^2 x + (a + 1)\tan x - (a - 3) < 0$, is true for at least one $x \in \left(0, \frac{\pi}{2}\right)$.

Solution : The required condition will be satisfied if

(i) The quadratic expression (quadratic in $\tan x$)

$f(x) = \tan^2 x + (a + 1)\tan x - (a - 3)$ has positive discriminant, and

(ii) At least one root of $f(x) = 0$ is positive, as $\tan x > 0, \forall x \in (0, \pi/2)$

For (i) Discriminant $> 0 \Rightarrow (a + 1)^2 + 4(a - 3) > 0$

$$\Rightarrow a > 2\sqrt{5} - 3 \text{ or } a < -(2\sqrt{5} + 3)$$

For (ii), we first find the condition, that both the roots of $t^2 + (a + 1)t - (a - 3) = 0$

($t = \tan x$) are non-positive for which

Sum of roots < 0 product of roots ≥ 0

$$\Rightarrow -(a + 1) < 0 \text{ and } -(a - 3) \geq 0 \Rightarrow -1 < a \leq 3$$

Condition (ii) will be fulfilled if $a \leq -1$ or $a > 3$... (2)

Required values of a is given by intersection of (1) and (2)

$$\text{Hence } a \in (-\infty, -(2\sqrt{5} + 3)) \cup (3, \infty)$$

Do yourself - 10 :

1. If α, β are roots of $7x^2 + 9x - 2 = 0$, find their position with respect to following ($\alpha < \beta$) :
(a) -3 (b) 0 (c) 1
2. If $a > 1$, roots of the equation $(1 - a)x^2 + 3ax - 1 = 0$ are -
(a) one positive one negative (b) both negative
(c) both positive (d) both non-real
3. Find the set of value of a for which the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3.
4. If α, β are the roots of $x^2 - 3x + a = 0$, $a \in \mathbb{R}$ and $\alpha < 1 < \beta$, then find the values of a .
5. If α, β are roots of $4x^2 - 16x + \lambda = 0$, $\lambda \in \mathbb{R}$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then find the range of λ .
6. For what values of a does the equation $(2 - x)(x + 1) = a$ possess real and positive roots ?

13. GENERAL QUADRATIC EXPRESSION IN TWO VARIABLES :

$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factors if ;

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad \text{OR} \quad \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Illustration 54 : If $x^2 + 2xy + 2x + my - 3$ have two linear factor then m is equal to -

- (A) 6, 2 (B) -6, 2 (C) 6, -2 (D) -6, -2

Solution : Here $a=1, h=1, b=0, g=1, f=m/2, c=-3$

$$\text{So } \Delta = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & m/2 \\ 1 & m/2 & -3 \end{vmatrix} = 0$$

$$\Rightarrow -\frac{m^2}{4} - (-3 - m/2) + m/2 = 0 \quad \Rightarrow -\frac{m^2}{4} + m + 3 = 0$$

$$\Rightarrow m^2 - 4m - 12 = 0 \quad \Rightarrow m = -2, 6$$

Ans. (C)

Do yourself - 11 :

1. Find the value of k for which the expression $x^2 + 2xy + ky^2 + 2x + k = 0$ can be resolved into two linear factors.
2. For what values of m will the expression $y^2 + 2xy + 2x + my - 3$ be capable of resolution into two rational factors ?
3. Find the value of m which will make $2x^2 + mxy + 3y^2 - 5y - 2$ equivalent to the product of two linear factors.
4. Find the condition that the expression $\ell x^2 + mxy + ny^2$, $\ell'x^2 + m'xy + n'y^2$ may have a common linear factor.
5. Find the condition that the expression $ax^2 + 2hxy + by^2$, $a'x^2 + 2h'xy + b'y^2$ may be respectively divisible by factors of the form $y - mx$, $my + x$.
6. If x and y are two real quantities connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then show that x will lie between 3 and 6. and y between 1 and 10.

14. THEORY OF EQUATIONS :

Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are roots of the equation, $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$, where $a_0, a_1, \dots, a_n$ are constants and $a_0 \neq 0$.

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_n)$$

$$\therefore a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = a_0(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$$

Comparing the coefficients of like powers of x , we get

$$\sum \alpha_i = -\frac{a_1}{a_0} = S_1 \text{ (say)}$$

$$\text{or } S_1 = -\frac{\text{coefficient of } x^{n-1}}{\text{coefficient of } x^n}$$

$$S_2 = \sum_{i \neq j} \alpha_i \alpha_j = (-1)^2 \frac{a_2}{a_0}$$

$$S_3 = \sum_{i \neq j \neq k} \alpha_i \alpha_j \alpha_k = (-1)^3 \frac{a_3}{a_0}$$

$\vdots$

$$S_n = \alpha_1 \alpha_2 \dots \alpha_n = (-1)^n \frac{a_n}{a_0} = (-1)^n \frac{\text{constant term}}{\text{coefficient of } x^n}$$

where S_k denotes the sum of the product of root taken k at a time.

Quadratic equation : If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$,

$$\text{then } \alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Cubic equation : If α, β, γ are roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$, then

$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} \text{ and } \alpha\beta\gamma = -\frac{d}{a}$$

Note :

- (i) If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$ is a factor of $f(x)$ and conversely.
- (ii) Every equation of n th degree ($n \geq 1$) has exactly n roots & if the equation has more than n roots, it is an identity.
- (iii) If the coefficients of the equation $f(x) = 0$ are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. **imaginary roots occur in conjugate pairs.**
- (iv) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in \mathbb{Q}$ & β is not a perfect square.
- (v) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x) = 0$ must have at least one real root between 'a' and 'b'.
- (vi) Every equation $f(x) = 0$ of degree odd and leading coefficient positive has at least one real root of a sign opposite to that of its non zero constant term.

Illustration 55: If two roots are equal, find the roots of $4x^3 + 20x^2 - 23x + 6 = 0$.

Solution : Let roots be α, α and β

$$\therefore \alpha + \alpha + \beta = -\frac{20}{4} \Rightarrow 2\alpha + \beta = -5 \quad \dots\dots\dots (i)$$

$$\therefore \alpha \cdot \alpha + \alpha\beta + \alpha\beta = -\frac{23}{4} \Rightarrow \alpha^2 + 2\alpha\beta = -\frac{23}{4} \text{ \& } \alpha^2\beta = -\frac{6}{4}$$

from equation (i)

$$\alpha^2 + 2\alpha(-5 - 2\alpha) = -\frac{23}{4} \Rightarrow \alpha^2 - 10\alpha - 4\alpha^2 = -\frac{23}{4} \Rightarrow 12\alpha^2 + 40\alpha - 23 = 0$$

$$\therefore \alpha = 1/2, -\frac{23}{6}$$

$$\text{when } \alpha = \frac{1}{2}$$

$$\alpha^2\beta = \frac{1}{4}(-5 - 1) = -\frac{3}{2}$$

$$\text{when } \alpha = -\frac{23}{6} \Rightarrow \alpha^2\beta = \frac{23 \times 23}{36} \left(-5 - 2 \times \left(-\frac{23}{6} \right) \right) \neq -\frac{3}{2} \Rightarrow \alpha = \frac{1}{2} \quad \beta = -6$$

$$\text{Hence roots of equation} = \frac{1}{2}, \frac{1}{2}, -6 \quad \text{Ans.}$$

Illustration 56 : If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$, find :

$$(i) \quad \sum \alpha^3 \qquad (ii) \quad \alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta)$$

Solution : We know that $\alpha + \beta + \gamma = p$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

$$\alpha\beta\gamma = r$$

$$(i) \quad \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma + (\alpha + \beta + \gamma)\{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)\} \\ = 3r + p\{p^2 - 3q\} = 3r + p^3 - 3pq$$

$$(ii) \quad \alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta) = \alpha^2(p - \alpha) + \beta^2(p - \beta) + \gamma^2(p - \gamma) \\ = p(\alpha^2 + \beta^2 + \gamma^2) - 3r - p^3 + 3pq = p(p^2 - 2q) - 3r - p^3 + 3pq = pq - 3r$$

Illustration 57 : If $b^2 < 2ac$ and $a, b, c, d \in \mathbb{R}$, then prove that $ax^3 + bx^2 + cx + d = 0$ has exactly one real root.

Solution : Let α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$

$$\text{Then } \alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 < 0$, which is not possible if all α, β, γ are real. So atleast one root is non-real, but complex roots occurs in pair. Hence given cubic equation has two non-real and one real roots.

Illustration 58 : If the sum of two roots of the equation $x^3 - px^2 + qx - r = 0$ is zero, then prove that $pq = r$.

Solution : Let the roots of the given equation by α, β, γ such that $\alpha + \beta = 0$, Then,

$$\alpha + \beta + \gamma = -\frac{(-p)}{1} \Rightarrow \alpha + \beta + \gamma = p \Rightarrow \gamma = p \quad [\because \alpha + \beta = 0]$$

But γ is a root of the given equation. Therefore,

$$\gamma^3 - p\gamma^2 + q\gamma - r = 0 \Rightarrow p^3 - p^3 + qp - r = 0 \Rightarrow pq = r$$

Illustration 59 : Find the condition that the roots of the equation $x^3 - px^2 + qx - r = 0$ may be in A.P.

Solution : Let the roots of the given equation be $a - d, a, a + d$.

Then,

$$(a - d) + a + (a + d) = \frac{-(-p)}{1} \Rightarrow a = p/3$$

Since a is a root of the given equation. Therefore,

$$a^3 - pa^2 + qa - r = 0 \Rightarrow \frac{p^3}{27} - \frac{p^3}{9} + \frac{qp}{3} - r = 0 \Rightarrow 2p^3 - 9pq + 27r = 0$$

This is the required condition.

Illustration 60 : If $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d$, then find the real root of

$$ax^3 + bx^2 + cx + d = 0.$$

Solution : $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d$. So roots of $x^2 + x + 1 = 0$ are also the roots of the equation $ax^3 + bx^2 + cx + d = 0$.

Let third roots of $ax^3 + bx^2 + cx + d$ is α .

Now we can write,

$$ax^3 + bx^2 + cx + d = a(x^2 + x + 1)(x - \alpha)$$

Comparing constant term on both sides, we get : $d = -a\alpha$

Hence, the real root of $ax^3 + bx^2 + cx + d = 0$ is $-\frac{d}{a}$.

Note : Roots of $x^2 + x + 1 = 0$ are complex

Do yourself - 12 :

- Let α, β be two of the roots of the equation $x^3 - px^2 + qx - r = 0$. If $\alpha + \beta = 0$, then show that $pq = r$
- If two roots of $x^3 + 3x^2 - 9x + c = 0$ are equal, then find the value of c .
- If α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$, then find the value of

$$(a) \sum \alpha^2 \quad (b) \sum \frac{1}{\alpha} \quad (c) \sum \alpha^2(\beta + \gamma)$$

EXERCISE (O-1)

Straight Objective Type

1. If the roots of the equation $x^2 - 5x + 16 = 0$ are α, β and the roots of the equation $x^2 + px + q = 0$ are $(\alpha^2 + \beta^2)$ and $\frac{\alpha\beta}{2}$, then -
- (A) $p = 1$ and $q = 56$ (B) $p = 1$ and $q = -56$
(C) $p = -1$ and $q = 56$ (D) $p = -1$ and $q = -56$

QE0001

2. If the roots of the equation $x^2 + px + q = 0$ are 8 and 2 and the roots of $x^2 + rx + s = 0$ are 3 and 3, then roots of $x^2 + px + s = 0$ are
- (A) -1, -9 (B) 1, 9 (C) 8, 3 (D) None

QE0002

- 3.** If α and β be the roots of the equation $(x - a)(x - b) = c$ and $c \neq 0$, then roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are -
 (A) a and c (B) b and c (C) a and b (D) $a + b$ and $b + c$

QE0003

4. If $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$ then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is -
- (A) $19/3$ (B) $25/3$ (C) $-19/3$ (D) none of these

QE0004

5. The value of a for which one roots of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is
- (A) $-2/3$ (B) $1/3$ (C) $-1/3$ (D) $2/3$

QE0005

- 6.** The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$, is-
- (A) 4 (B) 1 (C) 3

QE0006

7. If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its roots are-
 (A) 0, -1 (B) -1, 1 (C) 0, 1 (D) -1, 2

QE0007

8. If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals -
- (A) 1 (B) 2 (C) 3 (D) -2

QE0008

9. If $x^2 - A(x + 1) + C = 0$ has roots x_1 & x_2 , then the value of $x_1^2 + x_2^2 + (2 + A)x_1x_2$, is-
- (A) AC (B) $A^2 + AC$ (C) $A^2 - AC$ (D) $-AC$

QE0009

- 10.** The quadratic $x^2 + ax + b + 1 = 0$ has roots which are positive integers, then $(a^2 + b^2)$ can be equal to-
- (A) 50 (B) 17 (C) 29 (D) 53

QE0010

11. **Statement-1** : If sum of the roots of quadratic equation $2x^2 + bx + c = 0$ is equal to sum of the squares of the roots, then $b^2 + 2b = 4c$.

Statement-2 : If one root of quadratic equation $x^2 + px + q = 0$, is $4 + \sqrt{3}$, then other root is $4 - \sqrt{3}$.

- (A) Statement-1 is true and Statement-2 is false.
 (B) Statement-1 and Statement-2 both are true.
 (C) Statement-1 is false and Statement-2 is true.
 (D) Statement-1 and Statement-2 both are false.

QE0011

12. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of 'q' is-

- (A) 3 (B) 12 (C) 49/4 (D) 4

QE0012

13. The sum of the values of m for which the quadratic polynomial $P(x) = x^2 + (m + 5)x + (5m + 1)$ is a perfect square of linear in 'x' ($m \in \mathbb{R}$) is

- (A) 3 (B) 7 (C) 8 (D) 10

QE0013

14. If equations $x^2 - 5x + 5 = 0$ and $x^3 + ax^2 + bx + 5 = 0$ have common root, then value of $a + b$ ($a, b \in \mathbb{Q}$) is -

- (A) 4 (B) -4 (C) 0 (D) can't find

QE0014

15. If the equation $ax^2 + bx + c = 0$ has distinct real roots, both negative, then-

- (A) a,b,c must be of same sign
 (B) a,b must be of opposite sign
 (C) a,c must be of opposite sign
 (D) a,b must be of same sign and opposite to sign of c

QE0015

16. If $P(x) = x^2 - (2 - p)x + p - 2$ assumes both positive and negative value, then the complete set of values of 'p' is -

- (A) $(-\infty, 2)$ (B) $(6, \infty)$ (C) $(2, 6)$ (D) $(-\infty, 2) \cup (6, \infty)$

QE0016

17. If value of a for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assume the least value is -

- (A) 2 (B) 3 (C) 0 (D) 1

QE0017

18. If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in \mathbb{R}$, then

- (A) $-5 < a < 2$ (B) $a < -5$ (C) $a > 5$ (D) $2 < a < 5$

QE0018

28. Let r_1, r_2, r_3 be roots of equation $x^3 - 2x^2 + 4x + 5074 = 0$, then the value of $(r_1 + 2)(r_2 + 2)(r_3 + 2)$ is
(A) 5050 (B) -5050 (C) -5066 (D) -5068

QE0028

29. Let a, b, c are roots of equation $x^3 + 8x + 1 = 0$, then the value of

$$\frac{bc}{(8b+1)(8c+1)} + \frac{ac}{(8a+1)(8c+1)} + \frac{ab}{(8a+1)(8b+1)} \text{ is equal to}$$

- (A) 0 (B) -8 (C) -16 (D) 16

QE0029

30. Let $f(x) = x^3 + x + 1$ and $P(x)$ be a cubic polynomial such that $P(0) = -1$ and the roots of $P(x) = 0$ are the squares of the roots of $f(x) = 0$, then value of $P(9)$ is -
(A) 98 (B) 899 (C) 80 (D) 898

QE0030

EXERCISE (O-2)

Multiple Correct Answer Type

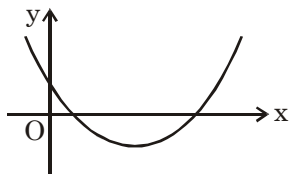
1. If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009}$ is greater than or equal to
 (A) -2 (B) -1 (C) 1 (D) 2
QE0031
2. The real value(s) of x satisfying the equation $2^{(x-1)(x^2+5x-50)} = 1$ is/are :
 (A) 1 (B) -5 (C) -10 (D) 5
QE0032
3. The value(s) of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is/are :
 (A) $\frac{1}{2}$ (B) 0 (C) 3 (D) 1
QE0033
4. The integral value(s) of α such that equation $x^2 - \alpha x + \alpha + 1 = 0$ has integral roots, is/are -
 (A) -1 (B) 4 (C) 3 (D) 5
QE0034
5. For the equation, $3x^2 + px + 3 = 0$, if one of the roots is square of the other, then the value(s) of p is/are -
 (A) $1/3$ (B) 1 (C) 3 (D) -6
QE0035
6. Let $p, q \in \mathbb{Q}$. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then :
 (A) $q^2 + 4p + 14 = 0$ (B) $p^2 - 4q - 12 = 0$ (C) $p = -4, q = 1$ (D) $p = 1, q = -4$
QE0038
7. If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from -1, then the value(s) of 'b' is/are :-
 (A) 0 (B) $\sqrt{3}i$ (C) $-\sqrt{3}i$ (D) 2
QE0041
8. Let $p(x)$ be a quadratic polynomial such that $p(0) = 1$. If $p(x)$ leaves remainder 4 when divided by $x - 1$ and it leaves remainder 6 when divided by $x + 1$; then :
 (A) $p(2) = 19$ (B) $p(-2) = 19$ (C) $p(-2) = 11$ (D) $p(2) = 15$
QE0042
9. If α & β ($\alpha < \beta$), are the roots of the equation, $x^2 + bx + c = 0$, where $c < 0 < b$, then -
 (A) $\alpha < 0$ (B) $\beta < |\alpha|$ (C) $\beta > 0$ (D) $\beta > |\alpha|$
QE0046

10. If $b > a$ and α, β ($\alpha < \beta$) are the roots of the equation, $(x - a)(x - b) - 1 = 0$, then -

- (A) $\alpha \in [b, \infty)$ (B) $\alpha \in (-\infty, a)$
 (C) $\beta \in (b, \infty)$ (D) $\alpha \in [a, b]$ & $\beta \in [a, b]$

QE0047

11. The graph of $y = ax^2 + bx + c$ is shown. Which of the following hold(s) good ?



- (A) $ab^2c^3 > 0$ (B) $ab^3c^2 < 0$ (C) $ab^3c^5 > 0$ (D) $b^2 > 4ac$

QE0048

12. The value(s) of x which satisfy the equation $|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0$, ($x > 0$), is/are :

- (A) 1 (B) 9 (C) 10 (D) 12

QE0049

13. The subset of the set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is/are -

- (A) $(-\infty, -2) \cup (2, \infty)$ (B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
 (C) $(-\infty, -1) \cup (1, \infty)$ (D) $(\sqrt{2}, \infty)$

QE0050

14. For the equation $|x|^2 + |x| - 6 = 0$, the correct statement (s) is (are) :

- (A) sum of roots is 0 (B) product of roots is -4
 (C) there are 4 real roots (D) there are only 2 real roots

QE0171

15. If α, β are the roots of $ax^2 + bx + c = 0$, and $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0$, (where $h \neq 0$), then

- (A) $\frac{a}{p} = \frac{b}{q} = \frac{c}{r}$ (B) $h = \frac{1}{2} \left(\frac{b}{a} - \frac{q}{p} \right)$ (C) $h = \frac{1}{2} \left(\frac{b}{a} + \frac{q}{p} \right)$ (D) $\frac{b^2 - 4ac}{a^2} = \frac{q^2 - 4pr}{p^2}$

QE0172

16. If a, b are non-zero real numbers and α, β the roots of $x^2 + ax + b = 0$, then

- (A) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$

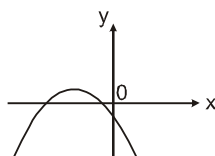
- (B) $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $bx^2 + ax + 1 = 0$

- (C) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $bx^2 + (2b - a^2)x + b = 0$

- (D) $(\alpha - 1), (\beta - 1)$ are the roots of the equation $x^2 + x(a + 2) + 1 + a + b = 0$

QE0173

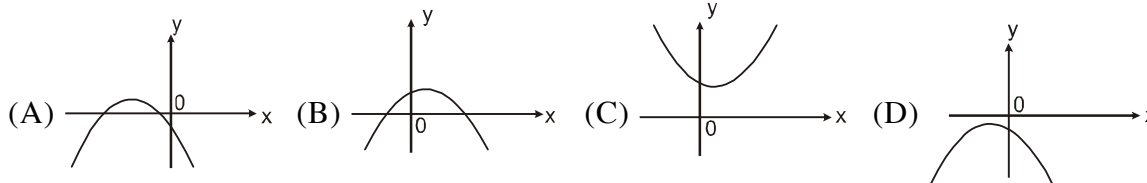
17. The graph of the quadratic polynomial $y = ax^2 + bx + c$ is as shown in the figure. Then :



- (A) $b^2 - 4ac > 0$ (B) $b < 0$ (C) $a > 0$ (D) $c < 0$

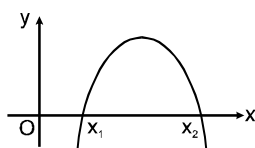
QE0174

18. For which of the following graphs of the quadratic expression $y = ax^2 + bx + c$, the product $a b c$ is negative ?



QE0175

19. The adjoining figure shows the graph of $y = ax^2 + bx + c$. Then



- (A) $a < 0$ (B) $b^2 < 4ac$
(C) $c > 0$ (D) a and b are of opposite sign

QE0176

20. $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d = 0$, then the real root of above equation is
($a, b, c, d \in \mathbb{R}$)

- (A) $-d/a$ (B) d/a (C) $(b - a)/a$ (D) $(a - b)/a$

QE0177

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Consider the expression $y = ax^2 + bx + c$, $a \neq 0$ and $a, b, c \in \mathbb{R}$ then the graph between x , y is always a parabola. If $a > 0$ then the shape of the parabola is concave upward and if $a < 0$ then the shape of the parabola is concave down ward. If $y > 0$ or $y < 0$ then discriminant $D < 0$.

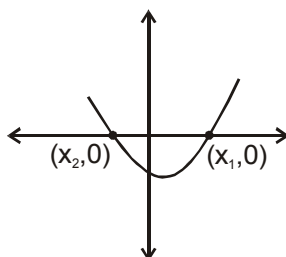
1. Let $x^2 + 2ax + 10 - 3a > 0$ for every real value of x , then –
 (A) $a > 5$ (B) $a < -5$ (C) $-5 < a < 2$ (D) $2 < a < 5$

QE0178

2. The value of $x^2 + 2bx + c$ is positive if –
 (A) $b^2 - 4c > 0$ (B) $b^2 - 4c < 0$ (C) $c^2 < b$ (D) $b^2 < c$

QE0179

3. The diagram show the graph of $y = ax^2 + bx + c$ then –



- (A) $a < 0$ (B) $c < 0$ (C) $b^2 - 4ac < 0$ (D) $b^2 - 4ac = 0$

QE0180

Paragraph for Question 4 to 6

Consider the equation $x^4 - \lambda x^2 + 9 = 0$.

4. If the equation has four real and distinct roots, then the possible value(s) of λ is/are
 (A) 7 (B) -7 (C) 8 (D) 6

QE0181

5. If the equation has no real root, then the possible value(s) of λ is/are
 (A) 6 (B) -7 (C) 0 (D) -6

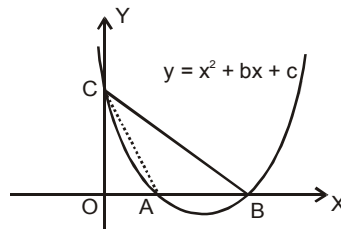
QE0182

6. If the equation has only two real roots, then which of the following value(s) of λ is/are Not possible
 (A) 7 (B) -7 (C) 0 (D) 6

QE0183

Paragraph for Question 7 and 8

In the given figure $\triangle OBC$ is an isosceles right triangle in which AC is a median, then answer the following questions :



7. Sum of roots of $y = 0$ is

QE0184

8. Minimum value of the quadratic expression corresponding to the quadratic equation whose roots are $(\alpha + \beta)$ & $(\alpha - \beta)$, where α, β ($\alpha > \beta$) are roots obtained in previous question occurs at $x =$

QE0185

Matrix Match Type

9.	List- I	List- II
(P)	If $\alpha, \alpha + 4$ are two roots of $x^2 - 8x + k = 0$, then possible value of k is	(1) 2
(Q)	Number of real roots of equation $x^2 - 5 x + 6 = 0$ are 'n', then value of $\frac{n}{2}$ is	(2) 3
(R)	If $3 - i$ is a root of $x^2 + ax + b = 0$ ($a, b \in \mathbb{R}$), then b is	(3) 12
(S)	If both roots of $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then 'k' may be equal to	(4) 10
(A)	$(P) \rightarrow (3), (Q) \rightarrow (1), (R) \rightarrow (4), (S) \rightarrow (2)$	
(B)	$(P) \rightarrow (1), (Q) \rightarrow (2), (R) \rightarrow (3), (S) \rightarrow (4)$	
(C)	$(P) \rightarrow (3), (Q) \rightarrow (2), (R) \rightarrow (3), (S) \rightarrow (2)$	
(D)	$(P) \rightarrow (4), (Q) \rightarrow (3), (R) \rightarrow (2), (S) \rightarrow (1)$	

QE0186

10. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

Match the expressions/statements in Column I with expressions/statements in Column II.

Column-I	Column-II
(A) If $-1 < x < 1$, then $f(x)$ satisfies	(p) $0 < f(x) < 1$
(B) If $1 < x < 2$, then $f(x)$ satisfies	(q) $f(x) < 0$
(C) If $3 < x < 5$, then $f(x)$ satisfies	(r) $f(x) > 0$
(D) If $x > 5$, then $f(x)$ satisfies	(s) $f(x) < 1$

QE0054

EXERCISE (JM)

1. Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$.
If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is : [AIEEE-2011]
(1) 18 (2) 3 (3) 9 (4) 6

QE0081

2. Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of x to get roots (3, 2). The correct roots of equation are : [AIEEE-2011]
(1) -4, -3 (2) 6, 1 (3) 4, 3 (4) -6, -1

QE0082

3. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is : [JEE-MAIN-2013]
(1) 1 : 2 : 3 (2) 3 : 2 : 1 (3) 1 : 3 : 2 (4) 3 : 1 : 2

QE0083

4. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to : [JEE-MAIN-2015]
(1) 3 (2) -3 (3) 6 (4) -6

QE0084

5. The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is :- [JEE-MAIN-2016]
(1) 5 (2) 3 (3) -4 (4) 6

QE0085

6. Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to : [JEE(Main)-2019]
(1) 512 (2) -512 (3) -256 (4) 256

QE0086

7. The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is : [JEE(Main)-2019]
(1) 2 (2) 5 (3) 3 (4) 4

QE0087

8. Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0,2) and its other root lies in the interval (2,3). Then the number of elements in S is : [JEE(Main)-2019]

(1) 11 (2) 18 (3) 10 (4) 12

QE0088

9. If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m-4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is : [JEE(Main)-2019]

(1) $2 - \sqrt{3}$ (2) $4 - 3\sqrt{2}$ (3) $-2 + \sqrt{2}$ (4) $4 - 2\sqrt{3}$

QE0089

10. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true ? [JEE (Main)-2020]

(1) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$ (2) $p_5 = 11$
(3) $p_3 = p_5 - p_4$ (4) $p_5 = p_2 \cdot p_3$

QE0090

11. The least positive value of 'a' for which the equation $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is

[JEE(Main)-2020]

QE0091

12. If $A = \{x \in \mathbf{R} : |x| < 2\}$ and $B = \{x \in \mathbf{R} : |x - 2| \geq 3\}$; then :

(1) $A \cup B = \mathbf{R} - (2, 5)$ (2) $A \cap B = (-2, -1)$
(3) $B - A = \mathbf{R} - (-2, 5)$ (4) $A - B = [-1, 2)$

[JEE(Main)-2020]

QE0092

13. Let p and q be two positive numbers such that $p + q = 2$ and $p^4 + q^4 = 272$. Then p and q are roots of the equation : [JEE(Main)-2021]

(1) $x^2 - 2x + 2 = 0$ (2) $x^2 - 2x + 8 = 0$
(3) $x^2 - 2x + 136 = 0$ (4) $x^2 - 2x + 16 = 0$

QE0189

14. The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ is equal to

[JEE(Main)-2021]

(1) $1.5 + \sqrt{3}$ (2) $2 + \sqrt{3}$ (3) $3 + 2\sqrt{3}$ (4) $4 + \sqrt{3}$

QE0190

15. The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is :

[JEE(Main)-2022]

- (A) 4 (B) 5 (C) 6 (D) 8

QE0191

16. Let $S = \left\{ x \in [-6, 3] - \{-2, 2\} : \frac{|x+3|-1}{|x|-2} \geq 0 \right\}$ and $T = \{x \in \mathbb{Z} : x^2 - 7|x| + 9 \leq 0\}$. Then the number of elements in $S \cap T$ is

[JEE(Main)-2022]

- (A) 7 (B) 5 (C) 4 (D) 3

QE0192

17. The number of integral values of k , for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other root lies in the interval $(2, 3)$, is :

[JEE(Main)-2023]

- (1) 2 (2) 0 (3) 1 (4) 3

QE0193

18. Let $S = \left\{ \alpha : \log_2(9^{2\alpha-4} + 13) - \log_2\left(\frac{5}{2} \cdot 3^{2\alpha-4} + 1\right) = 2 \right\}$. Then the maximum value of β for which the

equation $x^2 - 2\left(\sum_{\alpha \in S} \alpha\right)^2 x + \sum_{\alpha \in S} (\alpha + 1)^2 \beta = 0$ has real roots, is _____.

[JEE(Main)-2023]

QE0194

EXERCISE (JA)

1. The smallest value of k , for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is [JEE 2009, 4 (-1)]

QE0093

2. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is [JEE 2010, 3]

- (A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

QE0094

3. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is [JEE 2011]

- (A) 1 (B) 2 (C) 3 (D) 4

QE0095

4. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$

have one root in common is -

[JEE 2011]

- (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

QE0096

5. Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

[JEE 2015, 4M, -0M]

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

QE0097

PARAGRAPH

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

6. If $a_4 = 28$, then $p + 2q =$

(A) 14

(B) 7

(C) 12

(D) 21

[JEE(Advanced)-2017, 3(-1)]

QE0098

7. $a_{12} =$

(A) $2a_{11} + a_{10}$

(B) $a_{11} - a_{10}$

(C) $a_{11} + a_{10}$

(D) $a_{11} + 2a_{10}$

[JEE(Advanced)-2017, 3(-1)]

QE0098

8. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is

[JEE(Advanced)-2020]

(A) 0

(B) 8000

(C) 8080

(D) 16000

QE0169

9. For $x \in \mathbb{R}$, then number of real roots of the equation $3x^2 - 4|x^2 - 1| + x - 1 = 0$ is ____.

[JEE(Advanced)-2021]

QE0170

ANSWER KEY

Do yourself-1

- 1 (a) $-1, -2$; (b) 4 ; (c) $1 \pm \sqrt{2}$;
 2 $a, \frac{1}{a}$; 3. $\frac{7}{3}$ 4. $3, -\frac{1}{5}$
 5. $28x^2 - 20x + 1 = 0$ 6. $a_1 = -2, a_2 = 1$
 7. $a_1 = 2, a_2 = 9/2$ 8. $p = \pm 7$
 9. $k = \pm 3\sqrt{5}$ 10. $p = 0$
 11. $a = 2$ 12. $x_1^3 + x_2^3 = 3pq - p^3$
 13. $x_1^3 + x_2^3 = \frac{a(a^2 - 18a + 9)}{27}$ 14. $\frac{215}{27}$

Do yourself-2

1. (a) imaginary; (b) real & distinct; (c) real & coincident
 3. $a = 20 \pm 6\sqrt{5}$ 4. $k_1 = \frac{-22}{3}, k_2 = 2$ 5. $a = 4$
 6. $a = \pm 2$ 7. $k = 3$ 9. For all $m \in \left(\frac{1}{4}, +\infty\right)$
 10. For all $c \in [2, 4)$ 11. $k = 13$ 12. For all $m \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$
 13. For all $m \in \left(-\infty, -\frac{1}{7}\right) \cup (1, \infty)$ 14. $b = -4, c = 1$;
 15. 7 16. -15
 17. (a) $c = 0$; (b) $c = 1$; (c) $b \rightarrow \text{negative}, c \rightarrow \text{negative}$ 18. (ii), (iii) and (iv)
 19. (A) $\frac{b^2 - 2ac}{c^2}$ (B) $\frac{bc^4(3ac - b^2)}{a^7}$ (C) $\frac{b^2(b^2 - 4ac)}{a^2c^2}$
 20. $\frac{1}{x_1^3} + \frac{1}{x_2^3} = -\frac{27a^3 + 36a}{8}$ 21. $a_1 = 3/2, a_2 = 3$ 22. $a = -4$
 23. $a_1 = -\frac{3}{2}, a_2 = 6$. 24. $\{2, 18\}$ for $a = 6, \left\{\frac{2}{19}, \frac{18}{19}\right\}$ for $a = -\frac{6}{19}$
 25. $a_1 = -\frac{125}{8}, a_2 = \frac{27}{8}$ 26. $a_1 = 1/2, a_2 = 1$ 27. $p_1 = 0, q_1 = 0, p_2 = 1, q_2 = -2$

Do yourself-3

1. (a) $c^2y^2 + y(2ac - b^2) + a^2 = 0$; (b) $acx^2 - bx + 1 = 0$;
(c) $acx^2 + (a + c)bx + (a + c)^2 = 0$
2. $x^2 - 2(p^2 - 2q)x + p^2(p^2 - 4q) = 0$
3. $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

Do yourself-4

1. 1
2. $b = \frac{9}{2}, c = 5$
3. $c = 0, c = 6$
4. $a = -2$
5. $m = -2$

Do yourself-5

1. $x \in \mathbb{R}$
2. $x \in \mathbb{R}$
3. $x \in \left(-\infty, \frac{1}{2}\right) \cup (3, \infty)$
4. $x \in (-6, -3) \cup \left(\frac{1}{2}, 2\right) - \{1\} \cup (9, \infty)$
5. $[3, 7]$
6. $x \in \mathbb{R}$
7. $x \in (-\infty, -3) \cup (-2, 3)$
8. $\left(\frac{1}{2}, 3\right)$
9. $(2, 3)$
10. $(-\infty, -2) \cup \left(-2, -\frac{1}{2}\right) \cup (1, \infty)$

Do yourself-6

1. $x \in \mathbb{R} - (0, 1]$
2. $(1/2, 2]$
3. $(3, +\infty)$
4. $[1, 2) \cup (2, +\infty)$
5. $[20/9, 4) \cup (5, +\infty)$
6. $\left(\frac{1}{2}, \frac{5}{2}\right]$
7. $[0, 2]$
8. $x \in [-18, -2)$

Do yourself-7

1. $(-\infty, -10] \cup [10, \infty)$
2. $x \in [0, 10]$
3. $x \in [-2, 6]$
4. $\left[-\frac{7}{2}, 1\right] \cup \left[2, \frac{13}{2}\right]$
5. $(-\sqrt{2}, \sqrt{2})$
6. ϕ
7. $\left(-\frac{3}{2}, \frac{3}{2}\right)$
8. ϕ
9. $\left(1, \frac{7}{3}\right)$
10. $[-1, 3]$
11. $(-\infty, -2) \cup (2, \infty)$
12. $\left(-\frac{6}{7}, \frac{10}{7}\right)$
13. $x \in \mathbb{R}$
14. $\left(-\infty, -\frac{7}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$

Do yourself-8

- (a) 1 (b) -1
- (1) (i) $a < 0$ (ii) $b < 0$ (iii) $c < 0$ (iv) $D > 0$ (v) $\alpha + \beta < 0$ (vi) $\alpha\beta > 0$
 (2) (i) $a < 0$ (ii) $b > 0$ (iii) $c = 0$ (iv) $D > 0$ (v) $\alpha + \beta > 0$ (vi) $\alpha\beta = 0$
 (3) (i) $a < 0$ (ii) $b = 0$ (iii) $c = 0$ (iv) $D = 0$ (v) $\alpha + \beta = 0$ (vi) $\alpha\beta = 0$
- Third quadrant
- (a) $a > 9/16$ (b) $a < -2$
- $c < 0$
- For all $a \in (-6, 3)$
- $(-3, 5)$
- $k = 3$
- $\left[0, \frac{1}{2}\right)$
- $k = 5$
- For all $a \in (6, +\infty)$

Do yourself-9

- $(-\infty, 2] \cup [4, \infty)$
- $[0, 1)$
- $\left[\frac{1}{3}, 3\right]$
- $\left[-\frac{1}{11}, 1\right]$
- $(-\infty, 5] \cup [9, \infty)$
- $\frac{1}{3}$

Do yourself-10

- (a) α and β are greater than -3
 (b) '0' lies in between the roots α and β .
 (c) α and β are less than 1.
- C
- $a < 2$
- $a < 2$
- $12 < \lambda < 16$
- For all $a \in \left(2, \frac{9}{4}\right]$

Do yourself-11

- 0, 2
- 2
- ± 7
- $(n\ell' - n'\ell)^2 = (mn' - m'n)(\ell m' - \ell'm)$
- $(aa' - bb')^2 = -4(ah' + b'h)(a'h + bh')$

Do yourself-12

- 27, 5
- (a) $\frac{1}{a^2}(b^2 - 2ac)$ (b) $-\frac{c}{d}$, (c) $\frac{1}{a^2}(3ad - bc)$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	C	A	D	A	A	A	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	D	B	A	D	D	A	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	A	B	D	B	B	B	C	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	A,C,D	C,D	A,D	C,D	B,C	A,B,C	B,D	A,B,C	B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,D	A,B	A,B,D	A,B,D	B,D	B,C,D	A,B,D	A,B,C,D	A,D	A,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	D	B	A,C	B,C,D	A,B,C,D	3.00	2.00	A	
Q.10	A	B	C	D						
	p,r,s	q,s	q,s	p,r,s						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	254	191	24	5	7	4	4	3	3	20

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	1	1	2	3	3	1	2	4
Que.	11	12	13	14	15	16	17	18		
Ans.	8.00	3	4	1	C	D	3	25		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	
Ans.	2	B	C	B	A,D	C	C	D	4	